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4. United Nations Development Programme

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6. Project title: Benin National Child Project under the Africa Minigrid Program		
7. Country(ies): Benin	**8. Implementing Partner (GEF Executing Entity):** Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)	*9.* **Execution Modality**: 10. National Implementation Modality (NIM)
11. Contributing Outcome (UNDAF/CPD, RPD, GPD): 12. **UNDAF outcome 1:** By 2023, Benin's population, especially the most vulnerable, are more resilient and have a better quality of life through access to decent employment, food and nutrition security, clean energy, and the sustainable management of natural resources, the adverse effects of climate change, crises and disasters[1] 13. **CPD Output 1.3:** Solutions adopted for rural and urban populations to achieve access to clean, affordable and sustainable energy.[2]		
14. UNDP Social and Environmental Screening Category: 15. Substantial		**16. UNDP Gender Marker:** 17. 2
18. Atlas Award ID: 00142903		**19. Atlas Project/Output ID:** 00131156
20. UNDP-GEF PIMS ID number: 6658		**21. GEF Project ID number:** 10804
22. LPAC meeting date: 15 October 2022		
23. Last possible date to submit to GEF: 18 September 2022		
24. Latest possible CEO endorsement date: 19 December 2022		
25. Project duration in months: 48 months		
26. Planned start date: 10 October 2023		*27.* **Planned end date:** -30 September 2027
28. **Expected date of Mid-Term Review:** 1 March 2025		**29. Expected date of Terminal evaluation:** 1 February 2027
30. **Brief project description:** 31. As part of the UNDP-supported, GEF-financed Africa Minigrids Program (AMP), the Benin Project seeks to increase access to clean energy by increasing the financial viability, and promoting scaled-up commercial investment, in renewable energy (RE) minigrids, with a focus on cost-reduction levers and		

[1] Benin UNDAF, or Sustainable Development Cooperation Framework (link).
[2] Benin CPD (link)

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innovative business models. This will be achieved through five outcomes as follows: (i) Stakeholder ownership in a national minigrid delivery model is advanced, and appropriate policies and regulations are adopted to facilitate investment in low-carbon minigrids; (ii) Innovative business models based on cost reduction operationalized, with strengthened private sector participation in low-carbon/renewable energy minigrid development; (iii) Financial sector actors are ready to invest in a pipeline of low-carbon minigrids and concessional financial mechanisms are in place to incentivize scaled-up investment; (iv) Digitalization and data mainstreamed, across stakeholders, into local minigrid market development; and (v) Ensuring compliance with all mandatory monitoring and reporting requirements of the GEF.

32. The Project will increase knowledge, awareness and network opportunities in the minigrid market and among stakeholders, including benefitting from linkages to international good practice. The project is expected to bring about the direct commissioning of approximately 0.4 MW in solar photovoltaic (PV) generation capacity and 0.9 MWh of battery storage. The lifetime greenhouse gas (GHG) emissions reduction from project activities, particularly investment in minigrid pilots, is estimated at 9,000 metric tons of carbon dioxide equivalent (tCO2eq) (direct) and 180,000 tCO2eq (indirect). The number of direct beneficiaries is estimated at 12,000 people, of which at least 50% are women, as a result of 2,400 new and/or improved minigrid connections.

33. FINANCING PLAN

34. GEF Trust Fund grant	35. USD 1,326,147
36. UNDP TRAC resources	37. USD 200,000
38. Confirmed cash co-financing to be administered by UNDP	39. USD 0
(1) Total Budget administered by UNDP	**40. USD 1,526,147**
41.	
(2) Total confirmed co-financing to this project not administered by UNDP	**42. USD 33,000,000**
(3) Grand-Total Project Financing (1)+(2)	**43. USD 34,526,147**

44. SIGNATURES:

45. **Signature:** print name below **46.**	**47. Agreed by Government Development Coordination Authority**	***48.* Date/Month/Year:** *within 6 months of GEF CEO endorsement* 49.
50. **Signature:** print name below **51.**	**52. Agreed by Implementing Partner**	***53.* Date/Month/Year:** *within 6 months of GEF CEO endorsement* **54.**
55. **Signature:** print name below	**56. Agreed by UNDP**	**57. Date/Month/Year:** *within 6 months of GEF CEO endorsement*

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138. LIST OF ACRONYMS

AfDB	African Development Bank
ABERME	*Agence Béninoise de l'Electrification Rurale et de la Maîtrise de l'Energie*
AFD	*Agence Française de Développement*
AMDA	African Minigrid Developers Association
AMP	Africa Minigrids Program
ANCB	*Association National des Communes du Bénin*
ANM	*Agence Nationale de Normalisation, de Météorologie et du Contrôle Qualité*
ARE	*Autorité de Régulation d'Electricité*
BESS	Battery Energy Storage System
BOAD	*Banque Ouest-Africaine de Développement*
BPSNCE	Bureau for Policy and Programme Support, Nature, Climate and Energy
CAPEX	Capital Expenditure
CEB	*Communauté Électrique du Bénin*
CFP	Centre for Professional Formation
CO	Country Office
CoP	Community of Practice
DGRE	*Direction Générale des Ressources Energétiques*
DREI	Derisking Renewable Energy Investment
ACVDT	Living Environment Agency for Territorial Development
ECOWAS	Economic Community of West African States
ECREEE	ECOWAS Centre for Renewable Energy and Energy Efficiency

REEE	
EE	Energy Efficiency
EE	*Système d'Échonges d'Énergie Éleclrique Ouesl Africain (WAPP)*
EEOA	
EIB	European Investment Bank
EPC	Engineering, Procurement and Construction
ESCO	Energy Service Company
ESIA	Environmental and Social Impact Assessment
ESMAP	Energy Sector Management Assistance Program
ESMF	Environmental Safeguards Management Framework
ESMP	Environmental and Social management Plan
FER	*Fond d'Electrification Rurale*
FERER	*Fonds d'Electrification Rurale et des Energies Renouvelables*
FSP	Full Sized Project
GAP	Gender Action Plan
GDP	Gross Development Product
GEF	Global Environment Facility
GEFSEC	Global Environment Facility Secretariat
GHG	Greenhouse Gases
GIS	Geographic Information System
GIZ	German International Cooperation
GOB	Government of Benin
HVAC	Heating, Ventilation and Air Conditioning
IMF	International Monetary Fund
IP	Implementing Partner
IPP	Independent Power Producer

IR	Inception Report
IW	Inception Workshop
KPI	Key Performance Indicator
LCOE	Levelized Cost of Energy
LDC	Least Developed Country
LPG	Liquified Petrol Gas
MCC	Millennium Challenge Corporation
MCVDD	*Ministère de Cadre de Vie et du Developpement Durable*
ME	*Ministère d'Energie*
MEND	*Ministère de l'Economie Numérique et de la Digitalisation*
MG	Mini-Grid
MPP	Minigrid Pilot Plan
MTR	Mid-Term Review
MH(h)	Megawatt(hour)
M&E	Monitoring & Evaluation
NDC	Nationally Determined Contribution
NIM	National Implementation Modality
OFP	Operational Focal Point (GEF)
OGEF	Off-Grid Electrification Fund
OPEX	Operational Expenditure
O & M	Operation & Maintenance
PAG	*Programme d'Actions du Gouvernement*
PAYG	Pay-as-you-Go
PB	Project Board
PDE	*Plan Directeur de l'Electricité*
PDEH	*Plan Directeur de l'Electrification Hors Réseau*

PERHR	*Politique de l'Electrification Hors Réseau*
PFD	Program Framework Document
PIF	Project Identification Form (GEF)
PIR	Project Implementation Report (GEF)
PMC	Project Management Cost
PMU	Project Management Unit
PND	*Plan National de Développement*
PONAME	*Politique Nationale de Maîtrise de l´Énergie*
PONADER	*Politique Nationale du Développement des Energies Renouvelables*
POM	Project Operations Manual
POPP	Programme and Operations Policies and Procedures
PPA	Power Purchase Agreement
PPG	Project Preparation Grant
PPP	Public-Private Partnership
PSC	Project Steering Committee
PUE	Productive Use of Energy
PV	Photo-Voltaic
QAMF	Quality Assurance and Monitoring Framework
RBF	Results-Based Framework
RE	Renewable Energy
RMI	Rocky Mountains Institute
SBEE	*Société Béninoise d'Energie Electrique*
SDG	Sustainable Development Goal
SESP	Social and Environmental Screening Procedure

SP	
SEfor All	Sustainable Energy for All
SHS	Solar Home System
SME	Small- and Medium Enterprise
STAP	GEF Scientific Technical Advisory Panel
TBWP	Total Budget and Work Plan
TE	Terminal Evaluation
ToC	Theory of Change
ToR	Terms of Reference
T&D	Transmission & Distribution
UEF	Universal Energy Facility
UGEAP	Universal Green Energy Access Programme
UNDP	United Nations Development Programme
USD, US$	United States Dollar
VAT	Value Added Tax
WAPP	West African Power Pool
WB	World Bank

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384. II. DEVELOPMENT CHALLENGE

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1. The Republic of Benin (Benin) is located in the Gulf of Guinea in West Africa, bordering Burkina Faso, Niger, Nigeria and Togo. A majority of its 11.5 million population lives in the 125km long southern coastal region, which includes the port capital of Cotonou. The population is 52% rural, and predominantly employed in the agricultural sector, with cotton being a key export commodity driving 40% of the foreign currency income and employing 40% of the rural population.[3] Subsistence agriculture has a low level of mechanization and irrigation, but a significant potential for growth: arable land represents over 62% of the territory, but only 20% are currently cultivated.

2. Benin, a Least Developed Country, ranks among the countries with the highest levels of poverty and inequality in the world. Its Human Development Index in 2019 was 0.545, placing the country in the 158th position out of 189 countries and territories.[4] The national headcount poverty rate was estimated at 40.1% in 2015 as against 49.5% in 2008. The African Development Bank (AfDB) estimated annual GDP growth rate at 5.5% in 2017 but as a result of fast demographic expansion, per capita GDP growth is only 3.1%. As a consequence, poverty levels in Benin decline slower than hoped. The international Monetary Fund (FMI) points out that economic growth should be above 6% to enable a steady reduction of poverty levels.

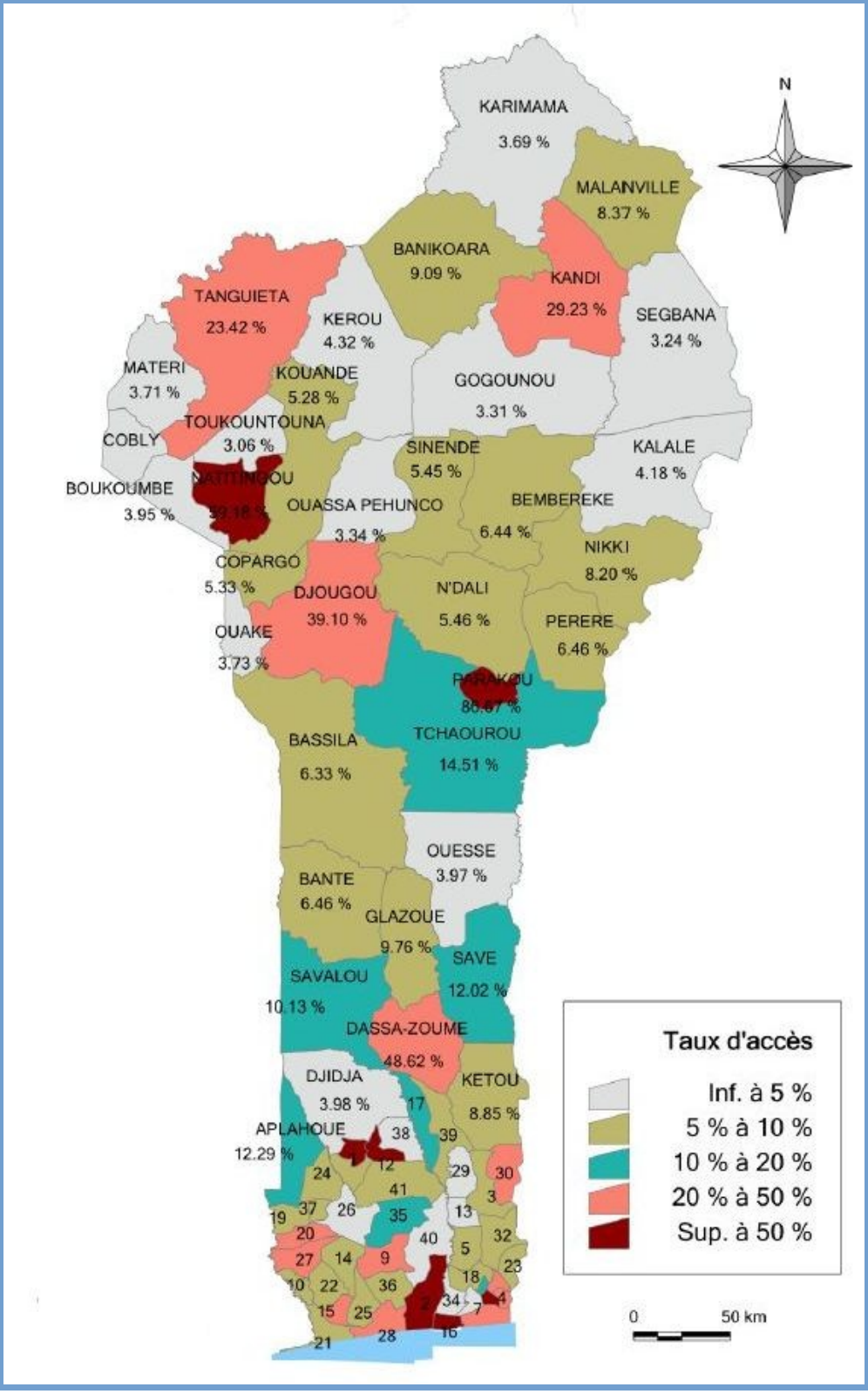
3. Global greenhouse gas (GHG) emissions in 2018 amounted to 16.9 million tons (Mton) CO2eq, or about 1.5 ton CO2eq per capita, excluding the LULUCF. Sector emissions are as follows: energy (58,09 %); agriculture (28,51 %); waste (5,38 %); industrial processes (1,22 %); and other sources including wildfires and HFC releases (6,80 %). Including carbon sequestration by the LULUCF sector, Benin's net emissions are calculated at 9.6 Mton CO2eq.[5] Primary energy supply (4,809 ktep, 2020) is dominated by traditional biomass (62%) and petroleum products (33%), followed by natural gas (3%), coal (1%), and electricity (1-2%). Energy Sector emissions (10.6 MtCO2 in 2020) are dominated by charcoal production (5.7 MtCO2) and biomass use by the commercial/institutional (4.2 MtCO2) and residential (8.1 MtCO2) sectors.[6]

4. Total final energy consumption in Benin is estimated at 0.39 ton oil-equivalent[7] (tep/yr) per capita (2020). Electricity consumption per capita grew by 1.58% over the period 2016-2020, reaching a meager 133 kWh per capita by 2020.[8] This figure demonstrates the situation of energy poverty that characterizes rural and peri-urban regions. Access to electricity has improved; the national electrification rate in 2020 is estimated at 30.4% and 57.4% in urban areas. This figure contrasts strongly with the rate in rural areas, which is as low as 5.7%,[9] implying that about 4.9 million people in rural Benin have no access to electricity.[10] The geographic disparities and the challenge to provide equitable access to electricity for all people in the country is demonstrated by the Figure below which provides additional information on the distribution of this access.

[3] https://www.wto.org/english/tratop_e/agric_e/01_benin.pdf
[4] http://hdr.undp.org/en/countries/profiles/BEN
[5] Benin Updated Nationally Determined Contributions (NDC), 2021.
[6] Source: National Energy Balance - *Chiffres Clés* 2021.
[7] Source: National Energy Balance - *Chiffres Clés* 2021, p.12 (1 tep = 41.868 GJ = 11.63 MWh).
[8] About 1/3 of the average in Sub-saharan Africa.
[9] *Rapport du Système d'Information Énergétique* (SIE) Benin, 2020.
[10] If electricity self-supply such as diesel and gasoline generators is taken into account, the figures increase to 33.4%, 60.3% and 8.7% respectively (INSAE data, 2020).

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386. **Figure 1 Geographic disparities in electricity access per commune in Benin. High electrification rates are only found in the urban centers while 20 communes still exhibit rates lower than 5%. Adapted from: National Energy Balance – *Chiffres Clés 2021*, p.16.**



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5. The Ministry of Energy (ME) is responsible for formulating and implementing national energy policy and regulations. The energy regulator is the aforementioned *Autorité de Régulation d'Electricité* (ARE), or Electricity Regulation Authority. The national utility is the *Société Béninoise d'Energie Electrique* (SBEE). The *Communauté Électrique du Bénin* (CEB)[11] is a generation and transmission utility owned by Benin and Togo. The new Electricity Law (2020) ended SBEE's monopoly and opened the sector to private investment in electricity generation, transmission, and distribution. It also reinforces the capacity of the ARE to ensure competition. With support from development partners, the Government of Benin (GOB) approved the construction of 50 MW on-grid solar PV through public-private partnerships (PPP). In September 2019, the GOB appointed a private company to manage SBEE for a period of four years to improve management and performance. The newly created Beninese Electricity Production Company (SBPE) will serve as a holding for public generating assets including future renewable energy (RE) power plants.

6. Benin imported 98.5% of its electricity in 2018. According to ARE, installed capacity was more than 60 MW below peak demand (240 MW). Electricity demand is growing (over 5% in 2015) due to expanding economic activities and population growth but the country faces challenges to develop and finance investment in the sector. Benin is member of the Economic Community of West African States (ECOWAS) and the *Système d'Échonges d'Énergie Éleclrique Ouesl Africain (EEEOA)* - the West African Power Pool (WAPP). Benin follows the ECOWAS approach in its adoption of the SEforALL Country Action agendas and RE and Energy Efficiency action plans.

7. Even so, large areas of the country will not be covered by the main grid in the foreseeable future; neither would the disperse population with low demand make such investments economically feasible. Arrears in upgrading of the sector have affected electricity supply to urban areas and industries, hampering prospects for inclusive economic growth in the country. The increase of biomass (2.91%) and decrease of hydroelectricity (-5.6%) also point at environmental challenges. The low level of energy inputs consumed by industry (4.1%), agriculture (0.3%), and services (10%) contrasts with households and transport (together 85%), which demonstrates the low level of mechanization of Benin's economy. Biomass is the main energy source for households, while transport sector is driven by gasoline and gasoil. Charcoal conversion efficiencies are estimated at 15% average. Solar PV production in 2019 was approximately 5.5 GWh.[12] Since 2016, Benin has added gas-fired power plants and, by now, over 70% of generating capacity is based on this source. It should be noted that the gas used in such plants is also imported.[13] Yet, Benin has vast natural gas reserves and RE potential, including hydro, biomass and solar energy. [14]

8. Benin has set the objective to attain electrification rates of 95% (urban) and 65% (rural), as well as a 24.6 % RE share of the national energy mix by 2025.[15] The commitment to RE is also reflected in the updated Nationally Determined Contribution (NDC), which seeks to bring the country on a low-emission pathway, including by promoting electricity access in rural areas and phase out fossil fuel (e.g. kerosene) and reducing reliance on charcoal and fuelwood. The ambitions set forth in the NDC 2020 are as follows: (i) interconnection of approx. 2,323 communities (villages) between 2021 – 2030; (ii) provision of electric lighting to 1,028,000 new households (replacing kerosene) in villages that will be served by the SBEE.[16] These plans imply a 4-fold acceleration compared to the period 2017-22, in which the (on-grid) Electrification Master Plan (PDE) proposed a goal of 495 communities in all inland departments, equivalent to about 80 connections per year. Off-grid

[11] Governed by an International Agreement instituting the Benin-Togo Electricity Code signed on 27 July 1968.

[12] The solar PV potential in most of the country is in the range 1400-1600 kWh/kWp, per year. https://www.irena.org/IRENADocuments/Statistical_Profiles/Africa/Benin_Africa_RE_SP.pdf

[13] Totalling 154,000 TEP in 2020. Source: National Energy Balance - Chiffres Cles 2021, p.19.

[14] https://www.iea.org/countries/benin

[15] https://www.se4all-africa.org/seforall-in-africa/country-data/benin/

[16] NDC Action "*Etendre l'accès des ménages à l'éclairage électrique en remplacement de l'éclairage au kérosène*."

17. *Volet raccordement au réseau, actions 4-5*.

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electrification envisions: (iii) access to electric lighting for households and public facilities by individual solar kits to reach 13,249 households by 2024 and another 100,000 between 2025-2030; (iv) access to electric light for social-community facilities (46 health centers and 26 police station; and (v) rural electrification through small solar PV power plants (261 communities).[17]

389. Legal framework for off-grid electrification

9. Acknowledging the technical and financial constraints of main grid extension, a rural electrification model by private companies covering fifteen territorial concessions (CER) was already established back in 2000. However private investors did not respond as hoped. Since then, rural electrification has been pushed forward through public investment, donor agencies, and contributions from the *Fond d'Electrification Rurale* (FER- Rural Electrification Fund), created in 2007. From 2014 onward, rural electrification programmes (which included minigrids) were developed and implemented through ANADER and its successor, the *Agence Béninoise de l'Electrification Rurale et de la Maîtrise de l'Energie* (ABERME), which is ascribed to the ME. The PRODERE project, funded by EUMOA, electrified 6 villages while PROVES, funded by the Government through an AfDB loan, targeted 105 communities, using solar PV technology. Despite progressive main grid expansion efforts, in 2017 it was estimated that over 2.3 million people in about 1,800 communities would not be reached by the 2028 deadline.[18]

10. Over the last decade, Benin has made significant progress towards modernization of its energy and electricity sector including the challenge to reach currently unserved and underserved population. A national policy and master plan (*Plan Directeur de l'Electrification Hors Réseau* – PDEHR) was approved in 2019 to complement the national grid expansion plan (*Plan Directeur de l'Electricité* – PDE). [19] Another important policy is the *Politique Nationale de Maîtrise de l´Énergie 2021-2030* (PONAME - National Energy Management Policy), which seeks promoting a national ecosystem for the production and rational use of energy including domestic RE sources such as biomass and small hydropower -alongside solar PV- and the development of qualified human resources, technology transfer, and innovation.

11. **Key challenges for rural electrification and minigrids** are captured within the above-mentioned PDEHR, and include the following:

    (i) mobilizing new actors,

    (ii) mobilizing new investors to leverage public investment and secure long-term management and expansion of local grids, and

    (iii) developing local competences for technical management, payment systems, and business operations.

Putting private operators at the heart of the model, the provision of an adequate regulatory framework is considered essential to reduce technical and financial risks. Under the above-mentioned programs, the ownership of the minigrids was ultimately transferred to the local community (through a mayoral office – the *Mairie)* with management, O&M, billing and customer services in the hands of a committee that performed as a local electric utility. Due to a complex of factors including but not limited to technical capacity, funding constraints, and lack of spare parts, this particular minigrid delivery model faced issues. Private sector involvement under a concession scheme can potentially address several of the issues encountered in the past. In 2020, a new Electricity Law was approved (*Loi No. 2020-05*, 1 April 2020) which integrates national on-grid and off-grid policies into one comprehensive framework in which off-grid electrification is pursued through a scheme of concessions led by the Ministry of Energy.[20]

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17 NDC, *Volet système électrique hors réseau, actions* 14-16.

18 PEHR, 2017.

19 The Plan Directeur d´Electrification Hors Réseau, prepared for the Ministry of Energy of Benin by IED Innovation Energie Développement (IED) consultants and Practical Action and funded through the Millennium Challenge Account Benin (MCA Benin II).

20 https://are.bj/wp-content/uploads/2017/09/LOI-N-2020-05-PORTANT-CODE-DE-LELECTRICITE-EN-REP-DU-BENIN_1.pdf. The law governs all aspects of electricity in Benin except: those subject to existing

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12. The following table summarizes the policy context for RE-based minigrids in Benin.

390. Table 1. Policy context for renewable energy minigrids in Benin

391. Policy / planning document		392. Name	393. Relevance
394. Sector Policies	395. Strategic Vision / Development Plan	*396. Plan National de Développement (PND) 2016-2025*	397. National Development Plan
		398. Etudes Nationales de Perspectives à Long Terme « Bénin Alafia 2025»	399. Supportive studies to the National Development Plan
		400. Programme d'Actions du Gouvernement «Bénin révélé» (PAG) 2021-2026	401. Five-year GOB investment plan
		402. Politique Nationale de Décentralisation et de Déconcentration.(PNDD)	403. National decentralization policy
	404. National energy policies and plans	*405. Politique Nationale de Maitrise d'Energie 2021-2030*	406. National Policy to improve national energy self-sufficiency
		407. Plan d'Action National pour la Politique d'Intégration du Genre dans l'Accès à 408. l'Énergie 2020 – 2024	409. National Gender Action Plan for energy sector
		410. La Politique Nationale du Développement des Energies Renouvelables (PONADER) 2020 – 2035	411. National RE development plan
		412. Politique Électricité Hors-Reseau (PEHR) 2018 – 2035	413. National Off-grid Electrification Policy
		414. Plan Directeur d'Électrification Hors Réseau (PDRHR) 2017	415. Off-grid Electrification Master Plan
	416. National policies and plans concerning environment and climate change sector	417. Updated Nationally Determined Contributions to the UNFCCC (NDC), 2021	418.
		419. Initial Nationally Determined Contributions to the UNFCCC (INDC) 2015	420.
		421. Plan National d'adaptation aux changements climatique (PNA) 2022	422. Updated National Climate Change Adaptation Plan
		423. Programme National de Gestion Durable des Ressources Naturelles (PNGDRN)	424. National Plan for the Sustainable Management of Natural Resources
425. Legislation	426. Overall legislation	*427. Constitution du 11 Décembre 1990 (Art 27),telle que révisée par la Loi n°2019-40 du 07 novembre 2019*	428. Assigns constitutional rights to all people in Benin to a health, adequate, and sustainable environment, as well as the obligation to preserve this. Further assigns the State with the duty to protect the environment.
		429. La Loi n° 2007-21, du 16 octobre 2007, portant sur la	430. Consumer protection law.

bilateral agreements, electricity supply systems for telecommunication installations, and thermal self-supply that is not commercialized (Art.2).

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391. Policy / planning document		392. Name	393. Relevance
		protection des consommateurs	
431. Laws	432. Energy Sector Laws and bylaws	*433. Loi n 2020-05, 1er avril 2020 portant Code de l'Electricité en République du Bénin, et ses textes d'application.*	434. National Electricity Law in vigor since 2020 governing on- and off-grid electricity sectors.
		435. Loi N° 2006-16, 27 mars 2007 (Code de l'Electricité du Bénin) et le décret 2008- 815 du 31 décembre 2008	436. Previous (2007) Electricity Law derogated by the 2020 Electricity Law
		437. Cadre National de Concertation Genre et Inclusion Sociale Energie (CC GISE) - Arrêté n 075/ME/DC/SG M/DAF/CGD/019SGG18/SA, 26 décembre 2018.	438. Provides the national framework for consultation of gender and social inclusion in the energy sector.
	439. Environmental legislation	*440. La loi N° 98-030 du 12 Février 1999 portant Loi-cadre sur l'environnement en République du Bénin*	441. Law establishing the framework for environmental legislation
442. Cross-cutting policy and regulation	443. Fiscal policy	*444. Loi n° 2021-15 du 23 décembre 2021 portant code général des impôts de la République du Bénin*	445. National fiscal law
	446. Customs	*447. Loi No 2014-20 du 12 septembre 2014 portant code des douanes; modifiée dans certains de ses articles par les lois de finances 2016-2021.*	448. National customs law
	449. Public-Private Partnerships Law or Framework	*450. La loi 2016 – 24 du 28 Juin 2017 portant cadre juridique du Partenariat Public – Privé en République du Bénin*	451. Legal framework for public-private partnerships
		452. Le Code des investissements au Bénin	453. National investment law

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455. Baseline initiatives and programmes

13. A variety of bilateral and multilateral agencies and donors support the GOB towards the attainment of its development objectives and the Sustainable Development Goals, (SDGs), including access to reliable, clean, and affordable electricity for the currently unserved and underserved population in Benin, as well as transitioning current energy systems to low-carbon ones.

14. As related to off-grid electrification, UNDP with its partners has supported the GOB through interventions including the *"Projet Commune du Millenaire de Bonou pour un Développement Durable"* (PCM Bonou)[21]; the *"Projet de Renforcement de la résilience du secteur de l'énergie aux impacts des changements climatiques au Bénin"* (PANA Energie)[22]; and the UNDP/GEF Project Promotion of Sustainable Biomass-based Electricity

[21] https://info.undp.org/docs/pdc/Documents/BEN/PRODOC%20PCM-Bonou.pdf

[22] https://energie.gouv.bj/page/projet-de-renforcement-de-la-resilience-du-secteur-de-lenergie-aux-impacts-des-changements-climatiques-au-benin-pana-energie

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Generation in Benin ("*Biomasse-Electricité*")[23]. UNDP's involvement has contributed to the adoption of climate change adaptation and mitigation measures in national policies and plans, including the PDEHR, the National Electrification Plan (PNE), the National Renewable Energy Policy 2020-2035 (PONADER), the PONAME, the Electricity Law, and the *Mécanisme de Soutien Financier* (MSF - Financial Support Mechanism). UNDP's focus thereby has been on small grids accelerating access to clean electricity for the most disadvantaged strata of the population.

15. Supported by UNDP/GEF project *Biomasse Électricité*, the GOB has boosted the utilization of decentralized RE technologies including biomass gasifiers using organic waste and agricultural residue streams to power isolated mini-grids. Outcomes of this project include contributions to the regulatory framework and notably the MSF. The latter is designed to provide payment guarantees to the private sector operators who will invest in the four (4) envisioned gasifier pilots under the project, totaling 4-MW. The experience can guide the design of financial instruments under the AMP.

16. The Millennium Challenge Corporation (MCC), through its Millennium Challenge Account-Benin II (MCA Benin II), is implementing an off-grid electricity access project consisting of 2 components: (i) the establishment of an enabling environment for off-grid electrification, which led to the adoption by the government in 2018 of its new regulatory framework; and (ii) the Off-grid Clean Energy Facility (OCEF), which selected 11 companies in a call for proposals in July 2020 for the construction of 8 solar PV minigrids (USD 60 million investment, composed of USD 24 million grant and USD 36 million private sector investment leveraged). The MCA-Benin II is entering into an additional 6th year of implementation (as of June 2022).

17. Sustainable Energy for All (SEforAll), in collaboration with several donors and partners (including the GIZ), is implementing the Benin window of its Universal Energy Facility (UEF).[24] In Benin, the UEF planned to disburse grant payments to deliver over 7,000 electricity connections based on a results-based incentive of USD 433 per connection, for a total amount of USD 3 million grant. The programme is leveraging Odyssey, an innovative digital platform using real-time operating analytics to support distributed energy data. As of September 2022, GIZ' activities have been extended for another year. The UEF has 5 developers onboard. GIZ is assisting GOB with the tendering of 66 minigrids; some developers have by now been selected. Some minigrids require refurbishment or an increase in capacity. GIZ has recently begun to assess digitalization options for minigrids with ABERME, which is aligned with the AMP approach.

18. The *Agence Française de Développement* (AFD) is focused on on-grid electrification and currently supports 4 projects (a fifth one completed recently), mostly on distribution, but also on production. The PRERA project connected 444,000 people and delivered power transformation stations. The PEDER project (also focusing on distribution) targeted 15 communes in 7 departments, providing 150,000 people with electricity access. DEFISSOL, backed by the EU developed a 25-MW solar PV plant and digital components for SBEE. The FORSUN project seeks to upgrade the capacity of this solar plant. So far, close to one million people obtained access to electricity through the AFD initiatives, at a total of € 251M investment.

19. The government-led *Projet de Valorisation de l'Énergie Solaire (PROVES)*, funded by the Government through an AfDB loan, started in 2016 and aimed at electrifying 105 villages through solar PV minigrids built by 6 different developers. More strategically, the AfDB is active in the overall sector through a US$66.5M loan for rural electrification from its ADB window which runs from 2019 to 2023.

20. Regional programs include the Universal Green Energy Access Programme (UGEAP) funded through the Green Climate Fund (GCF FP027)[25] offering equity to the private sector. Among other lines of action, it aims to provide financing for decentralized energy service companies for off-grid and mini-grid systems for rural households and communities and renewable energy for industrial players. The UGEAP is managed by Deutsche Bank and is

[23] GEF Project Promotion of Sustainable Biomass-based Electricity Generation in Benin (GEF IS 5752, UNDP PIMS 5115 ; GEF Budget US$ 3,872,602). The Projectis currently under implementation. See: https://www.thegef.org/news/building-sustainable-energy-future-benin

[24] A results-based financing scheme funded by the *Deutsche Gesellschaft für Internationale Zusammenarbeit* (GIZ) on behalf of the German Federal Ministry for Economic Cooperation and Development (BMZ).

[25] https://www.greenclimate.fund/project/fp027

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active in Benin. The *Banque Ouest-Africaine de Développement* (BOAD) implements thet Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs, also funded through the GCF (FP105)[26]. This 20-year program uses a blended finance approach to provide affordable long-term funding to solar projects and by providing tenor extension loans that will help de-risk projects, and crowd-in commercial and public banks in scaling up solar investments in the region, including Benin.

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[26] https://www.greenclimate.fund/project/fp105

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459. Barriers and risks to renewable energy minigrid development in Benin:

21. During the PPG and based on the Derisking Renewable Energy Investment (DREI) Methodology developed by UNDP, the following key barriers and risks to expansion of minigrids in the country were evaluated. Information sources were mainly stakeholder consultations, inception workshop and desk review.

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461. DREI Risk Category	462. Description/Underlying Barriers	463. Risk Level
464. **1. Energy market risk**	465. *Policy & regulatory framework, market access, competition and grid expansion:* 466. Benin's electricity sector is rapidly evolving allowing participation of private companies, unbundling of electricity generation from transmission and distribution (T&D), and the establishment of an independent regulator (ARE). A new Electricity Code was approved in 2020. There is a sense of urgency that reliance on electricity imports from neighboring countries must be reduced by exploiting domestic renewable energy sources. 467. In areas served by the national grid system, reliability of electricity services must further improve as a basis for economic growth (PONAME, 2019). Private concessions are the envisioned delivery model to reach the off-grid population (Electricity Law, Chapter IX). 468. National electrification plans exist for on-grid and off-grid areas; however, coordination issues exist which translate into a risk for minigrid investors. The model contract for minigrid concession includes provisions anticipating the arrival of the main grid.	469. Medium
	470. *Market understanding and outlook:* 471. The overall electrification rate in Benin being below 50%, opportunities for investment in the electricity sector exist in all areas. 472. The key challenges include: (i) weak public expenditure linked to Benin's position as an LDC; (ii) low purchase capacity affecting customers' capacity to pay for electricity service; (iii) nascent industrial sector including agri-processing; (iv) large unserved/underserved territory where national grid expansion is not economically feasible; and (v) insufficient collateral and/or guarantee schemes to secure (private) investment beyond electricity generation (IPP). By consequence, capital investments require careful planning and scenario analysis to ensure economic viability and reduce risks. 473. Recent investment in natural gas generation since 2016 may exacerbate Benin's reliance on imported fuels thereby weakening its commercial balance.	474. Substantial
	475. *Tariff and competing subsidies:* 476. Off-grid operators must propose a competitive, differentiated tariff scheme for their concession area, which will be subject to the Regulator for consent. It is understood that local payment capacity is too low for cost-reflective tariffs; therefore, the State will grant a subsidy to complement the tariffs, enable full recovery of operating costs plus a profit margin. 477. There is risk that the State will not be able to sustain such subsidies over the full concession period, unless guarantees can be offered.	478. Medium
	479. *Technical standards:* 480. All equipment used in Benin shall be compliant with technical standards. This is work in progress, although current projects will follow international standards. Benin lacks a test laboratory infrastructure for compliance verification	482. Medium

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461. DREI Risk Category	462. Description/Underlying Barriers	463. Risk Level
	481. Household appliances on the market are often from informal imports and may not meet applicable standards, and may lead to consumer dissatisfaction and safety issues.	
	483. *Data availability on electricity and energy end-uses:* 484. The relation between electricity supply and end-uses in rural households and communities in Benin is relatively well understood. Earlier minigrid projects faced challenges to meet demand, and field data from these projects have not been collected and analyzed as needed. By including after-sales services and appliances (sales, rental), current data gaps can be closed. 485. Fact sheets and best practices concerning PUE are work in progress in Benin. Business cases for fuel switch and electrification of traditional work methods (beyond water pumping) need further analysis and documentation.	486. Medium
487. 2. Social Acceptance risk	*488.* *Awareness of MG/RE systems:* 489. Many people in Benin are aware of minigrids and RE technologies. Small-scale PV systems and PV lanterns are sold in the street (often informally imported from Nigeria). There is a nascent but growing market for PV systems in urban areas and businesses. Hydropower and biomass power plants exist in the country but the potential remains under-exploited. Projects including *PANA Energie* aim to ensure operational feasibility of distributed RE technologies and promote market uptake. 490. Previous programs led by Government have increased awareness of the minigrid solution for off-grid electrification. However, performance issues have caused some households to seek alternative solutions such as gasoline generator or PV panels for self-supply (while others fell back to kerosene and batteries). Confidence in the minigrid model therefore has to be lifted again in these communities.	491. Medium
	492. *Willingness to pay for adequate (MG) electricity service:* 493. Prepaid payment have been implemented, including under the *PANA Energie* project. Surveys are needed to determine people´s willingness to pay, price elasticity, and acceptable tariff levels. Detailed studies and recommendations can help minigrid developers to prepare financially feasible proposals for concession.	494. Medium
	495. *Social stigmatization:* 496. In many countries and cultures, technologies have a status-bias which means that offering anything below the nominal solution (i.e. full grid power) is rejected. Benin's national policy integrates on-grid and off-grid electrification as a message that all people are served with the quality service they will need – the only difference being the delivery model.	497. Medium
498. 3. Hardware Risk	*499.* *MG hardware quality and performance risk:* 500. Although technically proven, procurement of minigrid equipment and installation, as other capital goods, shall consider warranties against defects, malfunctioning and underperformance to defer those risks that cannot or should not be borne by the operator. Approaches include extended warranties, service contracting, engineering, procurement and construction (EPC). **501.** Earlier minigrids have been identified which are in a deteriorated state and underperform, the causes thereof not being clarified. Caution is therefore needed to ensure long-term technical sustainability.	502. High
503. 4. Digital Ris	*504.* *Digital system design and delivery capacities:* 505. Digital technologies and telecommunication are governed by the *Ministère de l'Economie Numérique et de la Digitalisation (MEND)*, which seeks to increase equitable and inclusive access to communication services to all people and	506. Low

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461. DREI Risk Category		462. Description/Underlying Barriers	463. Risk Level
		businesses. Most people can use text (SMS) communication protocols, for example to access market information, or transmit system status data. Sector policies have been developed, for example in the field of e-agriculture including the implementation of digital platforms managing sector data and agents.[27] Notably, e-agriculture is viewed by public and private sector as a gamechanger for developing Benin's agricultural potential.	
		507. *Data management and purview:* 508. As the framework and procedures for digital tendering and monitoring of minigrids are not fully developed in Benin, project partners need to become familiar with aspects such as data quality, analysis and aggregation of data, confidentiality, and institutional set-up for data management. Coordination and capacity gaps expectedly exist, which need to be assessed, remedied and/or worked around.	509. Medium
510. **5. Labor Risk**		511. *Technical and professional skills:* 512. Operation and maintenance of minigrid systems requires fast responses to maintain service standards. Earlier minigrids under public programmes experienced difficulties to keep the systems up and running, probably due to a combination of factors including insufficient technical skills, lack of supplies and spare parts, and insufficient institutional back-up (including funding constraints). 513. Professional skills including background knowledge and proficiency with O&M of minigrids need to be maintained and updated regularly; similarly, management skills and competences need to be adequate to meet the challenges faced by minigrid operators. Contract terms require minigrid concessionaires to have a local representative; it may prove challenging to find such person.	514. Medium /High
515. **6. Developer Risk**		516. *Public sector expertise and track record:* 517. The public sector, specifically ABERME, has gained ample experience with minigrid development under public funding programmes. So far, the Government has limited experience with tendering of private minigrid concessions. Systemic challenges include: (i) public sector budget and capacity constraints which pose at risk consolidation of know-how and expertise within the public sector (ABERME); and (ii) transparency to assure of tender process quality; (iii) limited experience and procedures for monitoring of performance-based contracts. Automatization of the tendering process by means of digital technologies can help reduce the administrative burden on incumbent authorities, while improving transparency and accountability.	518. High
		519. *Private sector expertise and track record:* 520. Experience with private minigrids in Benin is currently being gained with funding from the MCA Benin-II and the SEforAll UEF. As such, the developer risk can be closely monitored over the next years and experiences incorporated into DREI analyses and resulting mitigation activities.	521. Medium
		522. *Business model risks:* 523. The business model for minigrid concessionaires in Benin is typical for the AMP. Since income levels in rural Benin are lower than in some other AMP countries, operators may face lower revenue streams at a similar CAPEX and OPEX level. The Off-grid Electrification Policy already anticipates on this circumstance by reserving the minigrid model to larger communities with development and PUE potential. Careful system design, including a modular approach, and cost-saving measures	525. Medium

[27] www.pgi-agriculture-benin.org.

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461. DREI Risk Category	462. Description/Underlying Barriers	463. Risk Level
	can help further reduce financial exposure. 524. A local anchor tenant can improve financial sustainability and subsidize other consumers during MG growth phase; however, such larger consumers are not widespread in rural Benin.	
	526. *Ownership risk:* 527. The ownership risk under the concession model appears well controlled. Concessionaires can make use of public (fixed) assets, which must be booked as asset and a corresponding liability. The concessionaire shall invest in additional fixed assets to deliver the service as stipulated. The State has the right to purchase the assets in case the concession is not renewed. Non-fixed assets remain in hands of the Concessionaire (but can be bought by the State). All assets are depreciated annually, and in case of public fixed assets, the Concessionaire is hold to reinvest to maintain their value/lifetime. Arbitrage is foreseen in case both parties do not agree on asset value.	528. Low
529. **7. End-user Credit Risk**	530. *Income level and collateral:* 531. Many people in rural areas are not credit-worthy as a result of poor purchase capacity and absence of collateral. However, they may wish to acquire electric appliances. Prepaid meters and SMS payment tools can be used to avoid end-user credit risks. The house wiring, according to a specified user profile, shall be included in the connection. 532. Promising opportunities to boost economic development in communities and leverage customers' money for new services and appliances, can be obtained from synergies between PAYG minigrids and telecom operators.	533. Medium
	534. *Financial products tailored to energy supply and services:* 535. Credit lines from commercial banks are available in the country. 536. There are no tools in place to assess end-users' ability to pay for electricity services and required appliances (such as credit score cards). 537. For productive uses, financiers in Benin do not accept the value of future production units (e.g., agricultural produce, workshop items, services) as collateral for lending, or they lack the knowledge and tools for proper assessment.	538. Medium
539. **8. Financing Risk**	540. *Country and sector risks and limitations:* 541. The *Fonds d´Électrification Rurale (FER)* was created by Law 2006-16 (27 March 2007) to promote investments linked to the (then) rural electrification concession (CER) model. The fund was fed by a 3 CFA levy for each kWh sold by the utility. This public funding model is insufficient to meet national rural electrification ambitions. The Government has identified private investment and operation as a way forward to reach currently unserved communities. However, low payment capacity in many communities will create challenges for minigrid investors to transit the "valley of death"[28] and become financially sustainable.	542. High
	543. *Business model and track record:* 544. Private minigrids have just started to build a track record in Benin. This may translate into a high perceived risk concerning the technical performance and the business model. The AMP Programme provides an opportunity for collecting experiences and gaining trust in the market.	545. High
	546. *Domestic investment and commercial banks:*	549. Medium

[28] The "valley of death" refers to the challenge for a viable technology or business to grow and reach the necessary scale to become financially sustainable. See also: IRENA Innovation Outlook Minigrids 2016, p.94.

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461. DREI Risk Category	462. Description/Underlying Barriers	463. Risk Level
	547. National financial institutions are not prepared to approve loans to private minigrid developers. Project proposals submitted in recent years recurred to international financiers and programs, not to the local bank sector. Government spending is backed up by multilateral and bilateral development banks and funds. 548. The local sector has limited capacity to assess applications for energy systems by customers, beyond standard consumer credit lines.	
550. 9. Currency Risk	551. *Exchange rate risk:* 552. The national currency in Benin is the African Financial Community Franc (FCFA). In use in the UEMOA, it is pegged to the Euro (1 FCFA = 0.0015 EUR). As such the exchange rate risk is the same as for USD (GEF grant currency) to EUR. Under normal market conditions, the UEMOA would not like devaluate the FCFA during the Project's time horizon and the lifetime of energy assets. Yet, the weakening of the EUR to USD exchange rate due to the war between Russia and Ukraine may induce EUMOA to reconsider its policy.	553. Low
	554. *Political risk:* 555. The country is relatively stable and responsive to the recommendations issued by its multilateral partners. Security issues have recently appeared including government overturns in Burkina Faso (2022) and in Mali (2021), and incursions from violent groups into the north of Benin. Security aspects may affect the deployment of AMP activities in the northern part of the country. They may also deter private companies from investing and give rise to a risk premium on capital.	556. Medium
557. 10. Sovereign Risk	558. Benin's systemic trade deficit is the result of the country's dependence on imports for energy needs and high value-added goods. Moreover, Benin's exports of agricultural products such as cotton and maize are highly vulnerable to the weather and price fluctuations.[29] The economy is also affected by the COVID-19 pandemic and the higher living costs resulting from the war in Russia-Ukraine. 559. This condition undermines Benin's ability to take loans to finance additional debt. The sovereign risk is mitigated by the international lending community. Benin has accessed the IMF's Extended Fund Facility (EFF) and the Extended Credit Facility (ECF) with a financial package to the tune of nearly US$650 million.[30] Without such support, public expenditures in Benin could not be continued.	560. High

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563. III. Strategy

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565. The Africa Minigrids Program (AMP)

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22. Programmatic approach. This project is part of the broader **Africa Minigrids program (AMP)**, a regional technical assistance program with the objective of supporting access to clean energy by increasing the financial viability and promoting scaled-up commercial investment in renewable minigrids, with a focus on cost-reduction levers and innovative business models. The programmatic approach aims to achieve greater impact by creating new minigrid markets across the African continent, which, in aggregate, will create scale and momentum, attracting private sector interest and investment. It will also allow for a broader sharing of knowledge and good practice, and create economies of scale in providing program services.

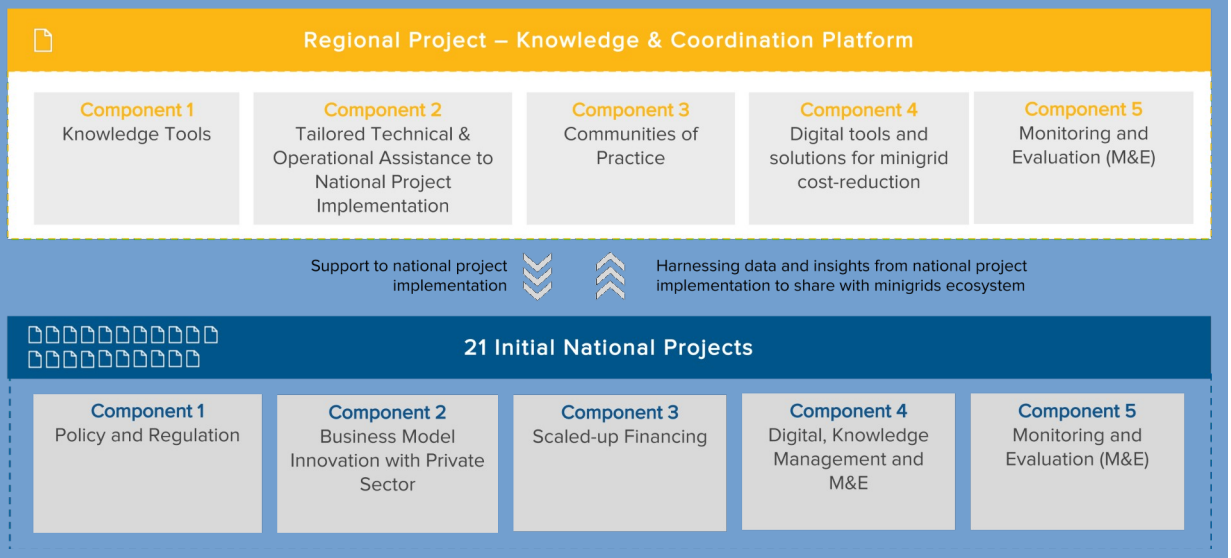
[29] https://tradingeconomics.com/benin/balance-of-trade
[30] https://www.imf.org/en/News/Articles/2022/07/14/CF-Benin-Taps-IMF-Facilities

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23. Program design. As shown in **Figure 2** below, AMP is comprised of two main elements: (i) a **Regional Project**, acting as the knowledge, advocacy and coordinating platform of the Program; and (ii) a current total of **21 AMP National Projects** (at various stages of development at the time of writing) that share a common approach, seeking to reduce minigrid costs via five country-level components: (i) policy and regulations, (ii) business model innovation with private sector, (iii) scaled-up financing, (iv) digital and knowledge management, and (v) monitoring and evaluation (M&E).

567.Figure 2. Africa Minigrids Program's Architecture

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24. The program is initially supporting three rounds of national projects, which together host an estimated total of **396 million people without electricity**,[31] or more than two thirds out of the 587 million total people without access to electricity in Africa. The initial AMP participating countries are shown in **Figure 3** below.

[31] IEA (20212, Tracking SDG7: The Energy Progress Report, 2022, IEA, Paris https://www.iea.org/reports/tracking-sdg7-the-energy-progress-report-2022

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569. **Figure 3. Africa Minigrids Program's participating countries list and map**[32]



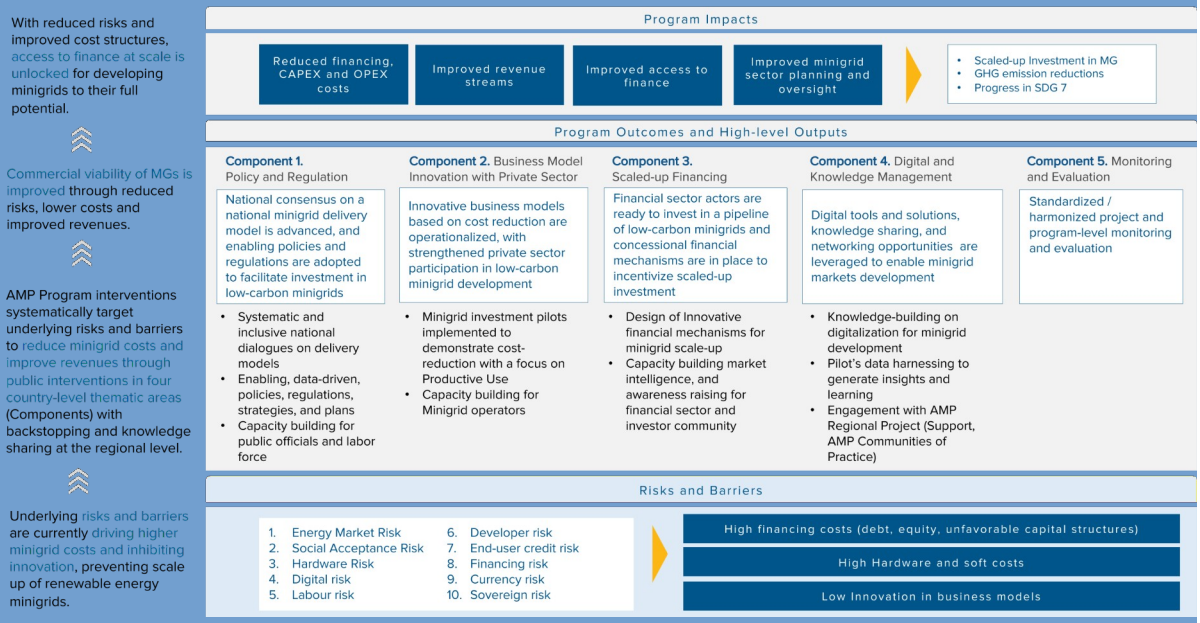
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25. Program's Theory of Change (ToC). This project will follow the AMP Theory of Change (ToC), developed in the Program Framework Document (PFD) and set out in Figure 4 below. This TOC is premised on a baseline context where, while good progress is being made, several risks and barriers are driving high financing costs (equity and debt) and reducing the competitiveness of minigrids with respect to fossil-fuel based alternatives. All else being equal, the need for higher returns that reflect these risks translates into higher energy prices that, in turn, adversely affect affordability for the end-user, or require larger subsidy requirements for rural electrification programs. As a result, renewable energy minigrids do not get financed and built at scale. By focusing on cost-reduction levers and innovative business models, the project can improve the financial viability of renewable energy minigrids which in turn can accelerate and scale up their adoption as part of the effort towards achieving universal energy access. When renewable energy minigrids are more competitive, private capital and commercial financing will then flow, resulting in various program benefits: investment at scale, GHG emission reductions, and electrification and lower tariffs for end-users.

[32] The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

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571.Figure 4. Africa Minigrids Program's Theory of Change



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26. Alignment with GEF focal areas. The proposed strategy is aligned with the GEF Strategic Focal Areas CCM-1-1 "Promote innovation and technology transfer for sustainable energy breakthroughs for de-centralized renewable power with energy storage", and CCM-1-3 "Promote innovation and technology transfer for sustainable energy breakthroughs for accelerating energy efficiency adoption".

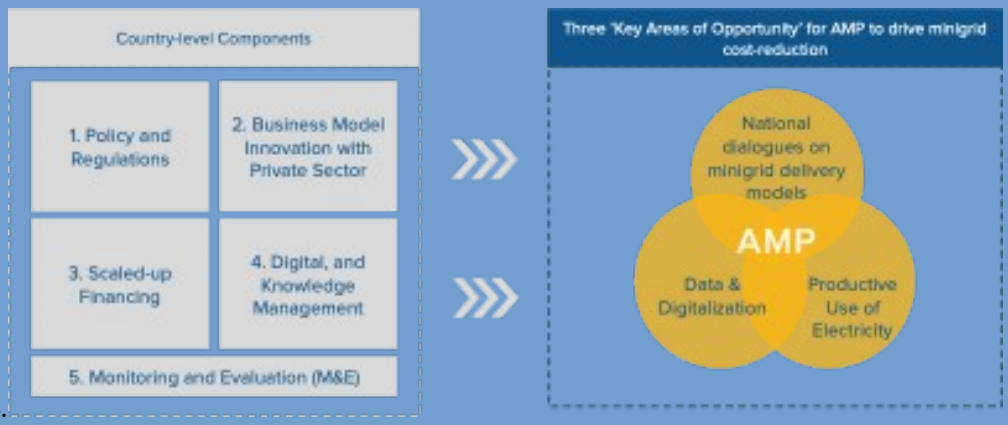
27. UNDP's Derisking Renewable Energy Investment (DREI). The Program's TOC draws on UNDP's Derisking Renewable Energy Investment (DREI) Framework by focusing on cost reduction levers across the themes of policy and regulation, business model innovation and private sector as well as innovative finance that can be employed to reduce risk (e.g. policy derisking), compensate for risk (e.g. financial incentives) or transfer risk (e.g. financial derisking). DREI is an innovative, quantitative framework to support policy makers to cost-efficiently promote private investment in renewable energy. In late 2018, UNDP expanded the DREI framework to include solar PV-battery minigrids, releasing open-source analytic and financial modelling tools to track investment risks, financing costs, and to support the private sector and policymakers in modelling levelized costs, tariffs and subsidies for minigrids. As regards AMP, UNDP's DREI framework will be applied either qualitatively and quantitatively at various points in the project cycle, both at the national level in each country, and then aggregated into regional knowledge products by the AMP Regional Project and disseminated widely. The DREI framework, both at the national and regional level (in aggregate), will act as the program's mechanism to harvest and disseminate data on changes in the financing costs, hard and soft costs, and resulting costs for minigrids.

28. Program's 'Key Areas of Opportunity'. The AMP has adopted a common architecture of four key components - a combination of enabling policy and regulations, business model innovation with private sector involvement, innovative financing and digital innovation - as the levers to lower investment risks, thereby reducing financing, hardware and soft costs while increasing revenues and improving system efficiencies. Within this architecture, AMP will emphasize - and seek to develop comparative advantages - in three 'key areas of opportunity': (i) an emphasis on advancing national dialogues on minigrid delivery models, (ii) promoting productive uses of electricity, and (iii) leveraging data and digital solutions for minigrid cost-reduction. Collectively these three areas can guide AMP's overall direction, creating a niche identity for the program. This approach, illustrated below in Figure 5, is structured to advance the program objectives of cost-reduction and innovation for minigrids and give effect to the TOC. The way in which this project will address these areas of opportunity is described in detail further below.

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573.**Figure 5. AMP's Key Areas of Opportunity**



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29. *National dialogues on minigrid delivery models*. A delivery model (refer concept in Box 1) that is suitable to country expectations and context for minigrids has proven critical to establish an enabling and attractive investment environment for minigrids. Equally, a delivery model that has not been defined or is consistent with the national context, will be a certain impediment to scaled-up investment. An important focus of the AMP is therefore to encourage a national dialogue between key stakeholders in support of a suitable delivery model being defined.

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576.Box 1. The Concept of a Minigrid Delivery Model

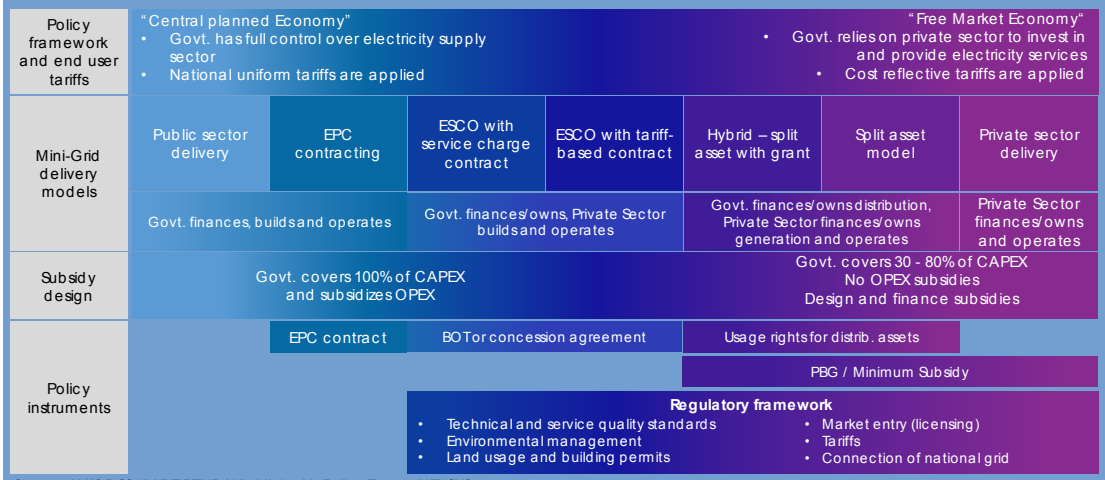
57. *The concept of a minigrid 'delivery model' is a key concept for the AMP. This text box seeks to set out a common understanding of the concept and its importance to the national projects and overall program.*

2. Definition: A minigrid delivery model, determined by the national government, is the cornerstone of a country's over-arching minigrid regulatory framework. It defines who finances, builds, owns and who operates and maintains the minigrids. Where applicable, it seeks to engage the private sector. A minigrid delivery model is closely associated to other key components of a minigrid framework, including tariff structures/mechanisms and subsidy levels/mechanisms.

3. In each country, identifying one (or more) delivery models will provide a framework for all sector stakeholders to plan for the longer term, particularly with regard to mobilizing private investment as one of the main objectives of the project. Figure 6 below describes the spectrum of design options for delivery models, across a number of different elements (ownership, policies, finance etc.)

4. Figure 6. Conceptual outline of minigrid delivery models

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Source: JAKOB SCHMIDT-REINDAHL, Mini-grids Policy Expert, INENSUS

6. This decision-making process around identifying a delivery model is complex and countries will be encouraged to establish a national dialogue for this purpose, involving all relevant stakeholders to varying degrees (different ministries such as energy, finance, health and environment, local authorities, the public, the media, the beneficiary communities, utilities, the private sector, and other key stakeholders) in order to build a national consensus on the basis of which large-scale deployment of mini-grids can be accelerated and have a sustainable impact.

30. *Digitalization (digital solutions and tools) and harnessing the data opportunity*. The emergence of minigrids as a viable solution to electrify remote and isolated communities relies strongly on digital tools and solutions[33]. Digital technologies and solutions are fundamental to enabling off-grid electrification and offer significant potential to lower minigrid costs, reduce risks, and address barriers to scale. Many of the opportunities around digitalization are related to leveraging the large amount of data generated by minigrid projects to surface insights, learning and optimization. Data is a tremendously valuable asset in the minigrid sector that remains underutilized. The programmatic approach allows the AMP to make an impactful contribution to growing a data asset and harmonized digitalization in the sector. Employing digital integration as catalyst for the minigrid sector reflects the UNDP digital transformation strategy that initiated a comprehensive process of connecting knowledge within the organization and across networks, creating opportunities, improving operational

[33] Solutions are commonly grouped into four categories: (i) digital planning, (ii) digital operations, (iii) digital aggregation platforms, and (iv) digital payments

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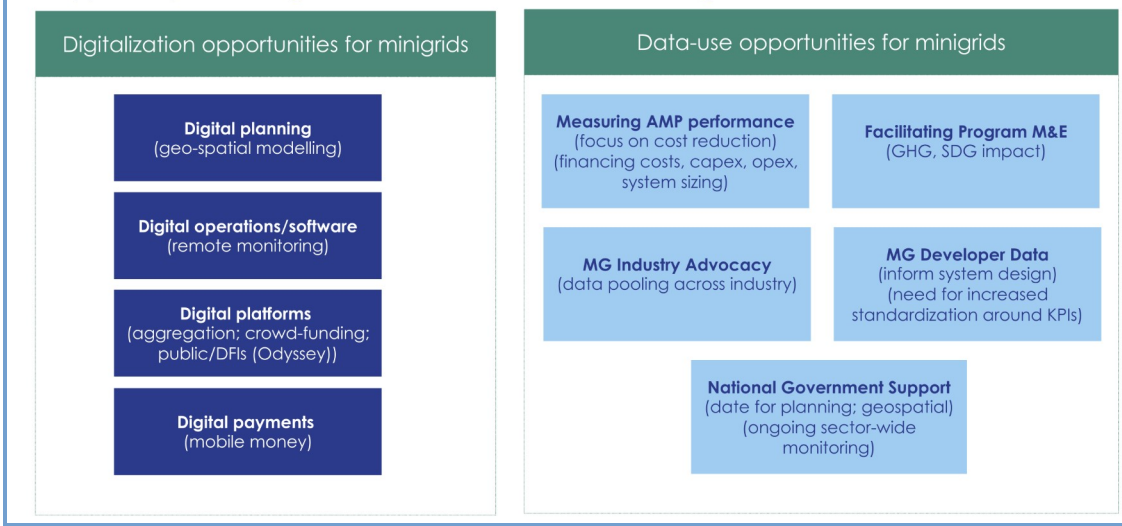
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efficiencies and building and maintaining partnerships and alliances. It also echoes the broader UN data-driven strategy and commitment to advance global "data action" with insight, impact and integrity.

Box 2: Digitalization and Minigrids

7. **Digital opportunity for minigrids** Figure 7 below shows different categories of digital solutions in the minigrid sector: (i) digital planning, (ii) digital operations, (iii) digital aggregation platforms, and (iv) digital payments. In common to all these is the potential of digital technologies – whether used by policy makers, financiers or minigrid developers - to lower minigrid costs, reduce risks, and address barriers to scale.

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9. **Figure 7. Digital and data opportunities for minigrids in the AMP**

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11. **Data use opportunity for minigrids.** Many opportunities around digitalization are related to leveraging the large amount of data generated by minigrid projects to surface actionable insights, learning and optimization to consolidate business models and technical solutions for scaling-up minigrids. For instance, the use of operational performance information from existing systems to forecast demand and design future minigrids can help avoid a very common pitfall of many minigrid systems which are significantly oversized and hence not financially viable.

12. **Opportunities across the Program, and with the AMP regional project.** The AMP provides a unique opportunity to develop a single set of metrics and guidelines for data collection, and use them to collect data from minigrid investment pilots across different national projects which the AMP Regional Project can then aggregate, derive insights from, and systematically disseminate knowledge with participating AMP countries and with the broader minigrids sector in Africa. At the same time, the link between the regional project and the various AMP national projects provides a unique 'distribution channel' opportunity across Africa for AMP to mainstream the use of digital tools and solutions for minigrids cost-reduction and scale-up.

31. *Productive uses of energy (PUE)*. While PUEs are widely recognized as a key element to improve the viability and sustainability of minigrids, the AMP focus is uniquely tailored, taking a deliberate, integrated approach across a broad ambit of the Sustainable Development Goals (SDGs)[34]. With the benefit of global experience and best practices, the AMP pursues solutions where productive uses are embedded in agricultural value chains or

[34] As described by SEforAll, access to energy is not the end in itself, but a means to many ends. Reliable and affordable energy is needed to improve living standards, increase rural incomes, support delivery of health and educational services, and improve gender and social inequality. It also enables access to clean cooking technologies and fuels – essential to reducing indoor air pollution and associated health risks. https://www.seforall.org/sites/default/files/Beyond-Connections-Introducing-Multi-Tier-Framework-for-Tracking-Energy-Access.pdf

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around which economic activity can be anchored. The AMP's emphasis on energizing agricultural production is based on the demonstrated impacts and amplified benefits resulting from (i) improved product quality and increased yields, (ii) contributions to value addition, (iii) increased value retention within the rural communities, and (iv) contributions to socio-economic developmental objectives for rural areas, which in turn has a positive effect on the minigrid revenue model. Further recognizing that these multiple benefits cannot be assumed with energy access, but depend on wider development programmes, the AMP approach combines the delivery of electricity infrastructure with innovative business models and various interventions aimed at encouraging economic activity, support business development and stimulate rural economic transformation with an emphasis on improved wellness, empowering women and youth as well as ensuring sound social and environmental stewardship.

32. Minigrid investment pilots contribution to the Program's Theory of Change (TOC). AMP National Projects include funds under Component 2 (Business model innovation and private sector) for supporting minigrid investment pilots seeking to demonstrate innovative business models and cost-reduction opportunities. Minigrid pilots have a key role within AMP by contributing to demonstrate cost-reduction which can be leveraged to improve the financial viability of renewable energy minigrids. Minigrid pilots are aligned with one or more of the three key areas of opportunity mentioned above by demonstrating: (i) a particular delivery model or elements of a delivery model around which the government wishes to build capacity and engage with minigrid developers; (ii) productive uses of electricity and their potential to reduce costs and enable minigrid development at scale; and/or (iii) opportunities around digitalization and the use of data for minigrid cost reduction. Feedback loops to other national project activities (e.g. national dialogues, capacity building) and with the AMP Regional Project (e.g. Community of Practice) are intended to actively disseminate the learnings from the pilots to inform both the policy and regulatory environment as well technical capacity building.

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579. AMP National Project in Benin

33. The objective of the Project is: "To support access to clean energy by increasing technical and financial feasibility and by promoting scaled-up commercial investment, in low-carbon minigrids in Benin, with a focus on cost-reduction levers and innovative business models." Specifically, the Benin national project aims to increase the supply of adequate, reliable, affordable, low-carbon electricity for unserved and underserved communities in Benin in support of the country's ambitious national off-grid electrification plan. The Project assigns a high degree of importance to the adoption and ownership of sustainable business models for minigrids as well as to raising the bar on the quality of installations, their operations and maintenance, two elements which may have been lacking in recent minigrids experiments.

34. The Project will foster a national dialogue on the delivery model, contribute to further articulate the regulatory framework and to streamline the subprocesses under the off-grid concession process cycle (from project development, application and permitting, to performance monitoring and reporting). In collaboration with the GOB and its development partners, this approach shall help accelerating the market uptake of low-carbon minigrids based on small-scale RE systems, the use of EE electric appliances and productive equipment, and adequate management and after-sales services. Through its focus on value creation from electricity inputs, the Project's approach is to transform the current cycle of low productivity and living standards in rural areas due to inadequate access to modern electricity and productive technologies, into a virtuous cycle, in which improved electricity services from minigrids in combination with sector development programs , enable higher productivity, which in turn triggers increased demand and generates the necessary revenues (increase in purchase capacity) to sustain the electricity service.

35. The Benin national Project will assume a supportive role to the GOB to assure quality along the minigrid value chain. This action is expected to increase confidence of investors and the general public in GOB's abilities to manage the off-grid concession delivery model. Supported by UNDP's DREI methodology, it will assess the technical, operational, institutional, and human capacity issues affecting present minigrid systems in the country and issue recommendations for their mitigation. The introduction of digital tools to facilitate data management and the planning, preparation and supervision of minigrid concessions, is expected to greatly reduce the administrative burden for the GOB while reducing transaction costs for applicants. This approach is

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also aligned with GOB initiatives for modernization of the public administration including the use of digital tools such as GIS platforms for planning and management by several sectors.

36. As such, the AMP is aligned closely with the present delivery model for minigrids in Benin (which is outlined in the following section) and seeks to make this more expedite and transparent, thereby reducing real and perceived risks and lowering transactions costs, while contributing to the establishment of robust, performance-based payment schemes that rely upon reliable information and report

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581. Current Delivery Model for Minigrids in Benin

37. The current delivery model for minigrids in Benin is based on concessions that are legally enabled as public-private partnerships. The Electricity Law 2020 (*Loi No. 2020-05*, 1 April 2020) integrates national on-grid and off-grid policies into one framework. This law governs all aspects of the electricity sector, including production, transport, distribution, commercialization, import, transit and export of electricity, and applies to connected and isolated grids. Among other goals (Art.3), it seeks to promote economic development and job creation, reduce dependency on imported energy supplies, diversify energy source and increase the share of RE in final energy consumption, warrant social and territorial cohesion by ensuring an affordable energy cost level for all Beninois; and promote gender equity and social inclusion. Sector development shall abide to the principles of economic and financial sustainability and respect for the environment (Art.4).

38. The Law confirms (Art. 7) the political responsibility for the sector by the Ministry of Energy (ME) and (Art.8) the autonomy of the regulator ARE. Among other roles, ARE is required to issue a positive advice for the technical design, hardware, sales contracts, and tariffs including for off-grid electrification projects (Art. 13). Electricity supply for public service is the responsibility of the State, who can defer as the *"autorité concédante"* to third parties including private entities, the responsibility to deliver this service on the terms set by sector regulation, primarily the Electricity Law 2020, and forthcoming decrees.[35] With a view on rural electrification, the Law identifies the "*exploitant d'électricité hors-réseau*", which holds the right to exploit an off-grid electricity system that may encompass production, distribution and supply (commercialization) to the end-users. Corresponding grid systems include (a) isolated grids supplied by RE sources (*"réseau de distribution isolé"*); (b) small-scale grids (*"pico-réseaux"*) serving customers from a central RE-powered productive use (such as a health center)[36]; and (c) stand-alone "*kits*" and other devices to deliver individual electricity services to disperse customers (*"sociétés de services électriques distribués"*).

39. An agency responsible for developing and implementing rural electrification is defined in Art.17. This role has been assumed by the Beninese Agency for Rural Electrification and Energy Management (ABERME), responsible for energy management programmes, developing pilot rural electrification projects and instituting incentive schemes to promote rural electrification and energy management. Funding for this agency (Art. 18) is secured through, among others, the *Fonds d´Électrification Rurale et des Énergies Renouvelables (FERER)*, previously the FER). The Law further confirms the figure of Independent Power Producers (IPP) in Art. 26 and the universal, non-discriminatory access to the national grid by consumers and producers, except for technical reasons (Art. 29). The involvement of third parties is based on annual work programs to be developed as public-private partnerships, which implies that listed projects are not subject to standard public procurement rules; instead, requirements and contractual conditions are set by decree (Art. 37). The system of concessions and contracts (Art. 37-45) includes a requirement for local content in public tenders (at least 15%, Art. 38).

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[35] *"Autorité concédante: personne publique investie de prérogatives de production, transport, distribution, commercialisation, transit, importation, exportation d'énergie électrique qui peut déléguer ses prérogatives à un tiers sous le régime de lo délégation de service public (...) à conclure avec des tiers des conventions ayant pour objet I'exploitation des activités de service public réglementées par la présente loi."* (Loi d'Eléctricité 2020, Art.5 Definitions.) See also. Art. 24.

[36] *"Pico-réseau ou mini-réseau: réseau de petite ou de très petite taille permettant d'acheminer l'énergie électrique des centres de production de capacité limitée vers les consommateurs d'électricité"* (Loi d'Eléctricité 2020, Art.5 Definitions).

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40. Chapter IX of the Law is specifically dedicated to renewable energy and off-grid electrification. A concession is required for generating capacities above 500-kW; below this value, an authorization is needed (Art. 61.2). The State offers subsidies, fiscal benefits or guarantees to societies and businesses involved in development and exploitation of RE-based power systems. The import of equipment and materials, including for isolated distribution grids, 100% exempted from import duties and local taxes (Art. 60:4). Capital investments and posterior spare part supplies under a PPP are exempted from taxes and levies (Art. 78.1) Renewable energy exploitations are further exempted from Value-Added Tax (VAT), *Taxe sur les Activités Financières* (TAF), and property taxes (*Taxe Foncière sur les Propriétés Bâties*, TFPB). Relevant is further the requirement that technicians involved in RE system installation shall be certified (Art 60.7) and projects shall preserve the environment (Art. 60.8).

41. The arrangements and contractual obligations resulting from the concession model are detailed in a Model Contract for minigrid concessionaires (see Error: Reference source not found). Note that model contract, specific regulation and ministerial decrees still need to be brought in alignment with the Electricity Law 2020.

42. It is important to note **that on-grid and off-grid electrification are both strategic propositions serving the same goal, i.e. to deliver adequate, affordable electricity services to the Beninois people in the entire territory**. The choice between on-grid and off-grid is determined by the gradual expansion of the national grid system and the recognition that in more remote areas, off-grid solutions may be faster to implement and even offer lower overall costs, especially where electricity demand is low. Both approaches are seen as highly compatible, complementary and in some cases, highly synergetic in building common best practices and national capacities.

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583.Box 3 Model contract for Off-Grid Concessionaires

5814. *Model contract for Off-Grid Concessionaires – Outline of arrangements and conditions.*

1. The contract signatory are the ABERME, on behalf of the State, and the Concessionaire. The contract is eligible for signature once the Concessionaire has obtained from the Regulator (ARE) the certificate authorizing exploitation of the minigrid. This involves acceptance of the proposed tariff scheme, approval of the installations, and validation of all permits and studies as required. The Concession Model is followed for capacities above 500-kW. Below this threshold, an Authorization can be granted, which is particularly relevant for local "spontaneous" initiatives typically built around a productive electricity user. Where appropriate, minigrid Concessionaires are allowed to sell RE-based surpluses to the national system, in which case they are responsible for financing and operation of the required 33-kV transmission line segment within their concession area, to connect to the SBEE or CEB main grid Art. 6 and 7)
2. The contracts under concession are issued for 15 to 25 years. Contracts bound to 15 years relate to concession areas with a higher likelihood to be incorporated into the national grid system. The concession grants the exclusive right, within a specified geographical perimeter to: (i) operate the installations; (ii) distribute and commercialize electricity; and (iii) provide electricity services. The Concessionaire is held to provide the services to an agreed number of household and other end-users, as stipulated, in a non-discriminatory manner in abidance to the Terms of Service that are part of the Concession (Art. 9 and 10) These include (disperse) households who may rely on RE pico-generators and/or solar lanterns.
3. The Concessionaire is bound to execute the internal installations (house-wiring) according to a standard outlay depending on the system size chosen, for which a fee is charged to the customer. (The standard layouts allow for a type approval by CONTRELEC (contrary to on-site inspections which would be unpractical and costly). The fee to be charged is subject to approval from ARE, Art 11). The service shall be metered and by default, billing is prepaid through convenient amounts adapted to the purchase capacity and needs of a variety of customers. Assistance shall be provided to customers who face difficulties to understand and use smart meters. Mechanisms to settle disputes and billing errors are foreseen and the Concessionaire has the right to suspend the service to customers who decline to pay or tamper with their connection (Art 13-16).
4. Various scenarios concerning the minigrid assets ownership are foreseen including: (i) distribution system built under public investment, for operation (and expansion) by the Concessionaire, who shall finance the connections with the end-users; (ii) assets financed and constructed by the private Concessionaire. The net present value of the assets is to be calculated annually, which shall allow for a fair transfer of fixed assets upon termination of the Contract (Chapter 9 of the Model Contract (Art 32-39). It is understood that the Concessionaire exploits his/her business to its own risk and exposure and assumes all related costs. He/she shall strive at delivering the service at the best cost price for the end-user while ensuring service quality, availabilty, and equitable access. **Remote surveillance is demanded to monitor performance and operational status of the minigrid** to enable preventive maintenance and optimize the service delivered Art.19). The service level (power level and guaranteed supply in hours per day) are defined in the Terms annexed to the Model Contract and shall be adhered to (Art.21).
5. As relates to the tariff setting, the Concessionaire shall present a proposal for approval by ARE according to a prescribed calculation model (Art.47). The model departs from a regulated value for the return on investment for the private Concessionaire. The model envisions a financial support (cash inflow) to the Concessionaire in to assure an end-user tariff that is compatible with their payment capacity while allowing the concessionaire to recover the actual costs of minigrid operation "cost plus pricing" (Art.47). The tariffs are revised periodically.

43. Table 2 summarizes the key aspects of the minigrid delivery model in Benin based on the current policy and regulatory framework.

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585.Table 2 Current Status Minigrid Delivery Models in Benin

586. Aspect	587. Current Status
588. Ownership and Operation	589. Fiscal spending limitations and the impossibility to reach the rural population by main grid expansion, has put private minigrid concessionaires at the heart of the rural electrification model. Concessions are issued for 15 to 25 years and grant the exclusive right, within a specified geographical perimeter to: operate the minigrid; distribute and commercialize electricity; and provide electricity services. Scenarios concerning the minigrid assets include: (i) public investment of grid assets, for operation (and expansion) by the Concessionaire; or (ii) assets financed and constructed by the private Concessionaire. Public (fixed) assets used shall be booked as an asset with a corresponding liability in the operator's balance sheet. The present value of assets is calculated annually, which shall allow for a fair transfer of fixed assets upon termination of the Contract.
590. Tariff mechanisms	591. The Concessionaire shall present a proposal for approval by ARE according to a prescribed calculation model. The model departs from a regulated value for the return on investment for the private Concessionaire. The model envisions a financial support (cash inflow) to the Concessionaire in to assure an end-user tariff that is compatible with their payment capacity while allowing the concessionaire to recover the actual costs of minigrid operation ("cost plus pricing"). The tariffs are revised periodically.
592. Subsidy mechanisms	593. Private developers are aided in several ways: (i) by making available to interested minigrid investors, information and consolidated data sets as a basis for proposal development; (ii) by public investment in minigrid infrastructure, to be operated by the Concessionaire, which reduces the need for, and exposure of, private equity; (iii) by providing a "cost plus" financial compensation to Concessionaires to top up the tariffs that can be paid by rural consumers; and (iv) by offering a range of fiscal exemptions and incentives to investors and operators of off-grid electricity grids (including import duty and VAT exemptions; no property taxes.
594. Regulations	595. The policy framework is provided by the *Loi d´Électricité* 2020 (Electricity Law) which integrates provisions and propositions from the National Off-grid Electrification Policy (PEHR) and corresponding master plan (PDEHR). Specific regulation and decrees still need to be developed and/or aligned with the Electricity Law.

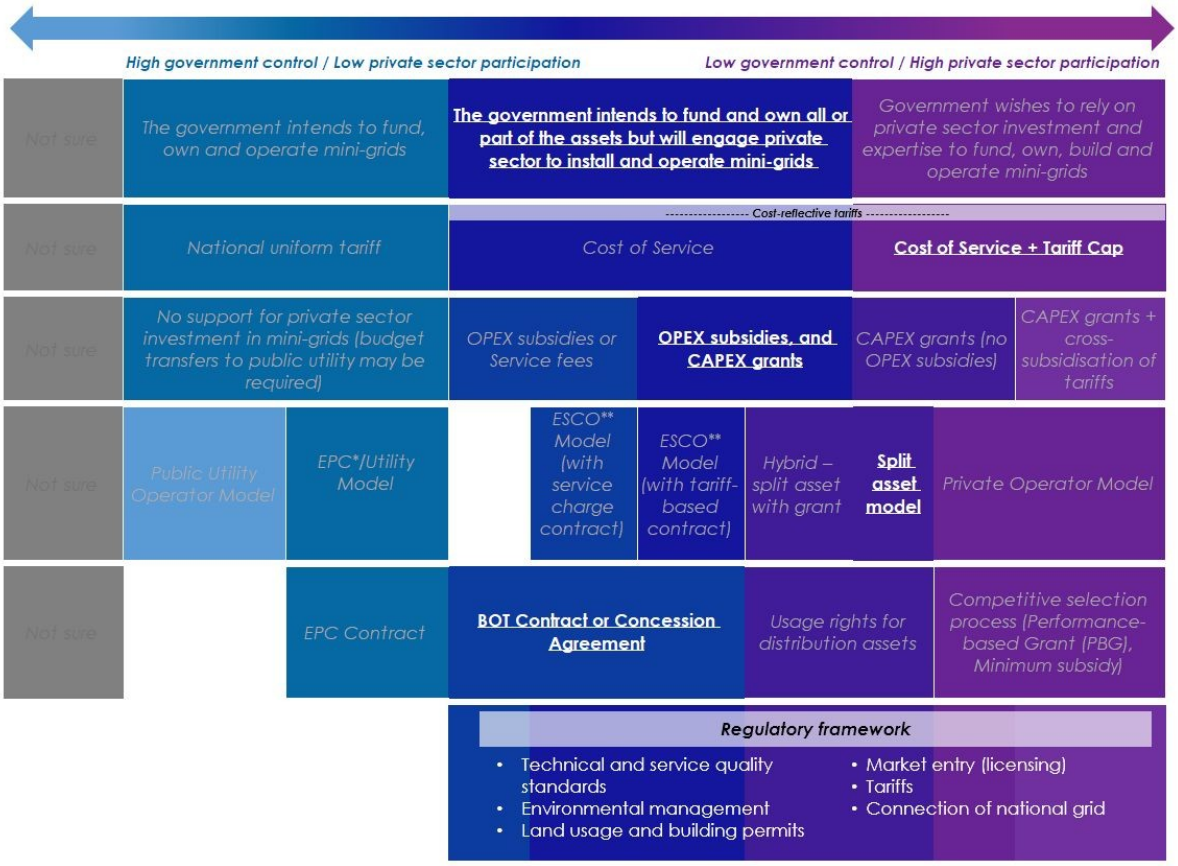
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44. To further position the country's current approach in the broader continuum of possible options and paths, the **Figure 8** below provides a snapshot based on a methodology by INENSUS to position the **key characteristics of the minigrid delivery model in Benin** (with the highlighted text representing the current situation in Benin).

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598.Figure 8 Preliminary identification of minigrid delivery model for Benin
599.(Figure and methodology from: Jakob Schmidt-Reindahl, INENSUS).

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605. AMP Project key principles and service offering

45. The AMP aims to implement minigrid pilots in all participating countries to collect operational data for fact-finding and extracting lessons towards strengthening of the business model and the attainment of cost reductions. The pilots provide an opportunity for governments and other stakeholders for learning-by-doing, especially in the less advanced markets. Specifically, the pilot provides a test case to start using digital management and tendering systems. To allow a comparison or results, the AMP has set a series of principles that shall be followed by all countries during preparation and execution of the minigrid pilots.

46. Minigrid Pilots Plan and key principles for minigrid pilot implementation: At the project preparation stage, considerable initial consultations, analysis and planning have been performed to advance the design of the minigrid pilots. This is described in the section directly below (Section IV) and associated annexes, and is to be read together with AMP's Key principles for minigrid pilot implementation (set out in Box 4 below). With regards to precise specifications and components of minigrid pilots, the **Project shall, in Year 1, produce a full Minigrid Pilot Plan for Benin** (see output 2.1). This Plan will notably detail the approach to implement the pilots and ensure alignment with the AMP key principles, as well as to keep it responsive to national priorities as formalized during Year 1. The Plan will be reviewed and cleared by UNDP (Benin CO and BPPS NCE) and shared with the Project Board.

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Box 4: Key principles for minigrid pilot implementation

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- **Principle 1. Digital platforms**. The use of digital platforms is a central element of the overall AMP and digital strategy for the project. With digital platforms emerging as critical enabler for procurement and operation of cost-effective and viable minigrids, using a digital platform for pilot projects provides an opportunity to build capacity of key stakeholders in using this facility which can then set the foundation for later using digital platforms for sector-wide, including large-scale tenders or results based financing programs. Digital platform software can manage the selection, Monitoring and Evaluation (M&E) and payments of pilots including capacity building of the Implementation Partner, Project Management Unit and minigrid developers.
- **Principle 2. Productive use: third party ownership model**. For pilots that will financially support the purchase of productive use equipment using an allocation under the GEF INV, it is required that the project will only provide its support via a third-party ownership model, as opposed to a self-ownership model. In the third-party ownership model, the minigrid asset owner purchases and owns the productive use equipment, and leases it back to the end-user as part of an "energy as a service" offer.
- **Principle 3. Clear methodological basis for additionality for calculating the level of (GEF INV and/or UNDP TRAC)/financial support for capital expenditures (CAPEX).** Grant funding for the pilots must be based on the principle of minimal concessionality of allocated CAPEX subsidy. Suitable methodologies for calculating the level of GEF investment support can be on the basis of achieving: (i) Levelised Cost of Electricity (LCOE) parity with a diesel mini-grid, (ii) LCOE parity with pre-existing residential tariffs; or (ii) LCOEs based on the willingness to pay of the end-users (via surveys etc). During implementation, the AMP regional project may provide updated guidance on suitable minimal concessionality methodologies.
- **Principle 4. Minigrid pilots data sharing**. Pilot beneficiaries (e.g. minigrid operators) receiving support from the project are required to share data on minigrid performance with the national and regional project. Specific terms and conditions for data-sharing and how best to operationalize the commitment and its adoption by the beneficiaries will be defined and agreed upon with minigrid operators during project implementation, with support from the AMP Regional Project.
- **Principle 5. Compliance with UNDP Social and Environmental Safeguards and Gender requirements**. Pilot projects receiving project funding are required to comply with all relevant national legislation and standards, as well as UNDP standards as pertaining to SES safeguards and gender equity. To this purpose, the Environmental Safeguards Management Framework (ESMF) and gender action plan are part of this ProDoc (Annexes 10 and 11). The ESMF provides a mandatory framework for developing the Environmental and Social management Plans (ESMP) for pilot projects. A critical aspect of the ESMF is the obligation for environmentally sound management of replaced equipment, including batteries, inverters and solar panels, after their usage. The responsible handling of waste, including recycling of batteries and other equipment, shall be clearly documented, budgeted and monitored in compliance with national and UNDP safeguards requirements.
- **Principle 6. Financial sustainability.** The business and delivery models of the pilots will be determined during project implementation as part of the minigrid pilot plan. A core principle to be upheld will be to that all business models ensure the full financial sustainability of the minigrid pilot for the entirety of its asset lifetime, including the maintenance and replacement of key equipment such as batteries and inverters. These financial sustainability considerations will be evaluated, and must be in place, before the provision of any GEF INV to a minigrid pilot.

47. Linkages to the AMP Regional Project: The Project will align with the AMP Regional Project to facilitate knowledge sharing, learning, and synthesis of experiences from the AMP Regional Project to the Benin child project and vice versa. The AMP Regional Project will make available best practices for regulation and policy design, innovative and inclusive business models, digitalization and finance, to all AMP beneficiary countries, while providing technical and operational support for national projects' on-the-ground implementation. Box 5 below provides a summary of the technical and operational support provided by the AMP Regional Project to the child projects. More details will be communicated at the Benin project's Inception Workshop (IW), including: areas of support and listing of firms and consultants under contract by the AMP Regional Project.

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The AMP Regional Project will set up protocols for requesting and accessing expertise and communicate these with the participating child projects including Benin.

Box 5: AMP Regional Project Indicative Service Offering

15. Digital, Knowledge management and monitoring and evaluation (M&E)

- **Knowledge building/sharing**. The regional project will curate, develop and share knowledge with the project on program's thematic areas (Policies and regulation, innovative business models, financing, digitalization).
- **Insight Briefs development and dissemination**. National projects will gather data and audio-visual content (video footage, photos, etc.) highlighting national project activities which will be the subject of an 'insight brief' to be developed and widely disseminated by the AMP Regional Project.
- **Communities of Practice**. One of the primary ways national project staff will interface with the regional project is via the 'Communities of Practice' (CoPs) and associated activities/platforms. While it is expected that many of the activities will be undertaken virtually (via internet-based platforms, webinars or digital platforms) it is also expected that the CoPs will include actual in-person workshops, meetings or training events that project staff will participate on.
- **Common M&E Framework/QAMF**. The AMP Regional Project will develop, with inputs from national projects, a Quality Assurance and Monitoring Framework (QAMF) for measuring, reporting and verification of the sustainable development impacts of all minigrid pilots supported by national projects, including GHG emission reductions.
- **Data aggregation platform**. The AMP Regional Project will deploy and use a web-based data management platform to aggregate data from all national project pilots based on the QAMF to track Results Framework indicators as well as program objectives, SDG impacts and GHG emission reductions.
- **Systematic data analytics and insights.** The regional project will harness data shared by the national projects to extract insights and learnings which will be disseminated across all national projects and within the broader minigrids ecosystem.

16. Technical and operational support for national projects' implementation

- **Access to specialized expert international consultants in selected areas hired, retained, contracted and paid for by the AMP regional project** and made available to all participating national project staff and selected beneficiaries on as needed basis. This support may range from virtual assistance to in-country missions.
- **Database of qualified international consultants and firms** provided for information purposes to the project in an effort to assist in identifying high-quality experts and firms who may be available for contracting by national governments under their own procurement rules and modalities.
- **Generic terms of reference (ToR)** for various standard activities will be provided to projects for information purposes.
- **Specialized advisory support for implementing UNDP's minigrid DREI analyses**. During project implementation, the UNDP DREI Core team, working with the regional project, will make available to national teams and consultants the resources and tools to conduct full quantitative DREI applications, and will provide ongoing support and quality assurance.
- **Operational support for national projects**. The AMP Regional Project will provide support to the project, on an ad-hoc and as-needed basis, through its PMU staff or by hiring or recommending subject matter experts, for the project to execute activities. Further details on specific support around M&E activities provided in Section VI (Monitoring and Evaluation Plan).

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611. IV. RESULTS AND PARTNERSHIPS

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48. Greenhouse Gas (GHG) Emissions Mitigated. The Project will result in GHG emissions reductions, which will be measured via GEF7 Core indicator 6: Greenhouse Gas Emissions Mitigated. This indicator captures the amount of GHG emissions expected to be avoided through the project's investment in RE minigrid pilots and will be measured above a baseline value considering that in the absence of the project, the end users would have been supplied by fossil-fuel-based mini-grid(s).

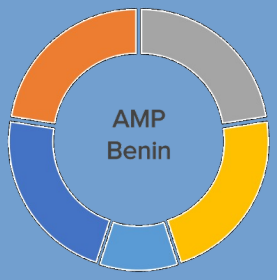
49. Mitigation benefits include both (i) direct emissions reductions attributable to the minigrid pilot investments made during the project's implementation period, totaled over the lifetime of the investments (20 years); and (ii) Indirect emissions reductions resulting from the increased uptake of minigrids for off-grid electrification of rural areas due to replication, scaling-up and market change to which the project has contributed by creating a general enabling investment environment for minigrid market development, and facilitating subsequent investment flows. Annex 12 describes the methodology used to define targets for direct and indirect GHG emissions mitigated.

50. 10% of the estimated indirect GHG mitigated of this project have been removed from the project and allocated to the AMP regional project, in line with the apportioning of the overall program budget and reflected in the PFD allocation of GHG emissions reductions across the different AMP national projects. This reflects the benefits of AMP national projects accessing the regional project's support which is expected to contribute and enhance the enabling conditions required for minigrids development across AMP countries.

51. Project components, outcomes, outputs and activities:

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Component 1	Component 2	Component 3	Component 4	Component 5
Policy and Regulation	Business Model Innovation with Private Sector	Scaled-up Financing	Digital and Knowledge Management	Monitoring and Evaluation (M&E)

615. Targeted outputs per component

616.	617.	618.	619.	620.
621. Output 1.1. An inclusive national dialogue in support of minigrid delivery models is facilitated through	622. Output 2.1. Pilots developed (preparatory phase), including on productive use and innovative appliances	623. Output 3.1. Innovative financing solutions for minigrid development are identified and implemente	624. Output 4.1. A project digital strategy is developed and implemented, including linkages to and	625.

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615. Targeted outputs per component				
articulation of national electrification strategies and updating of sector regulation	and modular hardware and system design, leading to cost-reduction in minigrids.	d with supporting human and institutional strengthening.	following guidance from the AMP Regional Project	
626. Output 1.2. Minigrid DREI techno-economic analyses carried out to propose most cost-effective basket of policy and financial de-risking instruments and contribute to AMP Flagship Report on Cost Reduction.	627. Output 2.2. Commissioning and monitoring of selected pilots in alignment with AMP principles.	628. Output 3.2. Domestic financial sector capacity-building on business and financing models for minigrids.	629. Output 4.2. Specification and implementation of Minigrids Digital Platform to track minigrid pilots and support scale-up and cost-reduction.	630.
631. Output 1.3. Capacity building provided to public officials (regulator, ministries) specifically to design procurement/ tender processes that incorporate cost-reduction levers and innovative business models.	632. Output 2.3. Enhancement of minigrid business model by strengthening operator and community capacities, development of PUE and other energy nexus, and the integration of local RE sources.	633.	634. Output 4.3. Adoption and operationalization of the project's Quality Assurance and Monitoring Framework (QAMF).	635. Output 5.1. Inception workshop is conducted and M&E plan is implemented.
636. Output 1.4. Domestication of quality standards for solar mini-grid components, and institutional	637.	638.	639. Output 4.4. Engage with regional project by participating in Communities of Practice	640. Output 5.2. Project Mid-Term Review is conducted.

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615. Targeted outputs per component

capacity of national standards organizations/ bureau strengthened.			and capturing and sharing of lessons learnt.	
641. Output 1.5. Support provided to establish the environmental and social policies and plans to ensure mini-grid risks are properly handled.	642.	643.	644.	645. Output 5.3. GEF Terminal Evaluation is conducted.

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Component 1. Policy and Regulation

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649.Outcome 1: Stakeholder ownership in a national minigrid delivery model is advanced, and appropriate policies and regulations are adopted to facilitate investment in low-carbon minigrids.

52. The specific objective of this component is to support the completion of the regulatory framework for off-grid electrification concessions in Benin, and to enable the incumbent authorities to lead the sector and implement the envisioned minigrid delivery model. Ownership is strengthened by inviting relevant stakeholders from public and private sector and civil society to become part of a national dialogue to maximize social and economic impact. The application of UNDP's Derisking Renewable Energy Investment (DREI) framework will support policy design to promote investment in RE-based minigrids. Complementary to baseline initiatives (e.g. funded through the MCA-II Benin), the GEF Project will provide support for developing specific technical, legal, social, and environmental aspects of minigrid planning and implementation. The results of the DREI exercise in Benin will be shared with the AMP Regional Project and participating countries.

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Output 1.1. An inclusive national dialogue in support of minigrid delivery models is facilitated through articulation of national electrification strategies and updating of sector regulation.

53. The following activities are envisioned:

Activity 1.1.1. **Participate in multi-stakeholder national dialogue group/platform.** The Project Management Unit (PMU), supported by the Project Board (PB), will facilitate a multi-stakeholder dialogue led by the Ministry of Energy (ME) to further advance Benin's national electrification agenda with a focus on rural electrification using minigrid solutions. This dialogue will support Benin to meet its electrification targets as set out in national policies. By facilitating thematic working groups and contracting specialized studies, as deemed necessary, the GEF Project will contribute to address identified challenges and workplaces (*"chantiers"*) including: (i) updating of legal, technical, and administrative regulation, guidelines, and processes for minigrid concessions, in line with the new Electricity Law (2020); (ii) assessment of mechanisms for long-term financing of tariff subsidies for off-grid electricity concessionaires; (iii) pertinence and scope of technical standards for minigrid systems (see output 1.4); (iv) capabilities and technical requirements for effective purview of off-grid concessionaires,

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including the need for digital support tools (see Outcome 4); (v) other de-risking measures (as informed by the DREI analysis); and (vi) linkages of electrification strategies with other sectors ("nexus") including public health, water and sanitation, SME development, digital and e-commerce support, and agriculture.

The dialogue offers a platform for evaluating the off-grid delivery model and monitoring the response from the market (private investors and developers). It further offers a space for evaluating model contracts for concessionaires, terms of service, quality and sustainability issues, and receive feedback and inputs from civil society. Participants include national Government partners, sector entities, local authorities, civil society, private sector, multilateral organizations, and other stakeholders. In accordance with the Gender Action Plan, one specific working group will be set up to assure gender equity and equality during all phases of the AMP child project in Benin[37]. The Project Manager (PM) will specifically be involved in liaison between stakeholders, problem analysis and drafting of recommendations, drafting of Terms of Reference for subcontractor studies and services (if needed), presentation on Project progress and thematic issues; and is available to act as the penholder for the dialogue platform.

Activity 1.1.2. **Provide inputs for dialogue and decision-making.** This is achieved through gap analyses, best practices, recommendations for delivery models, and expert advice on the implications of choices made taken for the overarching framework. Ensure a feedback loop between the dialogue and the experiences and preliminary results delivered by the Project. The inputs will be developed by the Project Manager (PM) with technical backstopping from the Technical Advisor(s) (TA) and Gender Expert (GE). The TA position is a "placeholder" for one or more international consultants who will be hired by the Project on a part-time base (likely short-listed by the AMP Regional Project and shared between various AMP country projects). The GE will be a national expert hired by the Project for screening and advice on gender aspects.

Activity 1.1.3. **Update pertinent regulation, guidelines, and processes for minigrid concessions, in line with the new Electricity Law (2020).** This activity aims to bring existing regulation for off-grid electrification through minigrid concessionaires in congruence with the new Electricity Code 2020. The scope of this activity shall be determined by the Project Board following adaptive management principles. GEF funding will be incremental, complementing baseline activities by the Government and other development partners. Specific elements that may require updating and/or enhancement include: model contract and terms of service; the "cost plus" tariff model methodology; inventory of public assets (land, grid infrastructure) used by concessionaires including status monitoring and periodic value taxation; procedures for performance assessment of concessionaires; assignment of responsibilities for sector data collection and management; and arbitrage protocols.

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Output 1.2. Minigrid DREI techno-economic analyses carried out to propose most cost-effective basket of policy and financial de-risking instruments and contribute to AMP Flagship Report on Cost Reduction.

54. This output will assist the GOB to identify appropriate instruments for de-risking of investments in RE assets and related infrastructure. Since 2018 the DREI includes solar PV-battery minigrids to track investment risks and financing costs, and to model levelized energy costs (LCOE), tariffs and subsidies for minigrids. The AMP envisions the application, either qualitative or quantitatively, of the DREI framework at various points in the project's implementation. The results will be aggregated and incorporated into knowledge products by the AMP Regional Project for wider dissemination.

55. This output entails the following activities:

[37] The Gender Action Plan suggests "Diversity and gender balance in [the] national dialogue, with concerns of women and men being addressed equally, leading to gender-sensitive (i.e. not gender-blind or neutral) policies and regulations"

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Activity 1.2.1. **Conduct a full DREI application.** The DREI framework delivers quantitative insight into the financial costs of electricity supply options in a country. The financial model that is used in the DREI framework, captures CAPEX (project development costs, costs of equipment, civil works and commissioning) costs of capital, and operational costs over lifetime (OPEX) to determine the levelized cost of electricity (LCOE) of a given technology. The DREI exercise assesses the impact of policy instruments on lowering the LCOE as well as the cost of such instruments for the state. Instruments can be geared towards reducing the risk profile of the investment as well as improving process efficiencies including scale benefits. The DREI framework will be the AMP's mechanism to track and share information on the cost level of minigrids across all participating countries, and make visible opportunities for cost reduction.

This activity will be carried out during Project Year 1 by a tandem of a national DREI consultant and an international expert recommended and shortlisted by the AMP Regional Project (both funded by the GEF Project). Together they will conduct interviews, collect quantitative data, perform analysis, generate reports and knowledge products, present the results to the GOB partners, and be available for further discussion and specific advice. The Regional Project will make available tools (such as spreadsheet models) to conduct the DREI analysis. Initial Terms of Reference (TOR) for both consultants are attached (ProDoc, Annex 20 and 21). The IP will assign a team of national stakeholders to participate in the process and share information and viewpoints with the DREI consultants. The Project Manager (PM) will act as focal point.

Activity 1.2.2. **Present the findings of DREI application.** The findings of the DREI analysis will be presented during the first half of Project Year 2 and include recommendations for de-risking in the policy, regulatory and financial domains. These include, but are not limited to: (1) specific regulation for concessionaires of minigrids and distributed electricity services; (2) enhancement of the business model including mitigation of operational risks; (3) technical standards for equipment and installation; (4) costs and sustainability of tariff subsidies; and (5) incentives and guarantees for investors. The results will be evaluated by the Project Board to align the Project's work plans to ensure relevance and best use of Project resources (adaptive management). This activity is funded by the national project.

Activity 1.2.3. **Share results of DREI with the AMP Regional Project.** The findings and results of the DREI analysis in Benin will be shared with the AMP Regional Project to feed into the "Flagship AMP Knowledge Product" (second half of Project Year 2). The Flagship Product constitutes a mechanism for south-south learning by disseminating the DREI findings and results across the AMP countries. The activities will encompass 3-4 round-table workshops over a 12-month period, involving government, private sector and other key stakeholders. This activity is funded by the AMP Regional Project.

Activity 1.2.4. **Develop Knowledge Product on AMP DREI application.** In Project Year 4, a 'light quantitative DREI analysis' will be performed for Benin using updated cost figures and financial parameters. The result will be presented as an Update Note (2-5 pages), which will contribute to the equally updated Flagship Knowledge Product (see above). This Product will provide an overview of the evolution of mini-grid costs across AMP countries by closure of the AMP Regional Project. The Regional Project will organize and fund the DREI update. The Benin child project will: (i) facilitate the DREI update exercise; (ii) disseminate the findings of the national DREI update note; (iii) collaborate with the AMP Regional Project team to update the Flagship Product.[38]

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Output 1.3. Capacity building provided to public officials (regulator, ministries) specifically to design procurement/tender processes that incorporate cost-reduction levers and innovative business models.

56. This output seeks to strengthen capacities of sector agencies complementary to baseline support (specifically from GIZ and MCC). Key entities involved include the Ministry of Energy (ME, DGRE), the implementing agency, and he regulator ARE. The Project team will accompany the GOB on its path towards full ownership of its

[38] As such two deliverables will be produced and funded by the AMP Regional Project – the full AMP DREI Flagship Knowledge Product report in Year 2, and the Update Note in Year 4.

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mandate to lead the off-grid electrification process, in accordance with national policies as laid out in the Electrification Law 2020 and resulting regulation and provisions. UNDP's PCAT further suggested "Enhanced monitoring & assurance activities", together with trainings with technical support for the implementation partner, which could be performed through the activities below. The Project Board for the Benin Project is well positioned to evaluate issues and recommendations that emanate from the national dialogue and propose specific actions under the GEF Project in response.

57. Activities foreseen under this output include:

Activity 1.3.1. **Provide training and knowledge products with support from AMP Regional Program.** This activity seeks to enhance knowledge and competences relevant to minigrid policy, regulation, and minigrid business approaches. Specific disciplines include rural development, economics and engineering, among others. The Project seeks to leverage partnerships between the IP and training institutions to create competences in the public sector for design, preparation and monitoring of rural electrification programmes in Benin. On-site and remote training will be offered by the Project team on-site as well as through modules on the regional AMP learning management platform. Training materials from project partners will be made available including in French (such as AfDB's Green Mini Grid initiative, IRENA, SE4All).

Activity 1.3.2. **Support overall quality of minigrid concession process by institutional strengthening.** The Project team will accompany the GOB on its path towards full ownership of its mandate to lead the off-grid electrification process, in accordance with national policies as laid out in the Electrification Law 2020 and resulting regulation and provisions. The Project Board for the Benin AMP project is well positioned to evaluate issues and recommendations that emanate from the national dialogue and propose specific actions under the GEF Project in response. The GEF project can offer human resources (staff time) for liaison, analysis and presentations to complement baseline GOB actions assigned to ME, ARE, and ABERME. Adaptive management will be applied to identify specific needs and budget specific activities in the Project's Annual Work Plans (AWP).

Output 1.4. Domestication of quality standards for solar mini-grid components, and institutional capacity of national standards organizations/bureau strengthened.

58. This output will complement baseline work on quality and performance standards for electric equipment and ancillary systems used in electric minigrids. The scope of the equipment includes, indicatively: (i) solar-PV panels and controllers; (ii) battery energy storage systems; (iii) devices for remote monitoring and data acquisition; (iv) electric appliances and lighting for residential and commercial application; (v) AC/DC cabling, connectors, grounding and lightning protection; (vi) rack mountings, casings, containers; (vii) design and materials for civil works; (viii) supplier and installer certifications including ISO9000, ISO14000, and ILO conventions. The Project shall seek harmonization with common standards in the ECOWAS/UEMOA market. The standards will provide a basis for efficient procurement and quality assurance and contribute to compliance with relevant UNDP and GEF safeguards.

59. This output will, indicatively, entail the following activities:

Activity 1.4.1. **Perform a detailed assessment of technical and operational status of existing minigrids.** This activity concerns a detailed technical audit of one or two existing minigrids to obtain a clear understanding of the status of existing minigrids in Benin, including technical causes for malfunctioning. The Project will commission an independent firm to carry out a rigorous technical analysis of components, installation quality and practices, functional design, technical specifications and material choices, operation and maintenance practices, compliance with specification and procedures, and potential weaknesses during the commissioning and reception process. The activity builds forth on findings gathered by the UNDP/GEF PANA Energy project and field visits by the PPG team. The study shall provide the necessary information and recommendations to assure technical and operational sustainability of minigrids projects in Benin, including the selection of appropriate technical standards.

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Activity 1.4.2. **Propose technical and performance standards.** The status of national standards development for relevant equipment, installation quality, and appliances will be assessed and remaining gaps identified. As appropriate, a work programme will be set up for specific support by the GEF Project; this can be informed by thematic working groups under the national dialogue platform (Output 1.1). For small grids, opportunities for simplified products standards to achieve cost reductions will be explored. Departing from identified end-user needs and customer preferences, an approach will be followed to avoid discriminatory effects of standards, enabling low-income households, men and women, and vulnerable people to equally benefit from the Project. Opportunities may exist to introduce appliances designed for individuals with visual or motor disabilities.

Activity 1.4.3. **Strengthen in-country capacity for compliance verification.** Strengthening of the capacity of the national standard body *Agence Nationale de Normalisation, de Météorologie et du Contrôle Qualité (ANM)* in cooperation with international partners. This includes the identification of shortlisted, regional certified test laboratories for equipment verification, definition of procedures for compliance verification including field inspections. It further extends to making accessible relevant information on minigrid equipment and appliances to national customs officers.

Output 1.5. Support provided to establish the environmental and social policies and plans to ensure mini-grid risks are properly handled.

60. The purpose of this output is to: (i) support government efforts to establish policies and plans to ensure social and environmental aspects of minigrid development are properly handled, and (ii) assure the Project's Environmental and Social Management Framework (ESMF) is effectively put in place and implemented, complementing baseline efforts where needed.

61. The following activities will be carried out:

Activity 1.5.1. **Provide recommendations for enhancement of the ESIA framework for electric minigrids.** Benin has policy and strategies in place to prevent the release into environment of solid waste and harmful substances and devices. The waste sector is supervised by the *Ministère de Cadre de Vie et du Developpement Durable (MCVDD)*; waste management executed by the ACVDT. The waste management system in the larger cities is being upgraded including an institutional reform through the creation of regional waste management companies such as SGDS-GN in Grand Nokoué.[39] Actions to modernize the sector include the acquisition of garbage trucks and collectors, awareness raising campaigns at schools (*"Programme éco-citoyenneté"*), centers for waste separation, revalorization and logistics distributed over the territory (Natitingou, Parakou, Bohicon, Abomey, Lokossa), the elimination of open waste dumps, and the implementation of a geo-referential data management system (PAG 2021-2026)[40]. Prevention of electronic waste is also demanded by the *Code Numérique* (Loi 2017-20, Art. 32).

Progress under the baseline provides opportunities for minigrid concessionaires to connect to national waste management systems to prevent the release of electronic[41] components – specifically discarded PV panels, batteries and appliances – into the environment. Moreover, minigrid operators may help building environmental awareness among remote communities and put in place incentives for people to hand in broken equipment (*Extended Supplier Responsibility - ESR*).

Activity 1.5.2. **Periodically review and update the Project's ESMF.** The Project will cover the cost of services of a social & environmental safeguards expert for periodic review and updating of the SESP and EMSF of the overall project, as needed. The expert will bring in experiences from the AMP Regional Project and share with the incumbent authorities, relevant approaches and methodologies as related to social and

[39] See: https://www.afrik21.africa/benin-la-facturation-de-la-collecte-de-dechets-entrera-en-vigueur-en-janvier-2023/. Note that urban citizens will now pay a fee for the waste collection service.

[40] Programme d'Actions du Gouvernement (PAG) 2021-2026, p. 196-210. Source: https://beninrevele.bj/pag-2021-2026/

[41] The "*Déchets d'équipements électriques et électroniques*" (DEEE).

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environmental aspects of minigrids and the management and recycling of electronic waste. This shall ensure compliance and alignment of the Project with best practices (SESP and ESMF aspects) and open opportunities for learning-by-doing for authorities and developers through exposure to (and abiding by) ESMF-required studies and measures such as Strategic Environmental Assessments, minigrids-specific Environmental and Social Management Plans, etc.

Activity 1.5.3. **Manage proactively climate change-related risks.** The effects of climatic variability as well as manifestations of climate change potentially affect RE minigrids assets and infrastructure. Special consideration will be given to this risk during planning, design, construction and operation of mini-grids. This activity will draw on findings and practices gathered in related interventions supported by UNDP (e.g., *PANA Energie*, *PCM Bonou*, and *Biomasse Electricité*) which identified climate-related risks including violent winds, prolonged periods of temperature rise, flooding threatening mini-grid project.

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Component 2. Business Model Innovation with Private Sector

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656.Outcome 2: Innovative business models based on cost reduction are operationalized, with strengthened private sector participation in low-carbon/renewable energy minigrid development.

62. Against the backdrop of a growing portfolio of private minigrid concessions aided by bilateral and multilateral investment funds[42], this component pursues the objective to integrate and demonstrate various aspects of the minigrid business model, including: (i) adequacy of the delivery model including community's capacity to effectively support minigrids; (ii) overall quality assurance of all stages of the concession process; (iii) adequacy of technical standards and performance benchmarks; and (iv) optimized system sizing to achieve cost reductions. The minigrid pilots will be selected through a competitive tender mechanism; the specific scope and terms of these to be defined by the Project in dialogue with the GOB, during the first year of Project execution.

63. Selection of the one or more pilots will take place under output 1.1. It is observed that Benin's Off-Grid Electrification Master Plan (PDEHR) presents a classification of rural communities according to their size and remoteness to the main grid. A methodology is applied to determine the social and economic potential of each community according to a set of parameters (including population size, economic activity, among others). This approach allows for a ranking of communities enabling a prioritization in time. Under guidance of ABERME, the country has progressed in mapping rural communities in the entire country, including surveys of socio-economic and terrain parameters.

64. The Government of Benin classifies potential sites in three main groups according to their size: (i) villages above 1,500 people which it envisions to serve by RE-based, or hybrid minigrids (*"mini-réseaux"*); (ii) villages between 800 and 1,500 inhabitants powered by RE-based micro/pico power plants (*"mini-centrales"*) to serve social and public services (water, health, education, civil protection; and productive uses) and community spaces and activities, to be connected through a microgrid; (iii) villages with less than 800 people are served by stand-alone technologies including solar PV kits for homes and community facilities, solar lanterns and PV street lights. Smaller villages tend to be more disperse, making the full minigrid solution more costly. The project's Minigrids Pilots Plan will take stock of these sites and their characteristics. However, it should be noted that in no cases will GEF funding be used to finance interventions to grids where the diesel capacity is increased, and in no cases GEF funding would ever be invested in fossil fuel power generation assets.

65. The country also has a legacy of nearly 100 solar PV with battery minigrids implemented through ABERME, for which the technical and financial management has been handed over to the corresponding *Mairies*. **A subset of these minigrids have experienced – or are currently experiencing - technical and operational challenges**. These minigrids did not manage to grow and become financially sustainable. Instead, performance declined over time and customers moved away from the systems, potentially resulting in a loss of confidence in the

[42] Reference to current programs UEF, MAC, etc.

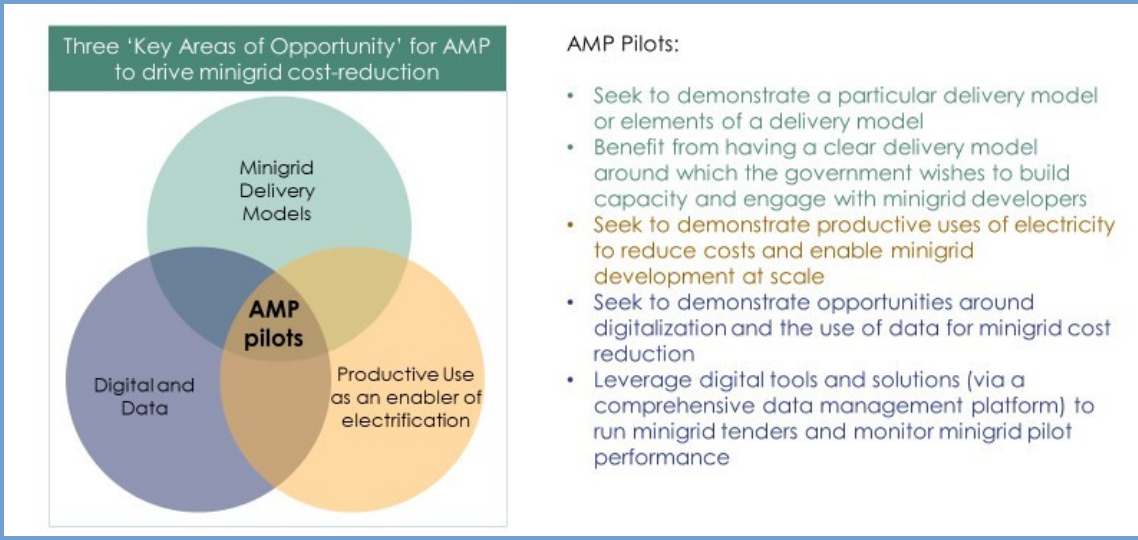
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envisioned off-grid market. A transfer of the technical and financial management of (clusters of) these grids to a private concessionaire would fit into the present delivery model outlined in the Electricity Law 2020; while the role of the community committee would become one of monitoring service quality and as a conduit to defend community interests.

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658.Figure 9 Objectives of Pilots in AMP National Projects.

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66. Productive Uses of Electricity (PUE) as an enabler for electrification and a vector of sustainability of the minigrids are considered in the above-mentioned prioritization methodology. The incorporation of PUE is therefore not a key objective for the Benin Project, however, opportunities for PUE will be assessed to maximize local social and economic impact within the pilot area. Similarly, and in alignment with the concession model, the use of other RE technologies can be assessed and may help reducing the LCOE for a specific minigrid. Note that, as per the delivery model, concessionaires must propose a competitive tariff.

Table 3. Minigrid pilot types considered by the project.

660. Type of pilot	661. Description	662. Scenario without project (baseline)	663. Use of GEF funds
664. Greenfield minigrids	665. 1) RE-based minigrids (“*mini-réseaux*”) in rural communities with a population size of typically above 1,500 people. 666. 2) Smaller villages (800-1,500 people) are typically targeted by RE-based micro-grids (“*mini-centrales*”) serving community uses and PUE. Households would	667. Without access to basic electricity service. Some small busines	668. GEF funds can be used to support development costs of selected minigrid pilots and as a subsidy to upgrade or complement public assets (distribution grid components, some PV/battery capacity).

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	initially be connected through prepaid charging services for solar lanterns and cell phones.		ses may use a gasoline generator.	
669. Hybridization of existing diesel minigrids	670. Only hybridization of existing diesel minigrids would be considered (i.e. displacing actual consumption). The Minigrids Pilot Plan would further document the justification (if any) of hybridizing instead of replacing existing minigrids for sites suggested by the Government of Benin.		671. Some existing diesel minigrids may continue to operate at a loss (operating costs).	*672. GEF funding will never be used to finance interventions to grids where the diesel capacity is increased, and in no cases GEF funding would ever be invested in fossil fuel power generation assets.*
673. A productive use overlay to an existing or planned minigrid	674. This type is not pursued explicitly by the AMP in Benin. However, productive use of electricity is specifically targeted in the Greenfield minigrid scheme.		675. Limited familiarity with electricity leaves opportunities for PUE under-exploited, limiting economic development and electricity demand growth.	676. As part of pilot design and community engagement, some GEF funds can be used for assessing PUE, creating users´ interest, and assess feasibility of electrification of specific end-uses (several electricity nexus).

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67. An approach (to be explored further as part of output 1.1) could be for GOB to **rehabilitate and expand an existing minigrid** as a "*Projet Phare*" ("Lighthouse Project") following a strict technical and operational due diligence process covering aspects including: (i) hardware specification including climate risks; (ii) technical compliance and boundary conditions for proper functioning; (iii) quality of civil works, distribution network, and indoor installations; (iv) community engagement and awareness; (v) clarification of roles and responsibilities between system operator, *Mairie*; (vi) technical support and availability of spare parts; (vii) system sizing and strategies to promote demand growth; and (viii) integration of productive uses of electricity (PUE) and local RE sources. By addressing these aspects, the pilot could serve as a reference for rolling out inclusive minigrid systems in Benin, and as a benchmark for the GOB to set performance standards for concessionaires. GEF incremental action thereby enables the GOB to address a series of risks (informed by the DREI methodology), thereby lowering the costs and accelerating the market uptake of minigrids in the country. The PMU and consultants will guide the pilot design and implementation process, with some GEF funding available to support the investment, probably as a CAPEX subsidy (output 2.2).

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68. The Project further seeks to enhance the social, economic and environmental benefits (output 2.3). Empowerment of end-users and communities is pursued by close engagement with the target communities by specialized national experts, who will perform specific activities such collection of information on energy needs and preferences by women and men; organization of communities for quality assurance and complaints handling; as well as awareness and information campaigns in the pilot community and the broader area. Conduits for building momentum in the communities include associations private businesses, agricultural development organizations, women and youth associations, charity and religious organizations, among others. Where appropriate, the Project will partner with initiatives from sectors including water and sanitation, health, SME development to maximize impact. The Project will identify PUE opportunities in the pilot area and identify the conditions to be met for successful electrification. Fact sheets will be prepared for identified business cases as input for community outreach activities and sharing across the AMP Project community.

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Output 2.1. Pilots developed (preparatory phase), including on productive use and innovative appliances and modular hardware and system design, leading to cost-reduction in minigrids.

69. This output will support the GOB to establish the basis for selection of one or more pilots. Tendering, contracting and performance tracking is foreseen to be automized through a Digital Platform (see Component 4). The pilots shall respond to specific objectives set forth in a Minigrid Pilot Plan (MPP) to be developed under output 2.1. The pilots will enable the incumbent authority to gain proficiency with the configuration and use of the Digital Platform. The MPP will define evaluation criteria for selection of candidate pilot sites, milestones, and go/no-go decision points. The MPP shall be finalized by the end of Year 1, with support from the AMP Regional Project. The terms of reference will consider, among other factors which the PMU will determine with support from the AMP Regional project, the following: (i) establishing a requirement and incentives for pilots to share data with the project; (ii) including incentives for the proposals to be gender-responsive and (iii) including a requirement for environmentally-sound collection, storage and disposal of all electronic and electrical waste, including rechargeable batteries, associated with off-grid renewable energy technologies.[43]

70. The following activities are envisioned under this output:

Activity 2.1.1. **Develop the Minigrid Pilot Plan (MPP) for advancing the Project's pilot(s).** The MPP is a working document containing a detailed project plan to guide the preparation, planning, and implementation of the envisioned minigrid pilot(s). It will describe the involvement and roles of public actors (including ABERME and ARE), private actors (suppliers and operators), and end-users (households, businesses and communities). The MPP will define evaluation criteria for selection of candidate pilot sites, milestones, and go/no-go decision points. The MPP shall be finalized by the end of Year 1, with support from the AMP Regional Project. The MPP shall be technically reviewed and cleared by UNDP (Benin CO and BPPS NCE), and (ii) shared with the Project Board.

[43] Note that the items (ii) and (iii) are also demanded by national Law.

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679.Box 6: Purpose and scope of the Minigrid Pilot Plan

680. 17. In order to enable a comparison of minigrid pilots across participating countries in the African Minigrid Program, the Minigrids Pilot Plan shall cover the following aspects:

(1) Define the objective and scope of the pilot(s) and secure problem ownership by all key stakeholders;
(2) Describe the delivery model to be demonstrated, including assumptions and ;
(3) Define pilot type: greenfield/existing grid, productive uses or productive use overlays;
(4) Carry out geo-spatial mapping, system sizing and covered end-users;
(5) Identify and budget for required site-specific assessments including SES, gender assessments, electronic and other waste management and disposal plans;
(6) Specify functional and technical (including hardware) requirements and arrangements for remote monitoring and data collection requirements;
(7) Define the approach to mobilize private and public cofinancing to leverage GEF investment grants, i.e. to strive at minimum concessionality;
(8) Assure that GEF grant transfers (if any) are compliant with UNDP's policies and financial rules;
(9) Define grid asset ownership arrangements and applicable contracts;
(10) Ensure that productive-use equipment is owned by a third party;
(11) Identification of any legal and moral liabilities, measures to address them; and updating of the Project's Environmental and Social Management Framework (ESMF) and associated plans. And:
(12) Identification and budgeting of operational support during start-up phase, and training and community outreach, to enhance pilot sustainability, performance, and impact

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Activity 2.1.2. **Design of tender process for pilot(s) using a digital platform.** Private sector engagement can involve minigrid ownership, private sector engineering, procurement and commissioning (EPC), and private sector O&M services. A competitive tender process will be executed using the **tendering feature** of the Digital Platform to be procured under Component 4. The precise scope of the tender will depend on the chosen locations (e.g., full-size minigrids, mini-centrales, "legacy" minigrids with public assets in place, distributed electricity services "kits") but shall, as a minimum cover the distribution, commercialization, and O&M in a concession area, as per Electricity Law 2020.

Under this activity, the PMU in dialogue with the IP and under guidance of the AMP Regional Project, will translate the approach set out in the MPP into the design of a customized tendering process on the digital platform, including requirements, specifications and evaluation criteria. The terms of reference will consider, among other factors which the PMU will determine with support from the AMP Regional project, the following: (i) establishing a requirement and incentives for pilots to share data with the project; (ii) including incentives for the proposals to be gender-responsive and (iii) including a requirement for environmentally-sound collection, storage and disposal of all electronic and electrical waste, including rechargeable batteries, associated with off-grid renewable energy technologies.[44] At the end of this activity, the tendering process on the digital platform will be ready to launch. The tender process itself should be launched before the end of Year 2.

Activity 2.1.3. **Execution of tender, contracting and payments to the selected pilot beneficiaries.** In year 2, the tender will be launched and executed according to the design finalized in activity 2.1.2, resulting in minigrid developers/operators being selected as pilot beneficiaries. Submissions to the tender will be assessed against evaluation criteria including the tariff proposal as demanded by Law, with the PMU supported by appropriate expertise. The tariff schedule shall be approved by the Regulator (ARE). Following selection of beneficiaries, the incumbent authority will enter into legal contracts with the selected concessionaires, supported by appropriate expertise, and make payments on pre-defined milestones. The digital platform will then validate the payment milestones.

[44] Note that the items (ii) and (iii) are also demanded by national Law.

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Output 2.2. Commissioning and monitoring of selected pilots in alignment with AMP principles.

71. This output will provide technical support for pilot implementation through its Project Engineer who will also act as a resource person to the *Mairie*. The Project Engineer will follow-up on permitting processes, and maintain close communication with local communities and beneficiaries. Note that, in response to a tender, a private concessionaire may take charge of installing the minigrid. In this scenario, the role of the MPP and the PE is to accompany the process to ensure that the pilot can be delivered timely to have a reasonable timespan before Project termination to collect meaningful data for fact-finding and learning. The pilot will be equipped with data logging and communication devices to collect operational data, including performance data and information to verify the status of hardware components and meters. GEF funding is available to enable financial viability of the tender, possibly through a CAPEX subsidy complementing GOB co-finance and parallel investment by the investor (to be detailed in the MPP).

72. This output comprises the following activities:

Activity 2.2.1. **Deliver technical designs and engineering plans.** This activity concerns support to the design phase of the pilot including detailed technical specifications and engineering of civil works and equipment, system integration and operational requirements including annual expenditures and staffing. For the considered pilot project, the *Mairie* will assign a focal person for engagement with the local community and the Project team. The Project will provide technical support through its Project Engineer hired with GEF budget, who will also act as a resource person to the *Mairie*. The Project Engineer will guide the minigrid project design, follow-up on permitting processes, and maintain close communication with local communities and beneficiaries. GEF budget is further available for procuring specific studies to support project development, if required.

Activity 2.2.2. **Construct and commission the minigrid pilot(s).** This activity involves the construction, commissioning, and initial operation of the envisioned minigrid pilot in alignment with the MPP and the proposed business model. GEF funding for investment can be used as a subsidy to build minigrid assets (distribution network, PV and battery capacity) which will be publicly owned and made available to the concessionaire. The Project Engineer (PE) hired by the Project will be in charge of process purview and be present during formal acceptance of equipment and works. The Technical Advisors, ideally recruited from an AMP Regional Project' shortlisting process (or at least vetting), will bring in quality advice and expertise gained across the participating countries to assure the quality of the minigrid pilots in Benin.

Note that, in response to a tender, a private concessionaire may take charge of installing the minigrid. In this scenario, the role of the MPP and the PE is to accompany the process to ensure that the pilot can be delivered timely to have a reasonable timespan before Project termination to collect meaningful data for fact-finding and learning.

Activity 2.2.3. **Collect operational data for analysis and sharing.** The pilot will be equipped with data logging and communication devices to collect operational data, including performance data and information to verify the status of hardware components and meters. Data will be stored, aggregated and shared through internet/GSM technology with the Digital Platform.[45] The latter will be hosted by the Ministry of Energy and technically operated with support from the GEF Project team (i.e., the PMU and technical consultants) – see Component 4. The collected data can be used for purposes including: (i) to track the performance of the minigrid in real-time; (ii) to monitor the evolution in time of load and RE capacity utilization; (iii) to monitor the number and type of connections/customers; and (iv) to validate the underlying assumptions for the business case. The ME will share data with the AMP Regional Project for aggregating and analyzing data across all AMP national projects. The Regional Project will use these data to produce insights and lessons learned, and to inform the development of knowledge products. The Project will cover the costs of data communication during the first pilot project year and provide technical support to assure adequate functioning of the system.

[45] Relevant data that cannot be collected remotely shall will be collected either by the minigrid operator as stipulated in the contract.

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73. The purpose of data sharing under the AMP is explained in Box 7.

682.Box 7: Data sharing for minigrid pilots

18. Beneficiaries from the AMP GEF Project (e.g. minigrid operators receiving support from the Project) will be required to share minigrid performance data with the national Project. Specific terms and conditions for data-sharing and how best to operationalize the commitment and its adoption by the beneficiaries, will be defined and agreed upon with minigrid operators during project implementation, including details of which data may and may not be collected and/or made public, based on consultations with industry stakeholders by the AMP Regional Project.

19. Specifications for data generation and collection will consult and follow guidance/standards provided by the AMP Regional Project. A standardized Quality Assurance and Monitoring Framework (QAMF) for application in all minigrid pilots, will be developed in Year 1 of the AMP Regional Project and disseminated among the national projects. Under Component 4, the QAMF will be adapted to the project context in Benin and implemented. The Digital Platform will be procured to serve a variety of purposes, including: (1) possibility to run tenders for off-grid electricity concessions; and (2) possibility to centralize relevant technical, performance, financial data of minigrid concessionaires; (3) tracking of infrastructure, equipment and other assets in public hands that are used by concessionaires. Other functionality enabling the effective implementation of Electricity Law 2020 and specific regulation, may be incorporated, to be specified by the Project with technical advice from the AMP Regional Project.

20. The Digital Platform enables the GOB to produce relevant information and data to develop tender documents and/or make available consolidated information to prospective minigrid developers and investors. The Digital Platform will incorporate best-in-industry tools for analyzing minigrids (e.g. demand forecasting, system optimization, distribution network design, detailed financial modeling at the site and portfolio level). The AMP Regional Program will roll-out activities across the participating countries to build capacities among government staff, minigrid developers and other stakeholders for the proficient use of analytical tools and data management technologies for

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74. Minigrid pilots are expected to share data with the Minigrid Digital Platform operated by the Project's Implementing Partner. This involves the use of data acquisition and communication hardware and standardized communication protocols, as sketched in Box 8. Technical specification shall be included in procurement documents.

684.Box 8: Indicative Specifications for Minigrid Pilots' Digital Hardware and Software (from AMP)

685. 21. Indicative minimum requirements and costs references for hardware/software for data-sharing include the following:

22. Offering	23. Details
24. Hardware requirements per site	• Inverter monitoring (monitoring & control) • Distribution monitoring • Optional current transformers for energy meter if more than 10 kW (single phase) or 30 kW (three-phase) • 24V power supply (50€) • Various data cables and installation material • Optional: 24V backup battery (50€) • Optional: Cabinet for the complete monitoring system • Industrial internet router • Industrial or high quality Ethernet Switches
25. Hardware requirements	• Smart meter

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Output 2.3. Enhancement of minigrid business model by strengthening operator and community capacities, development of PUE and other energy nexus, and the integration of local RE sources.

75. This output entails outreach and capacity building targeting communities, end-user groups and private sector, as well as additional studies and liaison to enhance social and economic impact and contribute to long-term sustainability. Activities that require sustained local presence, will be centered around the pilot site within the corresponding concession area. The following activities are envisioned:

Activity 2.3.1. **Empower end-users and communities.** This is pursued by close engagement with the target communities by specialized national experts, which will perform specific activities such collection of information on energy needs and preferences by women and men; organization of communities for quality assurance and complaints handling; as well as awareness and information campaigns in the pilot community and the broader area. Conduits for building momentum in the communities include associations private businesses, agricultural development organizations, women and youth associations, charity and religious organizations, among others. Where appropriate, the Project will partner with initiatives from sectors including water and sanitation, health, SME development to maximize impact.

Activity 2.3.2. **Promote employment in the minigrid sector.** Employment opportunities in the minigrid sector ultimately depend on the expansion of the market and the willingness of operators and developers to contract local personnel. In cooperation with the private sector, the Project will make an effort to draw interest from local individuals in the minigrid sector through engagement with youth and women organizations. At community level, the Project will seek and promote opportunities to employ unskilled and semi-skilled personnel and promote internships for women and men from communities with private and public sector entities. The Project will engage with educational institutes in the country including vocational schools to set up short technical courses on minigrid operation, maintenance, and management. Project funding will be selectively used for liaison and organization of events to complement baseline activities.

Activity 2.3.3. **Map and facilitate productive uses (PUE) in the pilot area.** Electricity demand from productive users enables minigrids to attain a positive cashflow in their early phases and successfully reach financial viability. However, larger consumers are not abundant in rural Benin as the majority of settlements consist of low—income families often dedicated to agriculture and cattle farming. Opportunities exist to electrify traditional production processes such as milling, wood crafting and water pumping. This activity will identify PUE opportunities in the pilot area and identify the conditions to be met for successful electrification. Fact sheets will be prepared for identified business cases as input for community outreach activities and sharing across the AMP Project community.

Activity 2.3.4. **Support the integration of local RE options.** The exploitation of domestic RE sources is one of the pillars of Benin's strategy to increase energy autonomy. Many unserved areas in Benin are well endowed with potential for decentralized hydropower and biomass systems, which has been mapped as part of Benin´s Off-Grid Electrification Master Plan (PDEHR, 2017). Run-off-river, pico/micro-hydropower plants and biomass gasifiers (in the range of 5-500 kW) can provide energy to minigrids (*"mini-réseaux"*) and community solutions (*"mini-centrales"*), thereby complementing solar PV/battery supply and diesel power. The Project will assist the GOB to explore opportunities for development and integration of such systems in the minigrid pilot area, through activities including prefeasibility studies, local capacity building, liaison with national and international technology providers, and promotion of investment opportunities.

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Component 3. Scaled-up Financing

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688. Outcome 3: Financial sector actors are ready to invest in a pipeline of low-carbon minigrids and concessional financial mechanisms are in place to incentivize scaled-up investment.

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76. Long-term concessions are foreseen in the Electricity Law as the framework allowing minigrid operators to generate, distribute, and commercialize electricity within a defined geographic perimeter. Given the low income level of most of Benin's rural population, end-user tariffs will not enable minigrid operators to fully recover operational costs. Operators applying for a concession shall propose a competitive tariff level. The GOB, through the incumbent authority, will provide a subsidy to the Concessionaire enabling full cost recovery plus a profit margin, to be calculated according to an established methodology and a positive advice by the regulator ARE (as per Electricity Law).

77. This specific objective of this component is to contribute to the development of financial mechanisms to sustain capital flows towards RE-based minigrids in Benin and foster investors' appetite in this sector. Given the low income level of most of Benin's rural population, end-user tariffs will not enable minigrid operators to fully recover operational costs. Operators applying for a concession shall propose a competitive tariff level. The GOB, through the incumbent authority, will provide a subsidy to the Concessionaire enabling full cost recovery plus a profit margin, to be calculated according to an established methodology and a positive advice by the regulator ARE. As public budgets are heavily constrained as the country largely depends on concessional funding from its development partners for investment and recurrent expenditures, there is substantial perceived risk that GOB could default on its commitment to subsidize electricity tariffs in the medium and long term. This (counterpart) risk deters private parties from investing in minigrid infrastructure, unless additional guarantees can be offered. To this purpose, the Project will closely engage with Benin's development partners push forward this agenda (*"chantier"*) to design and implement a long-term financial instrument to this purpose. Given the similarity of the challenge for most of the countries participating in the AMP, the Project will seek opportunities to address the finance barrier within the larger context of the minigrid market in Africa.

Output 3.1. Innovative financing solutions for minigrid development are identified and implemented with supporting human and institutional strengthening.

78. This output aims to contribute to the implementation of a financial facility for off-grid electricity systems in Benin, to be established through multilateral agency coordination in Benin and/or the region. The Project will support through participation in high-level negotiation and design panels. Where possible, leverage of partners and funding is sought through the AMP Regional Project thereby seeking economies of scale.

79. This output includes the following activities:

Activity 3.1.1. **Engage with international finance community.** Public sector agencies and international financiers will be engaged, including representatives of the *Fonds d'Electrification Rurale et des Energies Renouvelables (FERER)*, the SEforAll/GIZ Universal Energy Facility (UEF) and the MCA-Benin II supported Off-Grid Electrification Fund (OGEF), the African Development Bank (AfDB), BOAD, World Bank, Islamic Bank, and others. Private sector representatives include the African Minigrid Developers Association (AMDA). This output will review challenges and barriers in dialogue with the development partners of the GOB, with findings feeding into the national dialogue.

Activity 3.1.2. **Identify financing instruments and products.** Preliminary assessments and experience indicate that integrated financing solutions involving public and private agents could be more appropriately implemented in Benin through a second-tier bank or a national development bank. Therefore, this activity will be conducted in partnership with interested stakeholders and will focus on the need for – and capacity to roll out - financial products such as loans for procurement of RE systems (PV and battery storage systems) by minigrid developers and distributed energy companies. On the off-taker side, end-user-oriented finance instruments could include credits for EE appliances and small PV systems and loans for larger (EE) equipment used by commercial and agrifood businesses. The specific nature and blend of these products and mechanisms is to be designed.

Activity 3.1.3. **Liaise with digital payment providers.** The Project will liaise will the minigrid sector and telecom providers in Benin to identify opportunities for digital payment solutions which can boost local economic development and leverage end-users' money to invest in electric appliances thereby increasing demand. The analysis of Pay-as-you-Go (PAYG) prepaid solutions in combination with

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mobile payment solutions (e.g. Momo Pay)[46]_offered by national telecom provider MTN demonstrated the impact of new energy access as an enabler for customer behavior shift towards the increased utilization of digital services, including digital money.[47]

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Output 3.2. Domestic financial sector capacity-building on business and financing models for minigrids.

80. This output encompasses promotion and targeted training for national financial sector entities to understand and implement RE/EE financing schemes for households and commercial sector.

81. This output includes the following activity:

Activity 3.2.1. **Provide targeted training for bank officers.** Training is envisioned to enhance understanding and appraisal of energy investments solicited by minigrid operators and end-users. In this context, the Project will encourage strengthening of partnerships in the region enabling local banks to pool resources. In collaboration with MDBs, the Project will explore synergies with related fields, including investment in productive uses (such as by rural banks), insurance policies and terms to mitigate loan risks, collateral terms, and mitigation policies to address climate risks.

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Component 4. Digital and Knowledge Management

691. Outcome 4: Digitalization and data mainstreamed, across stakeholders, into local minigrid market development. Increased knowledge, awareness and network opportunities in the minigrid market and among stakeholders, including benefitting from linkages to international good practice.

82. The specific objective of this component is to twofold: (i) to enable the Project to monitor the minigrid pilot and share data and lessons with the AMP Regional Project; (ii) to support the Implementing Partner in its mandate to supervise the off-grid electricity sector, through the adoption of digital systems and skills allowing effective management of sector data, monitoring of minigrid concession contracts, cost and performance benchmarking, maintaining an updated inventory of the status of public minigrid assets.

83. Digital technologies and solutions are fundamental to enabling off-grid electrification. The viability of minigrids relies strongly on certain digital technologies including remote control and monitoring of minigrid operations and the collection of customers' payments, including the use of digital money. Digital solutions also have offer significant cost-reduction opportunities, thus contributing to the AMP's objectives. Opportunities around digitalization also relate to the analysis of large amounts of data ("big data") from minigrid projects to surface insights enabling learning and optimization.

The AMP Regional Project will develop and implement a digital platform for the aggregation of data shared by the AMP national projects. This platform will use common data protocols and standardized approaches for data analysis and for monitoring of the minigrid projects. The AMP Regional Project will make available specialized digital tools and solutions for the off-grid and minigrids sector in the participating countries, identify relevant cases, and assess value and social impact. It should be noted that according to the Gender Action Plan, the project should "Ensure that digital capabilities do not unintentionally discriminate, violate user privacy, or serve as tools of coercive control", which may be particularly relevant to women and other vulnerable groups.

Output 4.1. A project digital strategy is developed and implemented, including linkages to and following guidance from the AMP Regional Project

[46] https://www.mtn.bj/momo/

[47] Source: The Value of Pay-As-You-Go Solar for Mobile Operators – Insights from customer journeys in Benin and Cote d'Ivoire, GSMA, 2020. Funded by UK Aid. See: https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2020/05/Mobile-for-Development-Utilities-The-Value-of-Pay-as-you-go-Solar-for-Mobile-Operators.pdf

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84. This output entails the preparation of a digital strategy to improve minigrid scalability and oversight using specialized digital tools and solutions. To this purpose, the AMP Regional Project will develop a comprehensive digital strategy, which will encompass all facets of how data and digitalization is mainstreamed and embedded across all regional AMP activities. This exercise involves consultations with UNDP energy staff working on parallel data and digitalization tools such as Kobo and Data4Minigrids, as well as RMI-supported platforms such as Electrifying Economies. The national projects shall develop their own specific digital strategy guided by the AMP Regional Project.

85. This output includes the following activities:

Activity 4.1.1. **Develop and implement the Project Digital Strategy.** All AMP national projects will develop a Digital Strategy for the project in Year 1, which will be implemented thereafter. The Project Digital Strategy will be updated on an annual basis to reflect learnings from project implementation, guidance received from the AMP Regional Project on digital tools and solutions, and insights gained from minigrid pilot(s) data and experiences. The design of the Digital Strategy will take into account the institutional context of the IP, including the existence of other digital systems and staff competences.

Activity 4.1.2. **Assess digital technology needs in electricity-dependent services in communities**. The Project Team will carry out a survey of digital technology needs in the pilot project area (and other rural communities, if budget allows), such as: (i) remote monitoring of health facilities; (ii) daily status overviews of public health status to control epidemic outbreaks or calamities; (iii) local meteorological and hydrological data collection; (iv) early-warning data collection; (v) infrastructure surveillance data including for local water pumping; (vi) agricultural market data (e-agriculture); (vii) opportunities for e-learning.

The early identification of opportunities for digital solutions (even at moderate bandwidth), may enable high value services and unleash economic development potential in the minigrid area. The purpose of the survey is to anticipate on such opportunities by taking a strategic approach towards productive uses of electricity. The Project will make efforts to convoke GOB partners from other sectors, community stakeholders, and Benin's international development partners to shape digital policies and action plans.

Activity 4.1.3. **Develop recommendations for a national-level digital strategy for minigrid development.** Upon implementation of the project Digital Strategy and based on lessons learned related to leveraging of digital tools and solutions for minigrid sector development, the Project, as well as all other AMP national projects, will develop a set of evidence-based recommendations for rolling out digital solutions for minigrids at the national level. These recommendations will be used to update the Project's Digital Strategy and may inform the design, scope, and functionality of the Digital Platform. The recommendations will be shared with key national stakeholders as input for sector policy and planning.

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Output 4.2. Specification and implementation of Minigrids Digital Platform to track minigrid pilots and support scale-up and cost-reduction.

86. The implementation of the Minigrids Digital Platform is expected to result in improved minigrid design and system optimization, reduce uncertainties as related to input data and assumptions; provide transparency during the selection and contracting process; and lead to more competitive offers from bidders. The Digital Platform will be used to concentrate and manage technical and financial data related to minigrid sites. Incorporated information (e.g. demand forecasting, system optimization, distribution network design, inputs for financial modeling) and best-in-industry digital tools for analyzing minigrids enable the IP to provide project developers with necessary information to build robust business models while reducing transaction costs.

87. The Platform offers a series of benefits to the sector agencies (Implementing Partner) including: (i) validation and storage of data and characteristics of all distributed energy projects/programs at in a centralized database; (ii) interface for collection, management and aggregation of data from all minigrids and connected RE systems;

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(iii) possibility to run digitized tenders and administer grants (if considered by Government); (iv) performance verification of minigrids; (v) real-time monitoring and evaluation of electrification projects/programs; and (vi) advanced analytics of minigrid portfolios to generate critical insights. Procurement of the Platform includes on-site installation, configuration and hands-on training by the supplier.

88. This output includes the following activities:

Activity 4.2.1. **Analyze business operations and digital infrastructure.** The Project will support the host for the Digital Platform (envisioned: Ministry of Energy) to map and assess current business operations including recommendations to make such processes more effective and efficient. The exercise will consider factors including availability and competences of key staff; conditions required to attract and retain such staff; existent tools, including digital systems and services in use. The proposed Digital Platform shall minimize the burden on the incumbent authority to the extent possible. As part of the Digital Strategy, the Project will propose an exit strategy enabling the authority to keep using and expanding the Digital Platform post-project.

Activity 4.2.2. **Develop Terms of Reference (TORs) for procuring a Minigrids Digital Platform.** The project will use standardized TOR provided by the AMP Regional Project and tailor them to the specific needs of the host. Box 6 provides indicative specifications for the Digital Platform which the AMP regional project will develop into standardized TOR for all countries. The Project team in Benin will further adjust these TOR to meet the specific requirements and technical possibilities of the incumbent authority and/or the Project.

Activity 4.2.3. **Procure the Minigrids Digital Platform.** The project will procure a country-level mini-grids digital platform and set it up to: (i) enable convening and capacity building for key stakeholders (public/private), (ii) collect and manage technical and financial data related to minigrid pilot(s) in accordance with the project's Quality Assurance and Monitoring Framework (QAMF); (iii) exchange relevant data subsets with the AMP Regional Project for analysis and publication of aggregated results; (iv) allow the Project and the incumbent authority to carry out minigrid tenders; (v) enable the incumbent authority to monitor the performance of minigrid operators and their compliance with the Terms of their concessions; and (vi) enable the incumbent authority to carry out data analysis and benchmarking of minigrid operations.

Activity 4.2.4. **Deliver after-sales services, technical support and training.** Configuration of the Digital Platform and training of Government and project staff will be required at installation. Servicing and additional training is expectedly needed afterwards, specifically: (i) technical configuration of data communication modules once minigrid pilots are ready for data sharing; (ii) training on the use of data analysis and reporting modules; (iii) Q&A sessions and specific topics (webinar); and (iv) installation of software upgrades.

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693.Box 9: Indicative Specifications for the Project's Digital Platform

6527. The project digital platform will provide key functionality for the project in terms of acting as the (i) national digital convening platform for key stakeholders (public/private), (ii) providing ongoing data gathering and M&E on minigrids, including linking to the AMP regional project and (iii) acting as the mechanism for tenders for minigrid developers/sites.

28. Functionality	29. Details
30. National digital convening platform for key stakeholders	• Set up of a country-specific, web-based platform to manage all technical and financial data related to minigrid sites at the site and portfolio level • Single site register of minigrid sites, with geospatial views and technical/financial benchmarks for site assessment • Set of best-in-industry tools for analyzing minigrids, including demand forecasting, minigrid system design and optimization, and financial modeling • Capacity-building and in-depth training of key government and other stakeholders to use analytical tools and data management technologies
31. National monitoring and evaluation platform (remote monitoring & analytics)	• Direct integration with smart meters and remote monitoring systems for live data feeds and monitoring (with options to address lack of remote monitoring systems or other restrictions) • Big data analytics and customized reporting to calculate and report on standardized metrics for pilot performance, based on project QAMF • Quality assurance of data quality, accuracy, relevance, consistency • Interactive tools to analyze data, filter, and view at varying levels of granularity • All pilot-specific data can be rolled up into national view, and all country-specific data can be rolled-up into regional view
32. Financing platform for running tenders to select minigrid pilot beneficiaries	• Complete end-to-end management of e-tenders for mini-grids customized to specific project/pilot needs (e.g. customized technology solutions, customized workflow, customized KPIs for pilot monitoring) • Automated proposal analysis for quantitative proposal components • Remote verification of connections through smart meter integrations • Automated M&E analytics for all RBF program indicators (connections deployed, amounts paid, gender/environmental impact metrics, etc.)

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Output 4.3. Adoption and operationalization of the project's Quality Assurance and Monitoring Framework (QAMF).

89. This output envisions putting into operation a framework for measuring, reporting and verification of the sustainable-development impact of the supported pilots, including GHG emission reductions. A standardized QAMF for all supported pilots will be developed in Year 1 of the regional project for dissemination to the AMP national projects. This AMP-QAMF will build upon the minigrid Quality Assurance Framework (QAF), a set of

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technical and financial performance monitoring indicators, developed by the United States' National Renewable Energy Laboratory (NREL), the AfDB Sustainable Energy Fund for Africa (SEFA) and others, as well as data gathering, pooling and ongoing analytical work by UNDP's partners including the Rocky Mountains Institute (RMI), Sustainable Energy for All (SE4All) and the African Minigrid Developers Association (AMDA).

90. This output includes the following activities:

Activity 4.3.1. **Inputs and feedback from the Benin Project to the AMP Regional Project.** National project staff will provide inputs and feedback on the framework design, its operationalization and the required adoption by the supported pilots minigrid operators. Concerns around data privacy or sensitive data shall be considered in this respect.

Activity 4.3.2. **Adoption of the standardized Quality Assurance and Monitoring Framework (QAMF).** National project staff will provide both inputs and feedback on the development of this framework and its operationalization. Concerns around data privacy or sensitive data on the part of minigrid operators will be considered and addressed in each case. The adoption of the QAMF shall be demanded in the contracts with minigrid operators and other beneficiaries of the Benin Project. The purpose is to ensure that the AMP Regional Project can aggregate common data metrics and track a standardized set of key performance indicators across all minigrid pilots across all partner countries and report this data to the donor on a programmatic level.

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Output 4.4. Engage with regional project by participating in Communities of Practice and capturing and sharing of lessons learnt.

91. The AMP Regional Project will support and facilitate knowledge management and information sharing between the regional project and national projects, among national projects, and between the program and the broader minigrid community. Its aim is to ensure that the latest developments, cutting-edge guidance and good practices in minigrids, as they relate to the program's main thematic areas and areas of focus, are captured and made available to the project and to relevant stakeholders.

92. To that end, the project, as well as other AMP national projects, will engage with the AMP Regional Project by sharing information and lessons learned from project implementation, requesting and receiving operational and technical support for project implementation, participating in the Program's Communities of Practice, and sharing monitoring and evaluation data.

93. The Communities of Practice (CoP) are supported by RMI, which is UNDP's partner in the AMP. The COP will share knowledge and facilitate the development of solutions to common challenges within the African minigrid sector providing support to ministries, government agencies, and electric utilities, among others. The CoP will strengthen South-South cooperation and learning, drawing on the experiences of participating countries in minigrid cost reduction and deployment, with a focus on policy, regulation, finance, and new business models.

94. The CoP envisioned under the AMP Regional Project will work closely with other collaborative groups, including: (1) The Africa LEDS Partnership Africa Mini-Grids Community of Practice (AMGCOP), convened by SouthSouthNorth and the US National Renewable Energy Laboratory (NREL), with support from The German International Cooperation GIZ (since 2019); (2) The African Association for Rural Electrification (CLUB-ER), currently composed of 43 African national institutions (agencies, funds, regulators, departments of energy, utilities) from 32 African countries; and (3) The Mini-Grid Partnership, convened by SEforALL and chaired by the Africa Minigrid Developers Association.

95. The following activities are envisioned:

Activity 4.4.1. **Participate in AMP Communities of Practice (CoP).** One of the primary ways AMP national project staff will interface with the regional project is via the 'Communities of Practice' (CoPs) and associated activities/platforms. While it is expected that many of the activities under the Regional Project Component #3 will be undertaken virtually (via internet-based platforms, webinars or digital

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platforms) it is also expected that the CoPs will include presential "in-person" workshops, meetings and training events.

Activity 4.4.2. **Sharing of research and lessons learned with the AMP Regional Project.** Research and lessons learned will be systematically shared with the AMP Regional Project based on guidelines that will be defined by the regional project staff and shared at the Regional Project's Inception Workshop. Capacity building will be provided to the Project Management Unit to compile lessons learned and share knowledge effectively.

Activity 4.4.3. **Collaborate with the regional project on an Insight Brief.** This and every other AMP national project is expected (in the course of the four-year implementation cycle) to collaborate with regional project staff on the development of at least one (1) 'insight brief' capturing (in an accessible format) selected key highlights from a successful national project activity. The 'insight brief' can cover any activity of the project and take the form of a written brief or video brief. The Project will hire a consultant or local firm to gather data and produce audio-visual content (video footage, photos, etc.) for these briefs for sharing with the AMP Regional Project. The regional team will utilize this content to produce the 'insight brief' according to a standardized communications format for all AMP knowledge products for external audiences. The 'insight brief' will be produced in both the local/national language of the relevant national project as well as English for dissemination by the regional project to regional stakeholders and publishing on the AMP website.

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Component 5. Monitoring and Evaluation (M&E)

Outcome 5: Ensuring compliance with all mandatory monitoring and reporting requirements of the GEF.

96. This outcome will assist the Implementing Partner in establishing project oversight and monitoring systems, including the Project's Environmental and Social Management Framework (ESMF) and resulting Management Plans, the Gender Action Plan (GAP), the Mid-Term Review (MTR) and the GEF Terminal Evaluation (TE) of the Project. The Project's M&E Plan is built upon experiences during project preparation with a view on mitigating implementation and fiduciary risks.

Output 5.1. Inception workshop is conducted and M&E plan is implemented.

97. This output will assist the IP during the start-up phase of the Project to operationalize management tools including the M&E Plan, understanding project risks and assumptions, use of the Risk Log, preparation of the first Annual Work Plan, streamlining of project implementation processes with GOB procedures, and documenting thereof in a Project Operations Manual (POM). Another task is finalization of the Terms of Reference of key project staff and consultancies in dialogue with UNDP and the IP. A short-term consultancy is envisioned to guide the IP through the inception phase to put the Project on track. This output includes the following activities:

Activity 5.1.1. **Conducting inception workshop and preparing report.** A project inception workshop held to officially launch the project and, among other aims, familiarize key stakeholders with the detailed project strategy, roles and responsibilities of the project team, and project planning instruments such as the Total Budget and Work Plan (Section IX), Multi-year Work Plan (Annex 4), Monitoring Plan (Section VI), and the Procurement Plan (Annex 11), among others. The national inception workshop will be carried at the beginning of project implementation within 60 days of signing the UNDP Project Document of this project. The workshop will be organized by the PMU with support from the IP, and planned with support from the UNDP CO and the AMP Regional Project staff. Staff from the AMP Regional Project PMU will participate either remotely or in-person in the Inception Workshop and will provide support to the project PMU to plan the workshop, and develop materials and content that will facilitate project planning activities including the template for the Inception Workshop Report. The Inception workshop report will be prepared by the PMU and submitted to UNDP within 60 days of signing the UNDP Project Document of this project.

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Activity 5.1.2. **Ongoing project monitoring of Results Framework indicators**. As set out in the Monitoring and Evaluation Plan (Section VI), data on Results Framework Indicators will be systematically collected and analyzed to provide the Project Board, relevant project stakeholders and the GEF with: (i) information on progress in the achievement of agreed objectives and the use of allocated resources, and (ii) regular feedback on performance of projects and programs taking into account the external environment. Information from systematic monitoring serves as a critical input to ongoing PMU management decisions (adaptive management), evaluation, and learning. The GEF Core indicators (see Prodoc, Annex 16) are included in the Project's Results Framework (see Prodoc, Section V) for monitoring impact at the Project Objective level. Progress towards these indicators will be determined annually for Project Implementation Review (PIR) reporting to the GEF, as well as prior to the Mid-term Review (MTR) and Terminal Evaluation (TE).

Activity 5.1.3. **Ongoing project monitoring of key project plans**. The project is accompanied by various plans including Stakeholder Engagement Plan (Annex 8), mitigation plan for project risks (Risk Register in Annex 6), and Gender Action Plan (Annex 10). These plans will be reviewed according to the monitoring and evaluation requirements. According to the project's social and environmental risk rating, there is a need to carry out continuous monitoring of the social and environmental safeguards as proposed in the Environmental Social Management Framework (ESMF) and other SES frameworks/plans (Annex 9). The environmental and social management plan (ESMP) that will emanate from the application of the ESMF will also be monitored under this activity.

Output 5.2. *Project Mid-Term Review is conducted.*

Activity 5.2.1. **Conduct a Mid-term review (MTR) of the project**. An independent mid-term review (MTR) will take place at the half-way mark of project implementation and will be conducted according to guidance, rules and procedures for such evaluations as provided in the UNDP Evaluation Guidance for GEF Financed Projects. The MTR will be made available to all project stakeholders in the relevant language.

Output 5.3. *GEF Terminal Evaluation is conducted.*

Activity 5.3.1. **Conduct a Terminal evaluation (TE) of the project.** An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The project's terminal GEF PIR along with the TE report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lessons learned and opportunities for scaling up.

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699. Expected Results

98. The Project contributes to GEF-7 Core indicator 6: Greenhouse Gas Emissions Mitigated. Benefits include: (i) direct emissions reductions attributable to the minigrid pilot investments made during the project's implementation period, totaled over the lifetime of the investments (20 years) compared to the business-as-usual scenario, i.e. fossil-fuel-based mini-grid(s); and (ii) Indirect emissions reductions resulting from the increased uptake of minigrids for off-grid electrification of rural areas due to replication, scaling-up and market change, to which the Project has contributed by creating an enabling investment environment and facilitating subsequent investment flows. Annex 12 describes the methodology used to calculate the estimated GHG emission reductions. Note that 10% of the estimated GHG benefits from the child project are deducted and attributed to the AMP Regional Project, which reflects the direct contribution of the latter to the national projects.

99. As per Annex 12, the project is expected to bring about the direct commissioning of 0.4 MW in solar PV generation capacity and 0.9 MWh of battery storage. The lifetime greenhouse gas (GHG) emissions reduction from project activities, particularly investment in minigrid pilots, is estimated at approx. 9,000 metric tons of

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carbon dioxide equivalent (tCO2eq) (direct) and 180,000 tCO2eq (indirect). The number of direct beneficiaries is estimated at 12,000 people, of which at least 50% are women, as a result of 2,400 new and/or improved minigrid connections.

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702. Partnership and Stakeholder Engagement

100. Partnerships: The 21 countries participating in the AMP will collectively build a knowledge base concerning the implementation of minigrids in Africa. The Benin child project has access to specific know-how and advice, and can draw on experiences from more advanced countries in the region. The support and interaction between the participating countries is tracked through the Results Framework of the AMP Regional Project.

101. Benin's cooperation partners strive at coordinating programming under guidance of the GOB. In the energy sector, key partners include AfDB, World Bank, GIZ, AFD, BOAD, the World Bank and notably the Millennium Challenge Corporation (MCC) who have several projects and programs in the country targeting the electricity sector. National entities involved are, besides the Implementing Partner, the regulator ARE, ABERME, and the utility SBEE. In the regional context, one can identify ECOWAS, UEMOA, and ECREEE, among others.

102. Private sector's engagement in the project: The private sector assumes multiple roles in the Project and, in the broader sense, as a group of agents in Benin's off-grid electricity delivery model. These roles include: (a) RE equipment electricity grid hardware supplier; (b) project developer bringing in technological, managerial and process know-how; (c) operation and maintenance (O&M), largely relying on national businesses entering the market; (d) foreign and national capital providers and investors; (e) services including specialized companies (as well as non-for-profit entities) providing technical and financial services for project development, social and environmental assessments; (f) domestic banking sector; (g) minigrid operators and telecom businesses involved in innovative payment systems and derived services.

103. The private sector will be directly engaged through procurement of goods and services. Business events and consultations will be used to engage with the sector to have relevant feedback and foster interest in the minigrid market, and to learn about innovative technologies and solutions under development.

703. Co-financed activities

104. It is important to note that some of the project results will be realized by co-financing activities with resources that do not flow through UNDP accounts. In particular, the mini-grid pilots to be built in the projects could be funded through a CAPEX (partial) subsidy from the project budget (GEF funds and UNDP TRAC) while remaining Capital expenditures could be funded by third parties (who will likely be private sector developers, but those are not precisely defined yet). While the funds from third parties will not flow through UNDP accounts, they will directly contribute to the same mini-grid pilots the GEF and UNDP funds are contributing to and will be essential to realizing the project objectives. For this AMP child project, these are "co-financing activities included as project results". The precise sources and amounts of these co-financing activities will only be known at implementation stage. UNDP is accountable to monitor all project results, including results to be delivered by these co-financing activities, to ensure consistency with UNDP and GEF policies and procedures, including social and environmental safeguards policies and requirements (SES). This is further detailed in this project's ESMF.

105. For these co-financed activities included as project results with resources that do not flow through UNDP accounts, the following procedures will need to be applied before co-financing activities start:

1. The co-financing partner's capacities will need to be assessed through the Partner Capacity Assessment Tool (PCAT) and the co-financing partner will need to develop a risk management strategy if gaps are identified, for UNDP's approval and subsequent oversight/assurance.
2. The co-financing partner will need to sign a legal agreement with UNDP or the Implementing Partner to confirm accountabilities, mentioning in particular the following sentence: "*The co-financed activities will be undertaken in full compliance with [co-financing partner's] policies and procedures. However,*

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because the activities are included in the results of the project the [co-financing partner] commits to monitor these activities consistent with the UNDP Project Document. The Project Board and UNDP will also assume an oversight and assurance role to further ensure the project, including the co-financed activities covered by this letter, remains consistent with UNDP policies and procedures. These arrangements will be confirmed through [signature of Project Document OR signature of Responsible Party Agreement with reference to the Project Document].".

3. Risks stemming from and/or to co-financed activities – as with risks from/to all other project activities – will be included in the project risk register and monitored accordingly. The risk description will clarify relation to the specific co-financing.
4. Social and environmental risks associated with the co-financed activities will be identified during project design and included in the SESP and relevant safeguard management plans. Relevant safeguards instruments prepared by the co-financing partner will be reviewed by UNDP for consistency with UNDP's SES, during project development and implementation; any gaps will be resolved in discussion with the co-financier.

704. Once the co-financing activities will have started, risks will need to be monitored (as per item 3 above) and results achieved through co-financed activities will be monitored and reported in the annual GEF PIR, the independent mid-term review and the independent terminal evaluation.

106. Some partners have provided letters of co-financing for this project, as attached in Annex 14 to this project document. As further described in the co-financing table below, most of these co-financed activities correspond to funds not flowing through UNDP accounts and whose results are not included in the project results framework. In this case, UNDP is accountable to monitor the risk to realization of co-financing amounts and realization amounts annually in the GEF PIR, at mid-term and at terminal evaluation. Specifically, potential risks associated with co-financing that may affect the Project, including safeguards related risks that fall within the project context or area of influence, will be considered in safeguards due diligence and the project risk register and monitored accordingly. Risk management measures identified will be only those within the control of the UNDP project (e.g. managing reputational risk). See the ESMF (Annex 10) for more details on the management of risks related to the different types of co-financed activities in this project.

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706. Table 4. List of co-financed activities

707. Co-financing source	**708. Co-financing type**	**709. Co-financing amount**	**710. Included in project results?**	**711. If yes, list the relevant outputs**
712. Ministry of Energy (ME)	713. Public investment	714. USD 21,000,000	715. no	716. -
717. African Development Bank (AfDB)	718. Loan	719. USD 12,000,000[48]	720. no	721.

[48] Roughly equivalent to 20% of the overall envelope of AfDB's Rural Electrification project (USD 66.5 m)

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722. UNDP TRAC	723. Grant	724. USD 200,000	725. yes	726. Output PM 1.2;

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729. Risks

107. The overall risk profile of the Project has been assessed as "substantial". Based on the risk categorization assigned to the various country projects and the associated environmental and social risks, the following procedures for screening, assessing and managing those risks must be undertaken during project implementation of each country project. (1) Screening of social and environmental risks and impacts and determining applicable social and environmental standards and requirements (including UNDP SES). The screening process utilizes UNDP's SESP and develops a specific screening procedure for the forthcoming type of sub-projects/activities. (2) Appropriate types of social and environmental assessment to identify, document and address potential social and environmental risks and impacts. (3) Preparing and approving time-bound action plans for avoiding, and where avoidance is not possible, reducing, mitigating, and managing adverse impacts, including development of specific management plans according to applicable policies and regulations, including UNDP's SES (i.e. Environmental and Social Management Plans which would be completed post-assessment).

108. Specifically, the SESP identified 14 risks, 2 of which assessed as "substantial", 11 as "moderate", and 1 as "low". The DREI framework identifies 10 main risk categories which indicate an overall moderate market risk for minigrid development (see Prodoc §20, table). Social and environmental risks are primarily linked to human rights given the challenges to secure social inclusiveness and incorporate vulnerable people. Among other root causes, underlying factors include very low-income levels of rural people living in small communities which rely on subsistence farming and collection; undefined or absence of land tenure titles and associative structures enabling people to claim their rights; demographic pressure which may lead to local movements of people as a result of electrification, potentially separating them from their food sources.

109. While electricity supply is a socio-economic enabler, the nexus between productive uses and electricity supply needs to be further articulated – including according to gender. To address this risk, the Project design incorporates sustained community engagement during the preparation phase of the proposed pilots. This activity will draw upon country knowledge and positive experiences in other sectors (e.g., agricultural development and small-scale e-commerce) to make these available to the energy sector in Benin.

110. Environmental risks are essentially two: (i) potential adverse impacts on biodiversity and habitat; and (ii) dispersal of electronic waste including batteries, into the environment. As related to the former, Benin is highly vulnerable to the effects of climate change, exacerbated by human activity including over-exploitation of soils and aquifers, which can lead to loss of species, soil erosion, and degradation of rivers and aquifers. Electrification will lead to increased human activity; adverse effects are mitigated by proper planning, avoiding specific areas such as natural parks altogether, and promote best practices including changes in attitude among local dwellers. The community engagement shall result in an appraisal of site-specific risk and determination of mitigation options.

111. As concerns the latter, the PPG observed that waste management policy is advancing in Benin but facilities for adequate disposal, or transfer for recycling are often absent in rural areas. In particular non-degradable waste and objects (plastics, batteries, broken equipment and tools) are prone to be abandoned and become dispersed in the environment. The situation poses a challenge for the repair and end-of-life treatment of minigrid components including wiring and electric appliances. Increasing awareness of the local population and inducing a sense of ownership and responsibility is one line of action. A second one is the integration of electronic waste management into an Extended Supplier Responsibility (ESR) scheme. It is noted that the sector legislation in Benin requires off-grid electricity concessionaires to comply with social and environmentally sound practices as part of their contractual obligations.

112. For more details, reference is made to the ATLAS Risk Log, Annex 6.

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731.Stakeholder engagement and south-south cooperation:

113. The PPG phase consultations has identified the following core stakeholder categories: Government ministries and ascribed entities (ME, DGRE, MCVDD, ARE, ABERME); local government representatives (*Mairies*, ANCB); public service providers (SBEE); private sector (installers, prospective operators, commercial banks; academia and training institutes; civil society organizations; regional energy centers (ECREEE). The Stakeholder Engagement Plan is presented in Annex 7. This Plan will be used as a tool for reference and be further detailed during the Project's inception phase and updated annually. The Plan is a starting point for the design of the Project communication strategy and specific communication plans.

114. The project will contribute to UNDP's South-South Triangular Cooperation platform and facilitate dissemination through global ongoing South-South and global platforms, such as Africa Solutions Platform, the UN South-South Galaxy knowledge sharing platform and PANORAMA[49]. The project through the knowledge portal and other learning tools will make information, lessons learned, and practical experiences available for uptake to the UNDP's solution provider mechanism designed to leverage south-south exchange.

115. The Project will provide experts and stakeholders in Benin with access to and integrated lessons from UNDP's South-South initiative thereby guided by the AMP Regional Project. In addition, to bring the voice of Benin to global and regional fora, the Project will explore opportunities for meaningful participation by national representatives in specific events related to minigrid deployment, rural energy technologies, environmental management and conservation, among others.

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733.Gender equality and Women's Empowerment:

116. The project will adopt a participatory approach for maximum impact through the inclusion of all relevant social groups, including marginalized people (e.g. unemployed youth), with attention to the participation and inclusion of women whilst respecting the norms, values and customs of targeted communities. The project will endeavor to provide direct and indirect benefits to men and women in an equitable manner, wherever possible given the existing prevailing bias (e.g., there are fewer women in lead government roles).

117. A Gender Analysis and ambitious Gender Action plan were prepared during the PPG that are included in Annex 10. The four key aspects of this plan are to: (1) boost women's participation in program elaboration and implementation; (2) ensure electrification efforts are meeting the needs of a wide range of male and female end-users; (3) ensure anchor clients on the whole reflect a balance of men's and women's interests; and (4) address affordability in way that is equitable for both men and women. There are two gender indicators in the project results framework, one that summarizes the degree of participation throughout and another that seeks to ensure that gender analysis and mainstreaming continue to happen throughout the research, design and learning activities that will be pursued as part of the program Specifically, this involves recruitment, setup of consultation platforms, streamlining of government agencies, engagement of decision-makers, training, empowerment, risk assessments, and livelihood activities, among others. More specific gender indicators and targets have been suggested in the Gender Action Plan for consideration by the PMU at a later date. The project will ensure that women are effectively involved by contracting a specialized national institution or firm for community development with proven expertise in gender and rural energy issues (Output 2.3).

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735.Innovativeness, Sustainability and Potential for Scaling Up:

118. Innovation is at the core of the AMP Program given its focus on cost-reduction and enhanced business models to reduce minigrid cost to increase affordability of RE-based electricity. Business involvement is sought to access private capital to leverage public investment and grant funding. De-risking of RE investments in combination with cost reduction translates into lower financing costs for the investor and increases economic and financial sustainability. Hardware, project development, and operational cost reductions are sought through competitive selection of minigrid project proponents to set a trend towards lower unit energy costs (LCOE) in the region. The incorporation of productive energy uses provides opportunities for income generation in communities improving local capacity to pay for, and sustain, the service.

[49] https://panorama.solutions/en

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119. The focus on the utilization of digital technologies is another innovation of the AMP, harnessing the opportunities of digitalization for improved efficiencies, lower costs and risks, facilitate the flow of investment and revenue streams, and thereby contribute to sector development in the region. So far, the minigrid market has not fully exploited the potential of digital tools and solutions to accommodate multiple, decentralized and distributed 'data points' in the minigrid system. Finally, the regional approach enables national projects and implementation partners to engage, learn and share experiences and best practices as input for national and reginal policy agendas and programs.

120. Social and economic sustainability is pursued based on a human rights approach to secure access to clean and affordable energy and related services. RE-powered minigrids have proven to be superior to conventional diesel grids in terms of energy costs, operability, and maintainability; and are environmentally superior. The Benin Project aims to test and anchor the required boundary conditions for minigrids to actually deliver on this promise. These include: (a) thorough specification of applied hardware, including a definition of the appropriate product choice striking a balance between hardware design, serviceability, repairability, replacement, and overall costs and support infrastructure; (b) social acceptance including local willingness to pay and care for installed equipment; (c) promotion of productive energy uses and access to finance for appliances and equipment by end-users; (d) integration of environmental protection aspects in the business model (such as waste collection). Given beneficiaries' low income levels, the Project seeks securing financial and operational sustainability under Benin's concession model for off-grid electrification.

121. Potential for scaling up is significant at project level as well as strategically. The Project envisions accelerating market uptake of minigrids serving the over 2 million people that would not be reached by expansion of the main grid. This market potential is readily demonstrated by the presence and activity of multilateral and bilateral financiers in the sector (e.g., MCC, UEF).

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737. V. PROJECT RESULTS FRAMEWORK

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739. **This project will contribute to the following Sustainable Development Goal (s):**

- SDG7: Ensure access to affordable, reliable, sustainable and modern energy for all
  - SDG 7.1 By 2030, ensure universal access to affordable, reliable and modern energy services
  - SDG 7.2 By 2030, increase substantially the share of renewable energy in the global energy mix
- SDG13: Take urgent action to combat climate change and its impacts;
- SDG5: Achieve gender equality and empower all women and girls.

740. This project will contribute to the following country outcome (UNDAF[50]/CPD[51], RPD[52]):

741. UNDAF outcome 1: By 2023, Benin's population, especially the most vulnerable, are more resilient and have a better quality of life through access to decent employment, food and nutrition security, clean energy, and the sustainable management of natural resources, the adverse effects of climate change, crises and disasters.

742. CPD Output 1.3: Solutions adopted for rural and urban populations to achieve access to clean, affordable and sustainable energy.

743.	744. Objective and Outcome Indicators	745. Baseline	746. Mid-term Target	747. End of Project Target
748. Project Objective: **749.** *750.* **751.**	752. To support access to clean energy by increasing technical and financial feasibility and by promoting scaled-up commercial investment, in low-carbon minigrids in Benin, with a focus on cost-reduction levers and innovative business models.			
	753. *Indicator 1:* Greenhouse gas emissions mitigated 754. (GEF 7 Core Indicator 6.2) *755.* Unit of measure: metric tons of carbon dioxide equivalent (tCO2e)	756. 0 tCO2e 757. *758.* (Project has not started)	759. 0 tCO2e 760.	761. Direct: 9,056 tCO2e *762.* Indirect: 179,273 tCO2e
	763. *Indicator 2:* Number of direct beneficiaries benefitting from energy access via minigrids, disaggregated by gender and by customer segment (residential, social, commercial/productive use) as co-benefit of GEF investment 764. (GEF 7 Core Indicator 11) 765. Unit of measure: number of people	766. 0 people 767. *768.* (Project has not started)	769. 0 people	770. 12,152 people (of which 50% women) 771. ------- 772. 12,000 people (residential) 773. 32 people (social) 774. 120 people (commercial/PUE) *775.* 12,152 people (total)
	776. ***Indicator 3:*** **Increase in installed solar PV capacity and battery storage** 777. (GEF 7 Core Indicator 6.4) *778. Units of measure:* *MW (solar PV);* 779. *MWh (battery storage)*	780. 0 MW solar PV 781. 0 MWh (battery) 782. *783.* (Project has not started)	784. 0 MW solar PV 785. 0 MWh (battery) 786.	787. Solar PV: 0.368 MW *788.* Battery storage:0.892 MWh
	789. ***Indicator 4:*** **Local residents trained in different aspects of minigrid development and operation (e.g. sales, distribution, operations, management) disaggregated**	792. 0 people 793. 794. (Project has not started)	795. Female: 15 [people] 796. Male: 15 [people] 797. Total: 30 [people]	798. Female: 30 [people] 799. Male: 30 [people] 800. Total: 60 [people]

[50] United Nations Development Assistance Frameworks (UNDAF)
[51] Country Programme Document (CPD)
[52] Regional Programme Document (RPD)

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	by gender. *790.* 791. Unit of measure: number of people			
***801.* Project Component 1**	**802. Policy and Regulation**			
804. Outcome 1. Stakeholder ownership in a national minigrid delivery model is advanced, and appropriate policies and regulations are adopted to facilitate investment in low-carbon minigrids. 805. 806. 807.	**808. *Indicator 5:* A minigrid delivery model to enable minigrid development is endorsed/adopted by the national government through a consultative process involving key stakeholders (e.g. relevant ministries, local authorities, rural populations, private sector, media, etc.)** 809. 810. Unit of measure: binary (1/0)	811. 0 812. 813. (Project has not started)	814. 0 815. 816. (Multi-stakeholder, national dialogue platform on minigrid delivery models established and active.)	817. 1 818. 819. (At least one minigrid delivery model is identified and endorsed by the government through the work of the multi-stakeholder platform and dialogue.)
	821. *Indicator 6:* Number of policy derisking instruments for minigrid investments - whose development has been supported by the project - are endorsed/adopted by the national government 822. 823. Unit of measure: Number of policy derisking instruments	824. 0 825. 826. (no rural/off-grid electrification policy in place)	827. 1 policy derisking instrument(s) adopted 828. 829. The specific instruments will be informed by the full DREI analysis	830. 3 policy derisking instrument(s) adopted 831. 832. The specific instruments will be informed by the full DREI analysis
834. Outputs to achieve Outcome 1	835. Output 1.1. An inclusive national dialogue to identify minigrid delivery models is facilitated, clarifying priority interventions for an integrated approach to off-grid electrification. 836. Output 1.2. Minigrid DREI techno-economic analyses carried out to propose most cost-effective basket of policy and financial de-risking instruments and contribute to AMP Flagship Report on Cost Reduction. 837. Output 1.3. Capacity building provided to public officials (regulator, ministries) specifically to design procurement/tender processes that incorporate cost-reduction levers and innovative business models. 838. Output 1.4. Domestication of quality standards for solar mini-grid components, and institutional capacity of national standards organizations/bureau strengthened. 839. Output 1.5. Support provided to establish the environmental and social policies and plans to ensure mini-grid risks are properly handled.			
***841.* Project Component 2**	**842. Business model innovation with private sector.**			
844. Outcome 2. Innovative business models based on cost reduction operationalized, with strengthened private sector participation in low-carbon/renewable energy minigrid development 845.	**846. *Indicator 7:* Minigrid pilots implemented that demonstrate a delivery model, cost-reduction measure(s) and/or productive use of electricity** 847. 848. Unit of measure: binary (1/0) 849. 850.	851. 0 852. 853. (Project has not started) 854.	855. 1 856. 857. Minigrid Pilot Plan for advancing the minigrid pilots is developed, and cleared by UNDP and the Project Board. (1) 858. Any project tendering process, as applicable, for minigrid pilots is	859. 1 860. 861. The Minigrid Pilot Plan has been successfully executed and the pilots are delivered, operational, and being monitored. (1)

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			launched. (1)	
	863. Indicator 8: Capacity of minigrid developers and/or operators is enhanced to implement innovative business models and incorporate cost-reduction levers in minigrid projects. 864. 865. Unit of measure: binary (1/0) 866.	867. 0 868. 869. (The Project shall assess the baseline in Year 1)	870. 1 871. 872. Planned capacity building activities for year 1 and 2 are implemented. (1) 873. 874. The capacity of targeted recipients is assessed by survey towards the end of year 2. On a scale of 1 to 5, an average score of at least 2 is achieved. 875. 876. - 1 represents a low level of capacity 877. - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)	878. 1 879. 880. Planned capacity building activities for year 3 and 4 are implemented. (1) 881. 882. The capacity of targeted recipients is assessed by survey towards the end of the project. On a scale of 1 to 5, an average score of at least 4 is achieved. 883. 884. - 1 represents a low level of capacity 885. - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)
887. Outputs to achieve Outcome 2	888. Output 2.1. Pilots developed (preparatory phase), including on productive use and innovative appliances and modular hardware and system design, leading to cost-reduction in minigrids. 889. Output 2.2. Commissioning and monitoring of selected pilots in alignment with AMP principles. 890. Output 2.3. Enhancement of minigrid business model by strengthening operator and community capacities, development of PUE and other energy nexus, and the integration of local RE sources.			
892. Project Component 3	**893. Scaled-up financing**			
895. Outcome 3. Financial sector actors are ready to invest in a pipeline of low-carbon minigrids and concessional financial mechanisms are in place to incentivize scaled-up investment.	**896. *Indicator 9:* Capacity of financial institutions is enhanced through training, knowledge sharing, and/or awareness raising events aimed at increasing the financial sector's capacity to evaluate investments in minigrids.** 897. 898. Unit of measure: binary (1/0) 899.	900. 0 901. 902. (The Project shall assess the baseline in Year 1)	903. 1 904. 905. Planned capacity building activities for year 1 and 2 are implemented. (1) 906. 907. The capacity of targeted recipients is assessed by survey towards the end of year 2. On a scale of	911. 1 912. 913. Planned capacity building activities for year 3 and 4 are implemented. (1) 914. 915. The capacity of targeted recipients is assessed by survey towards the end of the project. On a scale of 1 to 5, an average score of at least 4 is achieved.

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			1 to 5, an average score of at least 2 is achieved. 908. 909. - 1 represents a low level of capacity 910. - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)	916. 917. - 1 represents a low level of capacity 918. - 5 represents a strong capacity to understand relevant issues and apply knowledge and skills to find effective solutions. (1)
	920. *Indicator 10:* Number of government- or impact investor-supported financing mechanisms offering concessional finance for low-carbon minigrids. 921. 922. Units of measure: binary (1/0)	923. 0 924. 925. (No financing mechanism in place)	926. 1 927. 928. At least one complementary funding instrument is designed and operational. (1)	929. 1 930. 931. At least one complementary funding instrument is designed and operational. (1)
933. Outputs to achieve Outcome 3	934. Output 3.1. Innovative financing solutions for minigrid development are identified and implemented with supporting human and institutional strengthening. 935. Output 3.2. Domestic financial sector capacity-building on business and financing models for minigrids.			
937. Project Component 4	***938.*** **Digital and Knowledge Management.**			
940. Outcome 4. Digitalization and data mainstreamed, across stakeholders, into local minigrid market development. Increased knowledge, awareness and network opportunities in the minigrid market and among stakeholders, including benefitting from linkages to international good practice. 941. 942. 943.	**944. *Indicator 11:* A project digital strategy is prepared and implemented by the PMU to contribute to project implementation and local minigrid market development.** 945. 946. Units of measure: binary (1/0)	947. 0 948. 949. (Project has not started)	950. 1 951. 952. The project digital strategy is developed and being implemented (1)	953. 1 954. 955. The project digital strategy is implemented. (1) 956. 957. Recommendations for rolling out digital solutions for minigrids at national level have been shared with key national stakeholders. (1)
	959. *Indicator 12:* Number of minigrid pilots sharing data on minigrid performance with the regional project and other stakeholders following best practices and received from the AMP Regional Project. *960.* 961. Units of measure: binary (1/0) 962.	963. 0 964. 965. (Project has not started) 966.	967. 1 968. 969. The project's 'Minigrids Digital and Data Management Platform' is procured and operational, ready for data	970. 1 971. 972. 100% of the planned minigrid pilots, as identified in the project's Minigrid Pilot Plan, are collecting and sharing data with the project's digital platform (1)

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			collection from the project's mini-grid pilot(s), and for data sharing with the AMP regional project's digital platform. (1)	
974. Outputs to achieve Outcome 4	975. Output 4.1. A project digital strategy is developed and implemented, including linkages to and following guidance from the AMP Regional Project. 976. Output 4.2. Specification and implementation of Minigrids Digital Platform to track minigrid pilots and support scale-up and cost-reduction. 977. Output 4.3. Adoption and operationalization of the project's Quality Assurance and Monitoring Framework (QAMF). 978. Output 4.4. Engage with regional project by participating in Communities of Practice and capturing and sharing of lessons learnt.			
980. Project Component 5	981. **Monitoring and Evaluation (M&E)**			
983. Outcome 5. Ensuring compliance with all mandatory monitoring and reporting requirements of the GEF.	984.			
986. Outputs to achieve Outcome 5	987. Output 5.1. Inception workshop is conducted and M&E plan is implemented. 988. Output 5.2. Project Mid-Term Review is conducted. 989. Output 5.3. GEF Terminal Evaluation is conducted.			

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997. VI. MONITORING AND EVALUATION (M&E) PLAN

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122. Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP (including guidance on GEF project revisions) and UNDP Evaluation Policy **The UNDP Country Office is responsible for ensuring full compliance with all UNDP project M&E requirements including project monitoring, UNDP quality assurance requirements, quarterly risk management, and evaluation requirements.**

123. Additional mandatory GEF-specific M&E requirements will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies[53]. The M&E plan and budget included below will guide the GEF-specific M&E activities to be undertaken by this project.

124. In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed – including during the Project Inception Workshop - and will be detailed in the Inception Report.

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1000. **Minimum project monitoring and reporting requirements as required by the GEF:**

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125. Inception Workshop and Report: A project inception workshop will be held within 2 months from the First disbursement date, with the aim to:

a. Familiarize key stakeholders with the detailed project strategy and discuss any changes that may have taken place in the overall context since the project idea was initially conceptualized that may influence its strategy and implementation.
b. Discuss the roles and responsibilities of the project team, including reporting lines, stakeholder engagement strategies and conflict resolution mechanisms.
c. Review the results framework and monitoring plan.
d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP and other stakeholders in project-level M&E.
e. Update and review responsibilities for monitoring project strategies, including the risk log; SESP report, Social and Environmental Management Framework (where relevant) and other safeguard requirements; project grievance mechanisms; gender strategy; knowledge management strategy, and other relevant management strategies.
f. Review financial reporting procedures and budget monitoring and other mandatory requirements and agree on the arrangements for the annual audit.
g. Plan and schedule Project Board meetings and finalize the first-year annual work plan. Finalize the TOR of the Project Board.
h. Formally launch the Project.

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126. GEF Project Implementation Report (PIR): The annual GEF PIR covering the reporting period July (previous year) to June (current year) will be completed for each year of project implementation. UNDP will undertake quality assurance of the PIR before submission to the GEF. The PIR submitted to the GEF will be shared with the Project Board. UNDP will conduct a quality review of the PIR, and this quality review and feedback will be used to inform the preparation of the subsequent annual PIR.

127. GEF Core Indicators: The GEF Core indicators included as Annex will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to MTR and TE. Note that the Project Team is responsible for updating the indicators' status. The updated monitoring data should be shared with MTR/TE consultants prior to required evaluation missions, so these can be used for subsequent ground truthing. The methodologies to be used in data collection have been defined by the GEF and are available on the GEF website.

[53] See https://www.thegef.org/gef/policies_guidelines

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1003. Box 10: GEF-7 Core Indicators

48. As reflected in the Results Framework, the project contributes to the following GEF-7 Core Indicators:

- **Core indicator 6: Greenhouse Gas Emissions Mitigated** captures the amount of GHG emissions expected to be avoided through the GEF project's investment in renewable energy minigrids. It should be measured above a baseline value. Mitigation benefits include:
  - **Direct emissions reductions** attributable to the investments made during the project's supervised implementation period, totaled over the respective lifetime of the investments.
  - **Indirect emissions reductions** that could result from a broader adoption of the outcomes of a GEF project plus longer-term emission reductions from behavioral change in the post-project period. Broader adoption of a GEF project proceeds through several processes including sustaining, mainstreaming, replication, scaling-up and market change.
- **Context Sub-indicator 6.4: Increase in installed renewable energy capacity per technology** captures the increase in renewable energy generation or storage capacity and should be disaggregate by type of renewable energy technology (biomass, geothermal, ocean, small hydro, solar photovoltaic, solar thermal, wind power, and storage).
- **Core indicator 11: Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment** captures the total number of direct beneficiaries including the proportion of women beneficiaries. Direct beneficiaries are all individuals receiving targeted support from a given project.

128. Independent Mid-term Review (MTR): A Mid-Term Review will be conducted by (DATE] and completed no later than 36 months after CEO Endorsement. The terms of reference, the review process and the final MTR report will follow the standard UNDP templates and UNDP guidance for GEF-financed projects available on the UNDP Evaluation Resource Center (ERC). The evaluation will be 'independent, impartial and rigorous'. The evaluators that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project under review. The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the evaluation process. Additional quality assurance support is available from the BPPS/NCE-VF Directorate. The final MTR report and MTR TOR will be publicly available in English and will be posted on the UNDP ERC by 1 March 2025. A management response to MTR recommendations will be posted in the ERC within six weeks of the MTR report's completion.

129. Terminal Evaluation (TE): An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance for GEF-financed projects available on the UNDP Evaluation Resource Center. TE should be completed 3 months before the estimated operational closure date, set from the signature of the ProDoc and according to the duration of the project. Provisions should be taken to complete the TE in due time to avoid delay in project closure. Therefore, TE must start no later than 6 months to the expected date of completion of the TE (or 9 months prior to the estimated operational closure date). The evaluation will be 'independent, impartial and rigorous'. The evaluators that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated. The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the BPPS/NCE-VF Directorate. The final TE report and TE TOR will be publicly available in English and posted on the UNDP ERC by 1 February 2027. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report's completion.

130. Final Report: The project's terminal GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

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131. Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information: To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP logo on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[54] and the GEF policy on public involvement[55].

132. M&E linkages to the AMP Regional Project.

1004. Box 11: M&E linkages to the AMP Regional Project

49. **National AMP Projects** will provide on an annual basis (and to the extent feasible if requested on an ad-hoc basis) the following M&E information to the AMP regional project staff:

- Standard reporting on all indicators in the results framework for aggregation and reporting to GEF (by the regional project) on the impacts of all participating national projects for the program as a whole; and
- Reporting on any and all additional Key Performance Indicators (KPIs) adopted by the project under the common M&E framework.

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51. The **AMP Regional Project** will provide support and guidance to the AMP National Projects for conducting M&E activities as follows:

- **Ongoing project monitoring**. The AMP Regional Project PMU will:
  a. Develop a 'common monitoring and evaluation (M&E) framework' against which GHG emission reductions and broader SDG impacts and program objectives can be measured, and work closely with AMP national projects to ensure operationalization and harmonization.
  b. Provide support to National Project PMUs for updating 'key project planning instruments' at least on an annual basis as required to comply with UNDP project monitoring, quality assurance, and risk management requirements, and ensure adequate project planning and adaptive management. This may entail developing common templates for 'key project planning instruments'.
  c. Review and provide feedback on reports submitted by the national project PMUs seeking to continuously improve the quality and ease of reporting by national projects.
  d. Aggregate M&E data from all national projects, including Results Framework and all additional Key Performance Indicators (KPIs) adopted by the project under the common M&E framework, and report back to GEF at the program level.
- **Evaluations (MTR and TE)**. The AMP Regional Project PMU will:
  a. Make available to national projects standardized terms of reference for MTR and TE as well as a roster of vetted evaluation consultants.
  b. Review and provide feedback on terms of reference and draft evaluation reports shared by the project PMU to ensure project-level evaluation will be undertaken in compliance with UNDP requirements.
  c. Make themselves available for interviews and consultation in the context of national project mid-term and terminal evaluations.

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[54] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/
[55] See https://www.thegef.org/gef/policies_guidelines

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1007. **Monitoring Plan:** The project results, corresponding indicators and mid-term and end-of-project targets in the project results framework will be monitored by the Project Management Unit annually, and will be reported in the GEF PIR every year, and will be evaluated periodically during project implementation. If baseline data for some of the results indicators is not yet available, it will be collected during the first year of project implementation. Project risks, as outlined in the risk register, will be monitored quarterly.

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1010. Results Monitoring	**1011. Indicators**	**1012. 1013. Mid-term Target**	**1014. End of Project Target**	**1015. 1016. Description of indicators and targets 1017.**	**1018. Data source/Collection Methods[56] 1019.**	**1020. Frequency 1021.**	**1022. Responsible for data collection**	**1023. Means of verification**	**1024. Risks/ Assumptions**
1025. Project objective		1026. To support access to clean energy by increasing technical and financial feasibility and by promoting scaled-up commercial investment, in low-carbon minigrids in Benin, with a focus on cost-reduction levers and innovative business models.							
1027. Objective Indicators	**1028. Indicator 1: Greenhouse gas emissions mitigated** 1029. 1030. Unit of measure: metric tons of carbon dioxide equivalent (tCO2e)) 1031.	1032. 0 tCO2eq 1033.	1034. Direct: 9,056 tCO2e 1035. Indirect: 179,273 tCO2e	1036. Direct: refer to Annex 12. 1037. 1038. Indirect: refer to Annex 12. 1039. 1040. (Note that 10% of indirect GHG reductio	1041. Baseline figures (diesel, present grid, if any) determined per MG site ex-ante 1042. Energy production figures via the digital platform. 1043. Site visits for	1045. Annually 1046. Reported in the GEF PIR	1047. PMU 1048. 1049. .	1050. Minigrids Digital and Data Management Platform 1051.	1052. Assumptions 1053. For assumptions and methodology used refer to Annex 12. 1054. Minigrid pilot operators share data via the

[56] Data collection methods should outline specific tools used to collect data and additional information as necessary to support monitoring. The PIR cannot be used as a source of verification.

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
				ns are to be allocated to the AMP regional project.)	verification and cross-correlation of data ex-post. 1044. irect and indirect emissions reductions as per AMP Model, refer to Annex 12.				digital platform 1055. 1056. Risks 1057. Development of the pilot projects delayed or not implemented.
	1058. Indicator 2: Number of direct beneficiaries benefitting from energy access via minigrids, disaggregated by gender	1061. 0 people	1062. 2,152 people (of which 50% women) 1063. ------ 1064. 2,000 people (residential) 1065.	1068. Total number of direct beneficiaries from the minigrid pilots disaggregated by gender and by customer segment	1069. roject reports 1070. inigrid technical and commercial reports	1071. Annually 1072. Reported in the GE	1073. PMU 1074. 1075. .	1076. Minigrids Digital and Data Management Platform. 1077. 1078. Periodic site	1082. Assumptions 1083. For assumptions and methodology used refer to Annex 12. 1084.

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	and by customer segment (residential, social, commercial/productive use) as co-benefit of GEF investment 1059. 1060. Unit of measure: number of people		2 people (social) 1066. 20 people (commercial/PUE) 1067. 2,152 people (total)	, as per Annex 12.		FPIR		visits. 1079. 1080. Interviews with customers. 1081.	1085. Risks 1086. Minigrid coverage less than foreseen, limiting the number of direct beneficiaries. 1087. 1088. Development of the pilot projects delayed or not implemented.
	1089. Indicator 3: Increase in installed solar PV	1092. 0 MW (solar PV) 1093.	1094. Solar PV: 0.368 MW	**1096.** Installed PV capacity and battery	1097. Project reports. 1098. Minigrid	1099. Annually 1100.	1101. PMU 1102. 1103.	1104. Minigrids Digital and Data	1109. Assumptions 1110. For

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	capacity and battery storage 1090. 1091. Units of measure: MW (solar PV); MWh (battery storage)	0 MWh (battery storage)	1095. Battery storage: 0.892 MWh	storage capacity in the minigrid pilots.	technical specifications.	Reported in the GEF PIR	.	Management Platform 1105. 1106. Periodic site visits. 1107. 1108.	assumptions and methodology used refer to Annex 12. 1111. 1112. Risks 1113. Development of the pilot projects delayed or not implemented.
	1114. Indicator 4: Local residents trained in different aspects of minigrid developm	1117. Female: 10 people 1118. Male: 10 people 1119. Total: 20	1120. Female: 30 people 1121. Male: 30 people 1122. Total: 60	1123. Number of people in country trained.	1124. Records of training activities.	1125. Annually 1126. Reported	1127. PMU	1128. Minigrids Digital and Data Management Platfo	1133. Assumptions 1134. A nascent market for minigrid development is

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	ent and operation (e.g. sales, distribution, operations, management) disaggregated by gender. 1115. **1116.** Unit of measure: number of people	people	people			nthe GEF PIR		rm. 1129. 1130. Interviews and appraisals by MTR/ RE evaluators. 1131. 1132.	present. 1135. 1136. Risks 1137. Local residents do respond weakly to get involved in minigrid development.
1138. Project Outcome 1	**1139. Indicator 5: A minigrid delivery model to enable minigrid development is endorsed/ adopted by the**	1142. Multi-stakeholder national dialogue platform on minigrid delivery models is	1143. At least one minigrid delivery model is identified and endorsed by the government	1144. Binary indicator: target fulfilled (1) or not (0). 1145.	1146. Project reports, Project Board minutes	1147. Annually 1148. Reported in the	1149. PMU	1150. Official Government publications, 1151. Electricity sector public	1152. Assumptions 1153. Minigrids are acknowledged as relevant and prioritized as such by

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	national government through a consultative process involving key stakeholders (e.g. relevant ministries, local authorities, rural populations, private sector, media, etc.) 1140. 1141. Unit of measure: binary (1/0)	established and active. (0)	through the work of the multi-stakeholder platform and dialogue. (1)			GEFPIR		ations and regulation.	Government and key stakeholders. 1154. 1155. Risks 1156. Policy making processes are slow or do not converge towards a suitable MG delivery model.
	1157. Indicator 6: Number of policy	1160. 1 policy deriskin g	1163. 3 policy derisking instrume	1167. Number (0-3) of endorse	1168. Project reports, Project	1169. Annuall y	1171. PMU	1172. Official Governmen	1174. Assumptions 1175. Minigrids are

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	derisking instruments for minigrid investments - whose development has been supported by the project - are endorsed/ adopted by the national government 1158. 1159. Unit of measure: Number of policy derisking instruments.	instrument(s) adopted 1161. 1162. The specific instruments will be informed by the full DREI analysis	nt(s) adopted 1164. 1165. The specific instruments will be informed by the full DREI analysis 1166.	d/ adopted policy instruments.	Board minutes	1170. Reported in the GEFPIR		t publications, 1173. Electricity sector publications and regulation.	acknowledged as relevant and prioritized as such by Government and key stakeholders. 1176. 1177. Risks 1178. Policy-making processes are slow or do not converge towards a suitable MG

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
									delivery model.
1179. Project Outcome 2	1180. Indicator 7: Minigrid pilots implemented that demonstrate a delivery model, cost-reduction measure(s) and/or productive use of electricity 1181. 1182. Unit of measure: binary (1/0) 1183.	1184. Minigrid Pilot Plan for advancing the minigrid pilots is developed, and cleared by UNDP and the Project Board. (1) 1185. 1186. Any project tendering process, as applicable, for minigrid pilots is	1187. The Minigrid Pilot Plan has been successfully executed and the pilots are delivered, operational, and being monitored. (1)	1188. Binary indicator: target fulfilled (1) or not (0)	1189. inigrids Digital and Data Management Platform. 1190.	1191. Annually 1192. Reported in the GEF PIR	1193. PMU	1194. Individual project technical reports. 1195. Energy contracts and metering data records. 1196. Expert site visits. 1197.	1198. Assumptions 1199. The minigrid pilots can be implemented as planned. 1200. 1201. Risks 1202. Technical and social factors lead to under performance of minigrids pilots. 1203. Hardware price levels in

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
		launched. (1)							the country are higher than expected. 1204. Main grid extension undermines the MG model.
	1205. Indicator 8: Capacity of minigrid developers and/or operators is enhanced to implement innovative business models	1210. Planned capacity building activities for year 1 and 2 are implemented. (1) 1211. 1212. The capacity	1213. Planned capacity building activities for year 3 and 4 are implemented. (1) 1214. 1215. The capacity of targeted recipients	1216. Binary indicator: target fulfilled (1) or not (0) 1217.	1218. Records of capacity building activities.	1219. Annually 1220. Reported in the GEFP	1221. PMU	1222. Appraisals by PMU, MTR/ RE evaluators. 1223. Publications by minigrid developers and	1224. Assumptions 1225. Minigrid developers demonstrate sustained interest in the market. 1226. 1227.

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	and incorporate cost-reduction levers in minigrid projects. 1206. 1207. Unit of measure: binary (1/0) 1208. **1209.**	of targeted recipients is assessed by survey towards the end of year 2. On a scale of 1 to 5, an average score of at least 2 is achieved[57] (1)	is assessed by survey towards the end of the project. On a scale of 1 to 5, an average score of at least 4 is achieved (1)			IR		operators.	Risks 1228. No specific risks identified.
1229. Project Outcome 3	**1230. Indicator 9: Capacity of financial institutions is enhanced**	1233. Planned capacity building activities for year 1 and 2	1236. Planned capacity building activities for year 3 and 4 are implemen	1239. Binary indicator: target fulfilled (1) or not (0)	1240. Records of capacity building activities. 1241.	1242. Annually 1243. Reported	1244. PMU	1245. Appraisals by PMU, MTR/ RE evaluators.	1247. Assumptions 1248. Financial institutions demonstrate

[57] As follows: 1 = lowest level of capacity; 5 = strongest capacity to understand relevant issues and apply knowledge and skills to find effective solutions.

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	through training, knowledge sharing, and/or awareness raising events aimed at increasing the financial sector's capacity to evaluate investments in minigrids. 1231. 1232. Unit of measure: binary (1/0)	are implemented. (1) 1234. 1235. The capacity of targeted recipients is assessed by survey towards the end of year 2. On a scale of 1 to 5, an average score of at least 2 is achieved.	ted. (1) 1237. 1238. The capacity of targeted recipients is assessed by survey towards the end of the project. On a scale of 1 to 5, an average score of at least 4 is achieved.			in the GEFPIR		1246. Publications by financial sector entities.	sustained interest in the market. 1249. 1250. Risks 1251. The financial sector in the country is very small and appetite among investors is traditionally low.
	1252. Indicator	1255. At least one	1256. At least	1257. Binary	1258. Publication	1259. Annua	1261. PMU, IP	1262. Official	1263. Assumptions

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	10: Number of government- or impact investor-supported financing mechanisms offering concessional finance for low-carbon minigrids. 1253. **1254.** Units of measure: binary (1/0)	complementary funding instrument is designed and operational. (1)	one complementary funding instrument is designed and operational. (1)	indicator: target fulfilled (1) or not (0)	s by Government, development partners and local banks	lly 1260. Reported in the GEFPIR		publications by Government entities and development partners.	1264. Government and multilateral financiers demonstrate sustained interest in the market. 1265. 1266. Risks 1267. Coordination between cooperation agencies, Government and multilateral financie

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
									rs proves more sluggish than expected.
1268. 1269. Project Outcome 4	1270. Indicator 11: A project digital strategy is prepared and implemented by the PMU to contribute to project implementation and local minigrid market development. 1271. 1272. Units of measure:	1273. The project digital strategy is developed and being implemented (1)	1274. The project digital strategy is implemented. (1) 1275. 1276. Recommendations for rolling out digital solutions for minigrids at national level have been shared with key	1277. Binary indicator: target fulfilled (1) or not (0)	1278. inigrids Digital and Data Management Platform 1279.	1280. Annually 1281. Reported in the GEFPIR	1282. PMU	1283. Project reports. 1284. Physical implementation of Digital Platform. 1285.	1286. Assumptions 1287. Government, stakeholders and private companies accept sharing information. 1288. 1289. Risks 1290. (1) Operation and management of the system

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	binary (1/0)		national stakeholders. (1)						may prove a burden to Government. 1291. (2) Stakeholders are reluctant to set up and share efficient data exchange protocols.
	1292. Indicator 12: Number of minigrid pilots sharing data on minigrid performance with	1295. The project's 'Minigrids Digital and Data Management Platform' is	1296. 100% of the planned minigrid pilots, as identified in the project's Minigrid Pilot Plan,	1297. Binary indicator: target fulfilled (1) or not (0)	1298. inigrids Digital and Data Management Platform	1299. Quarterly 1300. The AMP reg	1303. PMU 1304. 1305. AMP Regional Project M	1306. Examination of Digital Platform and data inputs from minig	1307. Assumptions 1308. All planned minigrids are implemented. 1309. Government, stakehol

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	the regional project and other stakeholders following best practices and received from the AMP Regional Project. 1293. **1294.** Units of measure: binary (1/0)	procured and operational, ready for data collection from the project's mini-grid pilot(s), and for data sharing with the AMP regional project's digital platform. (1)	are collecting and sharing data with the project's digital platform (1)			ional project will aggregated data form all pro	anager will confirmability to collect/ aggregate data form the project.	rid operators	ders and private companies accept sharing information 1310. 1311. Risks 1312. Technical, managerial, and financial issues may come up, with Government and AMP lacking funding and/or specific skills to address

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
						jects 1301. Reported in DOprogress tab of the GEF PIR			these.

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
						1302.			
1313. 1314. 1315. 1316. 1317. 1318. 1319. 1320. 1321. 1322. 1323. 1324. Gender action plan	1325. Indicator 13: Gender balance in implementation activities 1326.	1327. No more than 70% either gender, cumulative	1328. No more than 60% either gender, cumulative	1329. Gender ratios, cumulative, of all program event attendance, training completion, apprenticeships, job creation, & proposal acceptance.	1330. Record-keeping by program managers	1331. Quarterly	1332. PMU	1333. Verification of written records, limited phone-based audits	1334. Risks: 1335. One heavily attended workshop could skew the gender balance of smaller activities (like proposal support and apprenticeships) 1336.
	1337. Indicator 14: Gender	1338. Gender mainstreamed	1339. Gender mainstreamed in	1340. Binary indicator (y/n)	1341. Review of deliverables	1342. Annually	1343. PMU	1344. Review of deliverables	1345. Risks: 1346. Cursory

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
	mainstreamed in program analyses and design activities	in 100% of eligible activities in progress	100% of eligible activities completed	for whether gender was mainstreamed in each of the following when it was undertaken: National Dialog Discussions; DREI analysis; research and design activities; MG Pilot Plan; Financing analysis and design;	submitted				mentions of key words (e.g., "women," "gender") could be included in program deliverables to try and meet this target

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1010. Results Monitoring	1011. Indicators	1012. 1013. Mid-term Target	1014. End of Project Target	1015. 1016. Description of indicators and targets 1017.	1018. Data source/Collection Methods 1019.	1020. Frequency 1021.	1022. Responsible for data collection	1023. Means of verification	1024. Risks/ Assumptions
				Digital Strategy; Insight Brief					
1347. Stakeholder engagement plan	1348. Indicator 15: Engagement with stakeholders	1349. Stakeholders are annually engaged as planned representing all relevant sectors	1350. takeholders are annually engaged as planned representing all relevant sectors	1351. Binary indicator: target fulfilled (1) or not (0)	1352. ecord-keeping by program managers	1353. Quarterly	1354. PMU	1355. Verification of written records, limited phone-based audits	1356. Risks: 1357. Stakeholder representation may be biased towards public sector and government bodies, leaving end-users and vulnerable groups unheard.

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1361. **Monitoring and Evaluation Budget for project execution** *1362.*		
1363. GEF M&E requirements to be undertaken by Project Management Unit (PMU)	**1364. Indicative costs (US$)**	**1365. Time frame**
1366. **Inception Workshop and Report**	1367. US$ 11,000	1368. Inception Workshop, within 2 months of First Disbursement
1369. **M&E required to report on progress made in reaching GEF core indicators and project results included in the project results framework**	1370. N/A	1371. Annually, prior to MTR, and prior to TE
1372. **Preparation of the annual GEF Project Implementation Report (PIR)**	*1373.* N/A	1374. Annually, between June-August
1375. Monitoring all risks (Atlas risk log)	1376. N/A	1377. On-going
1378. **Monitoring of stakeholder engagement plan**	1379. N/A	1380. On-going
1381. Monitoring of ESMF and specific management plans (Output 1.5)	1382. N/A	1383. Annually before PIR and as requested
1384. Monitoring of Gender Action plan (Output 1.5)	1385. N/A	1386. Annually before PIR and as requested
1387. Supervision missions	*1388.* N/A	1389. As needed
1390. Learning missions	*1391.* N/A	1392. As needed
1393. Independent Mid-term Review (MTR)	*1394.* US$ 21,589	1395. 1 March 2025
1396. Independent Terminal Evaluation (TE)	1397. US$ 32,000	1398. 1 February 2027
1399. **TOTAL indicative COST**	**1400. US$ 64,589**	1401.

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1406. VII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

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1408. **Section 1: General roles and responsibilities in the projects' governance mechanism**

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133. Implementing Partner: The Implementing Partner for this project is the Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy). The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

134. The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Overseeing the management of project risks as included in this project document and new risks that may emerge during project implementation.
- Procurement of goods and services, including human resources.
- Financial management, including overseeing financial expenditures against project budgets.
- Approving and signing the multiyear workplan.
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

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135. Project stakeholders and target groups: Describe how project target groups will be engaged in decision making for the project.

136. Government stakeholders, notably the Ministry of Energy (ME), its Directorate-General Energy Resources (ME/DGRE), the Ministry of Living Conditions and Sustainable Development (MCVDD) and the Benin National Association of Communes (ANCB) are invited to take seat in the Project Steering Committee. Policy, technical and regulatory, and financial and tax policy aspects of minigrids and its broader context of a transition towards RE sources, will be reviewed in dedicated working groups that derive from the national dialogue. The Project will establish a technical working unit for preparing and implementing the minigrid pilots which will be composed of: Project Technical Advisor (TA), Project Engineer (PE), national Project MG/DREI expert (ME), and staff designated staff by the Implementing Partner.

137. The Project will make an ongoing effort to engage with representatives from beneficiaries, community development and gender experts, representatives of productive energy users, and others, to ensure inclusiveness and pro-gender action when required. Periodic consultations and events are scheduled in preparation of minigrid designs and calls for proposals. Specific outputs are programmed to strengthen capacities among national project developers and end-users. Community development is included as a key activity to maximize relevance and appropriateness of developed minigrid solutions and ensure long-term (social) sustainability.

138. UNDP: UNDP is accountable to the GEF for the implementation of this project. This includes overseeing project execution undertaken by the Implementing Partner to ensure that the project is being carried out in accordance with UNDP and GEF policies and procedures and the standards and provisions outlined in the Delegation of Authority (DOA) letter for this project. **The UNDP GEF Executive Coordinator, in consultation with UNDP Bureaus and the Implementing Partner, retains the right to revoke the project DOA, suspend or cancel this GEF project.** UNDP is responsible for the Project Assurance function in the project governance structure and presents to the Project Board and attends Project Board meetings as a non-voting member.

139. A firewall will be maintained between the delivery of project oversight and quality assurance performed by UNDP and charged to the GEF Fee and any support to project execution performed by UNDP (as requested by

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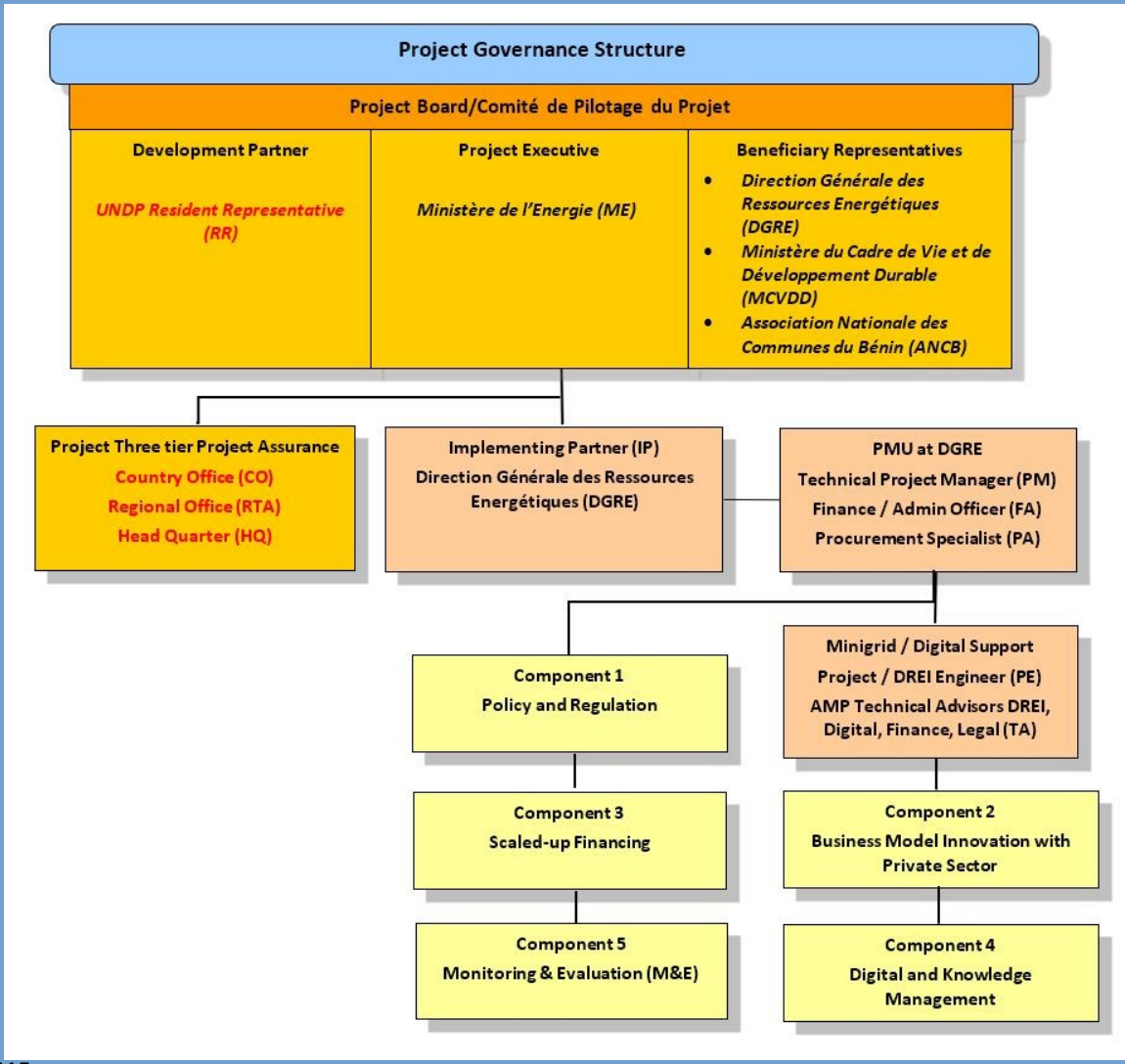
and agreed to by both the Implementing Partner and GEF) and may be charged to the GEF project management costs (only if approved by GEF). The segregation of functions and firewall provisions for UNDP in this case is described in the next section.

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1412. **Section 2: Project governance structure.**

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140. The UNDP Resident Representative (RR) assumes full responsibility and accountability for oversight and quality assurance of this Project and ensures its timely implementation in compliance with the GEF-specific requirements and UNDP's Programme and Operations Policies and Procedures (POPP), its Financial Regulations and Rules and Internal Control Framework. A representative of the UNDP Country Office will assume the assurance role and will present assurance findings to the Project Board, and therefore attends Project Board meetings as a non-voting member.

1416. **Section 3: Segregation of duties and firewalls vis-à-vis UNDP representation on the project board:**

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141. As noted in the Minimum Fiduciary Standards for GEF Partner Agencies, in cases where a GEF Partner Agency (i.e. UNDP) carries out both implementation oversight and execution of a project, the GEF Partner Agency (i.e. UNDP) must separate its project implementation oversight and execution duties, and describe in the relevant project document a: 1) Satisfactory institutional arrangement for the separation of implementation oversight and executing functions in different departments of the GEF Partner Agency; and 2) Clear lines of responsibility, reporting and accountability within the GEF Partner Agency between the project implementation oversight and execution functions.

142. In this case, UNDP is only performing an implementation oversight role in the project vis-à-vis our role in the project board and in the project assurance function and therefore a full separation of project implementation oversight and execution duties has been assured.

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1418. Section 4: Roles and Responsiblities of the Project Organization Structure:

143. **Project Board:** All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on monitoring and evaluation, and implementation issues to ensure quality delivery of results. The Project Board (also called the Project Steering Committee) is the most senior, dedicated oversight body for a project.

144. The two main (mandatory) roles of the project board are as follows:

1) **High-level oversight of the execution of the project by the Implementing Partner** (as explained in the "Provide Oversight" section of the POPP). This is the primary function of the project board and includes annual (and as-needed) assessments of any major risks to the project, and decisions/agreements on any management actions or remedial measures to address them effectively. The Project Board reviews evidence of project performance based on monitoring, evaluation and reporting, including progress reports, evaluations, risk logs and the combined delivery report. The Project Board is responsible for taking corrective action as needed to ensure the project achieves the desired results.
2) **Approval of strategic project execution decisions of the Implementing Partner** with a view to assess and manage risks, monitor and ensure the overall achievement of projected results and impacts and ensure long term sustainability of project execution decisions of the Implementing Partner (as explained in the "Manage Change" section of the POPP).

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145. **Requirements to serve on the Project Board**:

- ✓ Agree to the Terms of Reference of the Board and the rules on protocols, quorum and minuting.
- ✓ Meet annually; at least once.
- ✓ Disclose any conflict of interest in performing the functions of a Project Board member and take all measures to avoid any real or perceived conflicts of interest. This disclosure must be documented and kept on record by UNDP.
- ✓ Discharge the functions of the Project Board in accordance with UNDP policies and procedures.
- ✓ Ensure highest levels of transparency and ensure Project Board meeting minutes are recorded and shared with project stakeholders.

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146. **Responsibilities of the Project Board**:

- ✓ Consensus decision making:
  - The project board provides overall isocharge guidance and direction to the project, ensuring it remains within any specified constraints, and providing overall oversight of the project implementation.
  - Review project performance based on monitoring, evaluation and reporting, including progress reports, risk logs and the combined delivery report;
  - The project board is responsible for making management decisions by consensus.

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- In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.
- In case consensus cannot be reached within the Board, the UNDP representative on the board will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.

✓ Oversee project execution:

- Agree on project manager's tolerances as required, within the parameters outlined in the project document, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded.
- Appraise annual work plans prepared by the Implementing Partner for the Project; review combined delivery reports prior to certification by the implementing partner.
- Address any high-level project issues as raised by the project manager and project assurance;
- Advise on major and minor amendments to the project within the parameters set by UNDP and the donor and refer such proposed major and minor amendments to the UNDP BPPS Nature, Climate and Energy Executive Coordinator (and the GEF, as required by GEF policies);
- Provide high-level direction and recommendations to the project management unit to ensure that the agreed deliverables are produced satisfactorily and according to plans.
- Track and monitor co-financed activities and realization of co-financing amounts of this project.
- Approve the Inception Report, GEF annual project implementation reports, mid-term review and terminal evaluation reports.
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project.

✓ Risk Management:

- Provide guidance on evolving or materialized project risks and agree on possible mitigation and management actions to address specific risks.
- Review and update the project risk register and associated management plans based on the information prepared by the Implementing Partner. This includes risks related that can be directly managed by this project, as well as contextual risks that may affect project delivery or continued UNDP compliance and reputation but are outside of the control of the project. For example, social and environmental risks associated with co-financed activities or activities taking place in the project's area of influence that have implications for the project.
- Address project-level grievances.

✓ Coordination:

- Ensure coordination between various donor and government-funded projects and programmes.
- Ensure coordination with various government agencies and their participation in project activities.

147. **Composition of the Project Board**: The composition of the Project Board must include individuals assigned to the following three roles:

1. **Project Executive:** This is an individual who represents ownership of the project and chairs (or co-chairs) the Project Board. The Executive usually is the senior national counterpart for nationally implemented projects (typically from the same entity as the Implementing Partner), and it must be UNDP for projects that are direct implementation (DIM). In exceptional cases, two individuals from different entities can co-share this role and/or co-chair the Project Board. If the project executive co-chairs the project board with representatives of another category, it typically does so with a development partner representative. The Project Executive is: Ministry of Energy (ME), **Mr. Assan Todéman Fliso.**
2. **Beneficiary Representative(s):** Individuals or groups representing the interests of those groups of stakeholders who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. Often representatives from civil society, industry associations, or other government entities benefiting from the project can fulfil

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this role. There can be multiple beneficiary representatives in a Project Board. The Beneficiary representative (s) is/are: Directorate-General Energy Resources (ME/DGRE)

3. **Development Partner(s):** Individuals or groups representing the interests of the parties concerned that provide funding, strategic guidance and/or technical expertise to the project. The Development Partner(s) is/are: UNDP Resident Representative (RR)

148. **Project Assurance:** Project assurance is the responsibility of each project board member; however, UNDP has a distinct assurance role for all UNDP projects in carrying out objective and independent project oversight and monitoring functions. UNDP performs quality assurance and supports the Project Board (and Project Management Unit) by carrying out objective and independent project oversight and monitoring functions, including compliance with the risk management and social and environmental standards of UNDP. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. Project assurance is totally independent of project execution.

149. A designated representative of UNDP playing the project assurance role is expected to attend all board meetings and support board processes as a non-voting representative. It should be noted that while in certain cases UNDP's project assurance role across the project may encompass activities happening at several levels (e.g. global, regional), at least one UNDP representative playing that function must, as part of their duties, specifically attend board meeting and provide board members with the required documentation required to perform their duties. The UNDP representative playing the main project assurance function is/are: Mensah Azangnandji, programme officer

150. **Project Management – Execution of the Project:** The Project Manager (PM) (also called project coordinator) is the senior most representative of the Project Management Unit (PMU) and is responsible for the overall day-to-day management of the project on behalf of the Implementing Partner, including the mobilization of all project inputs, supervision over project staff, responsible parties, consultants and sub-contractors. The project manager typically presents key deliverables and documents to the board for their review and approval, including progress reports, annual work plans, adjustments to tolerance levels and risk registers.

The roles and responsibilities of the PMU members are detailed in the Annex 7. The PMU will be hosted by the DGRE. A designated representative of the PMU is expected to attend all board meetings and support board processes as a non-voting representative.

151. The primary PMU representative attending board meetings is: Project Manager (PM) (to be recruited by the Project)

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1428. VIII. Financial Planning and Management

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152. The total cost of the project is USD 34,526,147. This is financed through a GEF grant of USD 1,326,147 administered by UNDP; USD 200,000 UNDP TRAC funds in cash co-financing grant; to be administered by UNDP; and additional support ("co-finance") of USD 33,00,000. UNDP, as the GEF Implementing Agency, is responsible for the oversight of the GEF resources and the cash co-financing transferred to UNDP bank account only.

153. **Co-financing**: The actual achievement of project co-financing amounts will be monitored by the UNDP Country Office and the PMU on an annual basis in the GEF PIF and will be reported to the GEF during the MTR and TE process as follows:

1430. Co-financing source	**1431. Co-financing type**	**1432. Name of Co-financier**	**1433. Investment Mobilized**	**1434. Co-financing amount**
1435. Recipient Government	*1436.* Public investment	*1437.* Ministry of Energy (ME)	1438. Investment Mobilized	*1439.* US$ 21,000,000
1440. Donor Agency [58]	*1441.* Loan	*1442.* African Development Bank (AfDB) [59]	1443. Investment mobilized	*1444.* US$ 12,000,000
1445. Donor Agency	*1446.* Grant	*1447.* UNDP TRAC	1448. Investment mobilized	*1449.* US$ 200,000
		1450. ***Total***		1451. **US$ 33,200,000**

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154. **Budget Revision and Tolerance**: As per UNDP POPP, the project board may agree with the project manager on a tolerance level for each detailed plan under the overall multi-year workplan. The agreed tolerance should be written in the project document or approved project board meeting minutes. It should normally not exceed 10 percent of the agreed annual budget at the activity level, but within the overall approved multi-year workplan at the activity level. Within the agreed tolerances, the project manager can operate without intervention from the project board. Restrictions apply as follows:

[58] This concerns a USD 12 million grant from the African Development Bank through the African Development Fund (ADF) and Sustainable Energy Fund for Africa (SEFA) to fund the Energy Transition and institutional support (ETISP) which runs until 2025,

[59] This concerns a USD 12 million grant from the African Development Bank through the African Development Fund (ADF) and Sustainable Energy Fund for Africa (SEFA) to fund the Energy Transition and institutional support (ETISP) which runs until 2025,

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155. Should the following deviations occur, the Project Manager/IP through UNDP Country Office will seek the approval of the BPPS/NCE-VF team to ensure accurate reporting to the GEF. It is **strongly encouraged** to maintain the expenditures within the approved budget at the budgetary account and at the component level:

a) Budget reallocations must prove that the suggested changes in the budget will not lead to material changes in the results to be achieved by the project. A strong justification is required and will be approved on an exceptional basis. Budget re-allocations among the components (including PMC) of the approved Total Budget and Work Plans (TBWP) that represent a value greater than 10% of the total GEF grant.
b) Introduction of new outputs/activities (i.e. budget items) that were not part of the agreed project document and TBWP that represent a value greater than 5% of the total GEF grant. The new budget items must be eligible as per the GEF and UNDP policies.
c) Project management cost (PMC): budget under PMC component is capped and cannot be increased.

156. UNDP is not in a position to increase the total budget above the amount approved by the donor, therefore any over-expenditure would have to be absorbed from non-GEF resources by the Implementing Partner (GEF Executing Entity)

157. **Project extensions:** The UNDP-BPPS-NCE team Executive Coordinator must approve all requests for extension of the Project Completion Date and for other milestone extensions with hard deadlines. All extensions impose additional time and cost burdens at all levels and the GEF project budget cannot be increased beyond its originally approved amount. A single extension may be granted on an exceptional basis and subject to the conditions and maximum durations set out in the UNDP POPP. The project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs shall be covered by non-GEF resources; the additional UNDP oversight costs during the extension period must be covered by non-GEF resources, in accordance with UNDP's policy as set out in UNDP POPP.

158. For any extension request, UNDP CO and IP will consult and jointly present a clear plan indicating how and from which specific sources the additional oversight costs that will be incurred by UNDP will be covered during the extended period. The BPPS-NCE Executive Coordinator will consult the Regional Bureaux (RBX) and may reject the extension request if no (external co-financing by the IP or internal UNDP CO resources) can be identified.

159. All extension requests, along with all supporting documentation, shall be submitted by the IP to the UNDP CO in line with the requirements and within the deadlines set out in the UNDP SOPs and policies in UNDP POPP..

160. **Audit:** The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop. If the Implementing Partner is an UN Agency, the project will be audited according to that Agencies applicable audit policies. **Transfer or disposal of assets**: In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project, however, **must be done before the operational closure date**. In all cases of transfer, a transfer document must be prepared and kept on file . The transfer should be done before Project Management Unit complete their assignments.

161. **Completion Date:** The project completion date is the date of Project Document Signature plus project duration. This date can only be extended through a formal extension request. Prior to completion date, all UNDP-financed inputs must be provided and related activities for the Project completed. No activities, except for the final

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clearance of the Terminal Evaluation Report and the corresponding management response and the end-of-project review Project Board Meeting should take place after the Completion Date.

162. **Project Closure:** Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

- **Operational Closure: Operational closure must happen within 9 months from project completion date**. Prior to operational closure, the Terminal Evaluation must have been submitted and the corresponding TE management response and the end-of-project review Project Board meeting must have been completed. The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. Before Operational Closure, the project must have completed the transfer or disposal of any equipment that is still the property of UNDP.
- **Financial Closure: Financial closure must happen within 6 months of operational closure or after the date of cancellation.** The project will be financially closed when the following conditions have been met: a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

  Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to BPPS/NCE for confirmation before the project will be financially closed in Quantum by the UNDP Country Office.

163. **Cancellation and Suspension:** All projects considering going through cancellation or suspension must follow UNDP and GEF requirements. Guidance can be found in the UNDP POPP (SOPs for management actions of Vertical Fund projects escalated to the Executive Coordinator and Guidance for GEF project revisions).

164. **Refund to GEF:** Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/NCE team Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF. Unspent project balance is not permitted to be transferred to any other projects.

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1463. IX. TOTAL BUDGET AND WORK PLAN

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1465.	**Total Budget and Work Plan**		
1466. Quantum Business Unit	*1467.* UNDP-Benin		
1468. Quantum Project ID:	1469. 00131156	1470. Quantum Project Title:	1471. PIMS 6658 Africa MiniGrids-UNDP-BEN-00131156
1472. Quantum Award ID:	1473. 00142903 1474.	1475. Quantum Award Title:	1476. 1154157 BEN_PIMS_6658_AMP
1477. UNDP-GEF PIMS No.	1478. BEN10		
1479. Implementing Partner	*1480.* Directorate-General of Energy Resources (DGRE)		

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1485. Atlas Activity (GEF Component)	1486. Atlas Implementing Ag	1488. Atlas	1489. Donor Name 1490.	1491. Atla	1492. ATLAS Budget Account Description	1493. Amo	1494. Amo	1495. Am	1496. Am	1497. Total (USD)	1498. See Budget N
1499. COMPONENT 1 1500. Policy and Regulation	1501. IP	1502. 62	1503. GEF Trustee	1504. 712	1505. International Consultants	1506. 42,0	1507. 10,0	1508. 10,0	1509. 0	1510. 62,000	1511. 1
				1512. 713	1513. Local Consultants	1514. 8,00	1515. 13,0	1516. 5,00	1517. 0	1518. 27,000	1519. 2
				1520. 716	1521. Travel	1522. 3,00	1523. 6,00	1524. 5,00	1525. 3,00	1526. 17,000	1527. 3
				1528. 718	1529. Contractual Services – Indiv ImpPtnr	1530. 2,00	1531. 30,0	1532. 20,0	1533. 20,0	1534. 72,000	1535. 4
				1536. 721	1537. Contractual Services – Companies	1538. 58,0	1539. 30,0	1540. 15,0	1541. 0	1542. 104,000	1543. 5
				1544. 742	1545. Audio Visual & Print Prod Cost	1546. 2,00	1547. 2,00	1548. 1,00	1549. 0	1550. 5,000	1551. 6
				1552.	**1553. sub-total C1 (GEF)**	**1554. 117**	**1555. 91,0**	**1556. 56,0**	**1557. 23,0**	**1558. 287,000**	**1559.**
	1560. UNDP	1561. TRA	1562. UNDP	1563. 721	1564. Contractual Services – Companies	1565. 50,0	1566. 60,0	1567. 40,0	1568. 0	1569. 150,000	1570. 7
				1571.	**1572. Sub-total C1 (UNDP)**	**1573. 50,0**	**1574. 60,0**	**1575. 40,0**	**1576. 0**	**1577. 150,000**	1578.

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	1579.				**1580. TOTAL C1**	**1581. 167**	**1582. 151**	**1583. 86.0**	**1584. 23.0**	**1585. 437.000**	1586.
1587. COMPONENT 2 1588. Business Model Innovation with Private Sector	**1589. IP**	**1590. 62**	**1591. GEF Trustee**	1592. 712	1593. International Consultants	1594. 6,00	1595. 6,00	1596. 4,00	1597. 0	1598. 16,000	1599. 8
				1600. 713	1601. Local Consultants	1602. 0	1603. 10,0	1604. 10,0	1605. 8,00	1606. 28,000	1607. 9
				1608. 716	1609. Travel	1610. 2,00	1611. 5,00	1612. 5,00	1613. 4,00	1614. 16,000	1615. 10
				1616. 718	1617. Contractual Services – Indiv.ImpPtnr	1618. 18,0	1619. 25,0	1620. 25,0	1621. 15,0	1622. 84,000	1623. 11
				1624. 721	1625. Contractual Services – Companies	1626. 10,0	1627. 30,0	1628. 11,0	1629. 0	1630. 51,000	1631. 12
				1632. 723	1633. Materials & Goods	1634. 0	1635. 86,0	1636. 205	1637. 70,0	1638. 371,000	1639. 13
				1640. 742	1641. Audio Visual & Print Prod Cost	1642. 0	1643. 1,00	1644. 1,00	1645. 0	1646. 2,000	1647. 14
				1648. 745	1649. Miscellaneous	1650. 0	1651. 1,00	1652. 1,00	1653. 0	1654. 2,000	1655. 15
				1656.	1657. **Total C2**	**1658. 37,0**	**1659. 174**	**1660. 262**	**1661. 87,0**	**1662. 570.000**	1663.
1664. COMPONENT 3 1665. Scaled-up Financing	**1666. IP**	**1667. 62**	**1668. GEF Trustee**	1669. 712	1670. International Consultants	1671. 0	1672. 5,00	1673. 5,00	1674. 0	1675. 10,000	1676. 16
				1677. 713	1678. Local Consultants	1679. 0	1680. 8,00	1681. 8,00	1682. 0	1683. 18,000	1684. 17
				1685. 716	1686. Travel	1687. 0	1688. 3,00	1689. 2,00	1690. 0	1691. 5,000	1692. 18
				1693. 718	1694. Contractual Services – Indiv.ImpPtnr	1695. 12,0	1696. 10,0	1697. 10,0	1698. 0	1699. 32,000	1700. 19
				1701. 721	1702. Contractual Services – Companies	1703. 0	1704. 10,0	1705. 18,0	1706. 0	1707. 28,000	1708. 20
				1709. 742	1710. Audio Visual & Print Prod Cost	1711. 0	1712. 1,00	1713. 0	1714. 0	1715. 1,000	1716. 21
				1717. 745	1718. Miscellaneous	1719. 0	1720. 1,00	1721. 0	1722. 0	1723. 1,000	1724. 22
				1725.	1726. **Total C3**	**1727. 12,0**	**1728. 38,0**	**1729. 44,0**	**1730. 0**	**1731. 85,000**	1732.
1733. COMPONENT 4 1734. Digital and Knowledge Management	**1735. IP**	**1736. 62**	**1737. GEF Trustee**	1738. 712	1739. International Consultants	1740. 6,00	1741. 5,00	1742. 0	1743. 0	1744. 11,000	1745. 23
				1746. 716	1747. Travel	1748. 2,00	1749. 2,00	1750. 2,00	1751. 0	1752. 6,000	1753. 24
				1754. 718	1755. Contractual Services – Indiv.ImpPtnr	1756. 23,0	1757. 15,0	1758. 5,00	1759. 5,00	1760. 48,000	1761. 25
				1762. 721	1763. Contractual Services – Companies	1764. 15,0	1765. 10,0	1766. 0	1767. 0	1768. 25,000	1769. 26
				1770. 728	1771. Information Technology Equipmt	1772. 37,0	1773. 45,0	1774. 0	1775. 0	1776. 82,000	1777. 27
				1778. 742	1779. Audio Visual & Print Prod Cost	1780. 2,00	1781. 1,00	1782. 6,00	1783. 6,00	1784. 15,000	1785. 28
				1786. 745	1787. Miscellaneous	1788. 0	1789. 1,00	1790. 1,00	1791. 0	1792. 2,000	1793. 29
				1794.	1795. **Total C4**	**1796. 85,0**	**1797. 78,0**	**1798. 14,0**	**1799. 11,0**	**1800. 189.000**	1801.
1802. COMPONENT 5 1803. Monitoring and Evaluati	**1804. IP**	**1805. 62**	**1806. GEF Trustee**	1807. 712	1808. International Consultants	1809. 8,00	1810. 15,5	1811. 0	1812. 26,0	1813. 50,589	1814. 30
				1815. 713	1816. Local Consultants	1817. 0	1818. 2,00	1819. 0	1820. 2,00	1821. 4,000	1822. 31
				1823. 716	1824. Travel	1825. 2,00	1826. 4,00	1827. 0	1828. 4,00	1829. 10,000	1830. 32
				1831.	1832. **Total C5**	**1833. 11,0**	**1834. 21,5**	**1835. 0**	**1836. 32,0**	**1837. 64,589**	1838.

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1839. PROJECT MANAGEMENT	1840. IP	1841. 62	1842. GEF Trustee	1843. 716	1844. Travel	1845. 500	1846. 1,00	1847. 500	1848. 0	1849. 2,000	1850. 33
				1851. 718	1852. Contractual Services – Indiv ImpPtnr	1853. 23,0	1854. 25,0	1855. 25,0	1856. 20,0	1857. 93,000	1858. 34
				1859. 722	1860. Equipment and Furniture	1861. 1,55	1862. 0	1863. 0	1864. 0	1865. 1,558	1866. 35
				1867. 728	1868. Information Technology Equipmt	1869. 2,00	1870. 0	1871. 0	1872. 0	1873. 2,000	1874. 36
				1875. 741	1876. Professional Services	1877. 5,00	1878. 6,00	1879. 5,00	1880. 6,00	1881. 22,000	1882. 37
				1883.	1884. **sub-total PM (GEF)**	**1885. 32,0**	**1886. 32,0**	**1887. 30,5**	**1888. 26,0**	**1889. 120,558**	1890.
1891.	1892. UNDP	1893. TRA	1894. UNDP	1895. 718	1896. Contractual Services – Indiv ImpPtnr	1897. 10,0	1898. 10,0	1899. 10,0	1900. 5,00	1901. 35,000	1902. 38
1903.	1904.	1905.	1906.	1907. 757	1908. Training and Learning Costs	1909. 5,00	1910. 5,00	1911. 5,00	1912. 0	1913. 15,000	1914. 39
1915.	1916.	1917.	1918.	1919.	1920. **sub-total PM (UNDP)**	**1921. 15,0**	**1922. 15,0**	**1923. 15,0**	**1924. 5,00**	**1925. 50,000**	1926.
1927.	1928.	1929.	1930.	1931.	1932. **TOTAL (PM)**	**1933. 47,0**	**1934. 47,0**	**1935. 45,5**	**1936. 31,0**	**1937. 170,558**	1938.
1939.	1940.	1941.	1942.	1943. **PROJECT TOTAL**		**1944. 358**	**1945. 511**	**1946. 461**	**1947. 184**	**1948. 1,526,147**	1949.

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1954. **Summary of Funds:**	1955. 1961.	1956. 1962.	1957. 1963.	1958. 1964.	1959. 1965.
1960.	**1967. Amount**	**1968. Amount**	**1969. Amount**	**1970. Amount**	**1971. Total**
1966.	**1972. Year 1 (USD)**	**1973. Year 2 (USD)**	**1974. Year 3 (USD)**	**1975. Year 4 (USD)**	**1976. (USD)**

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1977. GEF grant administered by UNDP	1978. 294.059	1979. 436,588	1980. 406,500	1981. 189,000	1982. 1,326,147
1983. UNDP TRAC grant administered by UNDP	1984. 65,000	1985. 75,000	1986. 55,000	1987. 5,000	1988. 200,000
1989. Ministry of Energy (Public Investment)	1990. 4,000,000	1991. 5,000,000	1992. 8,000,000	1993. 4,000,000	1994. 21,000,000
1995. African Development Bank (Loan)	1996. 0	1997. 2,000,000	1998. 4,000,000	1999. 6,000,000	2000. 12,000,000
2001. TOTAL	**2002. 4,359,059**	**2003. 7,511,588**	**2004. 12,461,500**	**2005. 10,194,000**	**2006. 34,526,147**

2007. 2008. 2009. 2010. 2011. 2012.

2013.

2014.

2015. Table 5: Budget Notes

2016. Budget Note No.	**2017. Description**
2018.	**Component 1 (GEF Budget US$ 287,000; UNDP TRAC: US$ 150,000; Total US$ 437,000)**

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2016. Budget Note No.		2017. Description
2019.1		2020. 62k$ (1.1: 10k$ - 1.2: 42k$ - 1.3: 5k$ -1.4: 5k$) - One or more contracts with international consultant (shortlisted by AMP) for (i) technical backstopping on policy and regulation; (ii) implementation DREI framework; and (iii) advice for technical standards process.
2021.2		2022. 13k$ (1.5) – One national Social and Environmental Safeguards Expert for: (i) periodic supervision of ESMF implementation; (ii) periodic SESP rescreening; and (iii) systematization of lessons learnt and recommendations for enhancement. 2023. 14k$ (1.5) – One national gender expert for: (i) periodic supervision of Gender Action Plan implementation; and (ii) identification of issues and recommendations for enhancement.
2024.3		2025. 17k$ - Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (land travel, fuel, DSA).
2026.4		2027. 54k$ (1.1: 47k$ - 1.3: 5k$ - 1.5: 2k$) - One minigrid policy expert (P3-level for 4-yr period) to assume the role of Project Manager (PM) with responsibilities including: (i) lead consultant to the Executive (ME); (ii) lead consultant on topics related to regulation, institutional capacity, finance; (iii) engagement with GOB stakeholders, market actors and CSOs; (iv) drafting of Terms of Reference for consultancies and procurement of services; (v) quality assurance and overall supervision of contracted activities; (vi) engagement with AMP Regional Project partners for peer review of proposals, analysis of project approaches, and participation in AMP events in Benin and abroad; and (vii) compilation of proposals and presentations to stakeholders. 2028. 18k$ (1.2: 10k$ - 1.4: 8k$) - One national expert to act as Project Engineer (PE) for: (i) act as the national DREI consultant in tandem with the international DREI consultant provided by the AMP Regional Project; (ii) coordination of, and participation in, technical standards development process.
2029.5		2030. 30k$ (1.1) - One or more contracts with specialized consultancy firm or institution for drafting of pertinent regulation for minigrids and rural electrification in congruence with Electricity Law 2020. 2031. 14k$ (1.3) – One or more contracts with specialized firm or institution for tailored capacity building (courses, events, workshops, online webinars etc.) 2032. 15k$ (1.4) – One contract with specialized firm for drafting of technical standards proposals and recommendations. 2033. 45k$ (1.4) – One contract with specialized consultancy firm to carry out in-depth technical analysis of components, installation quality and practices, functional design, technical specifications and material choices, operation and maintenance practices, compliance with specification and procedures, and potential weaknesses during the commissioning and reception process.
2034.		2035. 5k$ - Printing of policy and regulation proposals; AV material for presentation to stakeholders;

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2016. Budget Note No.		2017. Description
6		printing of technical standard proposals documents.
2036. 7		2037. 150k$ (1.2) - One or more contracts with specialized firm or institution for developing of DREI-informed de-risking insruments and/or other forthcoming analysis, recommendations, and proposals for regulatory instruments.
		2038. Component 2 (Total GEF Budget US$ 570,000)
2039. 8		2040. 16k$ (2.3) - One contract with international consultant (shortlisted by AMP) for technical backstopping during the minigrid pilot.
2041. 9		2042. 11k$ (2.1) – Contracts with (envisioned: 2) local technicians to support minigrid pilot development and operation under supervision of the PE and in coordination with village committee and contractors. 2043. 17k$ (2.3) – One contract with national expert for training and community development in minigrid pilot areas.
2044. 10		2045. 16 k$ - Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (land travel, fuel, DSA).
2046. 11		2047. 52k$ (2.1: 27k$ - 2.2: 25k$) Project Engineer (PE) for: (i) team leader for minigrid pilot design and implementation in collaboration; (ii) drafting of Minigrid Pilot Plan; (iii) TOR for contracted services (studies related to feasibility analysis and ESIA); (iv) functional specification of minigrid equipment and ancillary systems; (v) participation in procurement and supplier selection process; (vi) supervision of products, goods and services delivered by contractors; (vi) lead consultant for monitoring of pilots and analysis of operational data; (vii) identification of operational issues and initiation of remedial actions; (viii) identification of opportunities for enhancement and/or upscaling of the pilots; and (ix) progress reporting to Project Board. 2048. 32k$ (2.3) - Project Manager (PM) with responsibilities including: (i) lead consultant to the Executive; (ii) organization and execution support to IP for tendering and selection process of minigrid pilots; (iii) drafting of Minigrid Pilot Plan with support from PR; (iv) drafting of Terms of Reference for consultancies and procurement of services; and (v) quality assurance and overall supervision of contracted activities.
2049. 12		2050. 25k$ (2.2) - One or more contracts with specialized firm for final engineering and installation works for envisioned minigrid pilots. 2051. 26k$ (2.3) - One or more contracts with national institutions or firms for community development and implementation of training and awareness raising activities including on business

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2016. Budget Note No.	2017. Description
	development, productive uses of electricity and renewable energies.
2052. 13	2053. 371k$ (2.2) - Equipment minigrid pilots as per technical specifications (PV panels, racks, batteries, T&D infrastructure and ancillary systems).
2054. 14	2055. 2k$ - Printing of technical design and studies; printing of communication leaflets and electronic media.
2056. 15	2057. 2k$ - Miscellaneous expenses (communication costs, insurances, supplies).
	2058. **Component 3 (Total GEF Budget US$ 95,000)**
2059. 16	2060. 10k$ (3.1) - One contract with international consultant (shortlisted by AMP) for advice on financial mechanism design and banking sector development.
2061. 17	2062. 18k$ (3.2) – Once contract with national finance expert for: (i) analysis and recommendations to enhance access to finance for minigrid investors and end-users; (ii) engagement with national banking institutes; and (iii) participation in workshops and training events.
2063. 18	2064. 5k$ - Costs of domestic travel (land travel, fuel, DSA).
2065. 19	2066. 32k$ (3.1: 24k$; 3.2: 8k$) - Project Manager (PM) for: (i) liaison with financial sector entities, development of workplans and participation in analytical work, promotion and communication events; (ii) drafting of Terms of Reference for consultancies; and (iii) quality assurance and overall supervision of contracted activities.
2067. 20	2068. 13k$ (3.1) – One contract with specialized firm or institute for design of proposal for innovative financing mechanisms for minigrids in Benin. 2069. 15k$ (3.2) - One contract with specialized firm or institute for: (i) design of training programme on minigrid finance for national banking sector; (ii) identification of microfinance opportunities addressing productive energy uses by women; and (iii) implementation of training activities (on-site, webinars).
2070. 21	2071. 1k$ - Printing of communication material and electronic media.
2072. 22	2073. 1k$ - Miscellaneous expenses (communication costs, insurances, supplies).
	2074. **Component 4 (Total GEF Budget US$ 189,000)**
2075.	2076. 11 k$ (4.1) - One contract with international consultant (shortlisted by AMP) for advice on digital

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2016. Budget Note No.	2017. Description
23	technologies and system integration.
2077. 24	2078. 6k$ - Mission costs (international travel and DSA) for participation in AMP Communities of Practice. Costs of domestic travel (land travel, fuel, DSA).
2079. 25	2080. 33k$ (4.1: 6k$; 4.3: 13k$; 4.4: 14k$) - Project Manager (PM) for: (i) lead of Digital Strategy drafting process including embedding in IP business context; (ii) drafting of Terms of Reference for consultancies; (iii) quality assurance and overall supervision of contracted activities; (iv) responsibility for QAMF implementation; and (v) for participation in AMP Communities of Practice. 2081. 15 k$ (4.2) - Project Engineer (PE) for: (i) development of functional and technical specification for Digital Platform in coordination with international AMP digital expert and suppliers.
2082. 26	2083. 25 k$ (4.2) – One contract with specialized firm for: (i) analysis of Digital Platform host's institutional context and business operations; (ii) assessment of existent digital infrastructure and systems; and (iii) drafting of recommendations and road map for implementation of AMP Digital Platform.
2084. 27	2085. 82k$ (4.2) – Procurement of AMP Digital Platform including: (i) hardware and software and ancillary systems and devices; (ii) installation and configuration; (iii) initial training for end-users; and (iv) after-sales services and warrantees as per specifications.
2086. 28	2087. 15k$ (4.4) – One or more contracts with national publishing company for preparation of content (including video material) as input for AMP "Insight Briefs".
2088. 29	2089. 2k$ - Miscellaneous expenses (communication costs, insurances, supplies).
	2090. **Component 5 (Total GEF Budget US$ 64,588)**
2091. 30	2092. 9k$ (5.1.1) - One international expert to support the IP during the Project's inception phase including: (i) detailing Project M&E Plan including indicators and milestones; (ii) update the first annual work plan (AWP) and procurement plan; and (iii) support IW preparation process. 2093. 15.589k$ (5.1.2) - One independent international expert to conduct the Mid-Term Evaluation. 2094. 26k$ (5.1.3) - One independent international expert to conduct the GEF Terminal Evaluation.
2095. 31	2096. 4k$ (5.2: 2k$; 5.3: 2k$) – Two local consultants for data collection and assistance during MTR and TE process.
2097. 32	2098. 10 k$ - Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (land travel, fuel, DSA).
	2099. **Project Management Costs (GEF Budget US$ 120,559; UNDP TRAC: US$ 50,000; Total US$ 170,559)**

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2016. Budget Note No.	2017. Description
2100. 33	2101. 2k$ - Costs of domestic travel (land travel, fuel, DSA)
2102. 34	2103. 31k$ - Project Manager (PM) for project management activities, as per terms of reference. 2104. 31k$ - Project Finance and Administrative Officer (PA), as per Terms of Reference (4 years, 2/5 part-time). 2105. 31k$ - Project Procurement Specialist (PS), as per Terms of Reference (4 years, 2/5 part-time).
2106. 35	2107. 1.558k$ - Office furniture for PMU
2108. 36	2109. 2k$ - Three (3) laptops, printer and digital camera for PMU.
2110. 37	2111. 22k$ - Professional services for annual auditing of project financial status, delivered outputs, and financial, asset and human resources management.
2112. 38	2113. 35k$ - Complementary funds for PMU staff salaries (see budget note 34)
2114. 39	2115. 15k$ - Cost of capacity building and learning to support project execution.

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2119. PROJECT RESOURCES AND BUDGET CATEGORIES (GEF BUDGET)		2120.	2121.	2122.	2123.
2124.	2	2126.	2127.	2128.	2129.
2130. Key Consultants	**2131.**	2132.	2133.	2134.	2135.
2136. Project Manager (1.1; 1.3; 1.5; 2.3; 3.1; 3.2; 4.1; 4.3; 4.4; PM)	2137. P	2138. To be recruited (NC)	2139. GEF budget	2140. $182,000	2141. 71800
2142. Project Engineer/DREI National Consultant (1.2; 1.4; 2.1-2; 4.2)	2143. P	2144. To be recruited (NC)	2145. GEF budget	2146. $85,000	2147. 71800
2148. Project Finance and Admin Officer	2149. P	2150. To be recruited (part-time) (NC)	2151. GEF budget	2152. $31,000	2153. 71800
2154. Project Procurement Specialist	2155. P	2156. To be recruited (part-time) (NC)	2157. GEF budget	2158. $31,000	2159. 71800
2160. AMP Technical Advisor(s) DREI, Digital, other areas	2161. T	2162. To be recruited (part-time) (IC)	2163. GEF budget	2164. $99,000	2165. 71200
2166.	2167.	2168.	2169.	**2170. $428,000**	**2171.**
2172. MG Project and Digital Platform	**2173.**	2174.	2175.	2176.	2177.
2178. MG procurement goods & services (2.2)	2179. G	2180. To be procured	2181. GEF budget	2182. $371,000	2183. 72300
2184. Digital Platform procurement (4.2)	2185. D	2186. To be procured	2187. GEF budget	2188. $82,000	2189. 72800
2190.	2191.	2192.	2193.	**2194. $453,000**	**2195.**
2196. Short-Term Consultants	**2197.**	2198.	2199.	2200.	2201.

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2119. PROJECT RESOURCES AND BUDGET CATEGORIES (GEF BUDGET)		2120.	2121.	2122.	2123.
2124.	2	2126.	2127.	2128.	2129.
2202. Inception Support Expert (5.1)	2203. IE	2204. To be recruited (IC)	2205. GEF budget	2206. $9,000	2207. 71200
2208. SE Safeguards Expert (1.5)	2209. S	2210. To be recruited (NC)	2211. GEF budget	2212. $13,000	2213. 71300
2214. Gender Expert (1.5)	2215. G	2216. To be recruited (NC)	2217. GEF budget	2218. $14,000	2219. 71300
2220. Minigrid Finance Expert (3.2)	2221. F	2222. To be recruited (NC)	2223. GEF budget	2224. $18,000	2225. 71300
2226. Minigrid Local Support Expert (2.1)	2227. M	2228. To be recruited (NC)	2229. GEF budget	2230. $11,000	2231. 71300
2232. Training and Community Development Expert (2.3)	2233. T	2234. To be recruited (NC)	2235. GEF budget	2236. $17,000	2237. 71300
2238. Data Collection Consultants (5.2; 5.3)	2239. D	2240. To be recruited (NC)	2241. GEF budget	2242. $4,000	2243. 71300
2244. GEF Evaluation Experts (5.2; 5.3)	2245. E	2246. To be recruited (IC)	2247. GEF budget	2248. $41,588	2249. 71200
2250.	2251.	2252.	2253.	**2254. $127,588**	**2255.**
2256. Short-Term Service Contracts – Companies	**2257.**	2258.	2259.	2260.	2261.
2262. Regulatory Framework Upgrading (1.1)	2263.	2264. Service contract	2265. GEF budget	2266. $30,000	2267. 72100

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2119. PROJECT RESOURCES AND BUDGET CATEGORIES (GEF BUDGET)		2120.	2121.	2122.	2123.
2124.	2	2126.	2127.	2128.	2129.
2268. Capacity development (1.3)	2269.	2270. Service contract	2271. GEF budget	2272. $14,000	2273. 72100
2274. Drafting Technical Standards (1.4)	2275.	2276. Service contract	2277. GEF budget	2278. $15,000	2279. 72100
2280. Minigrid Technical Assessments (1.4)	2281.	2282. Service contract	2283. GEF budget	2284. $45,000	2285. 72100
2286. Minigrid Engineering Studies (2.2)	2287.	2288. Service contract	2289. GEF budget	2290. $25,000	2291. 72100
2292. Minigrid Community Development (2.3)	2293.	2294. Service contract	2295. GEF budget	2296. $26,000	2297. 72100
2298. Financing Analysis and Advice (3.1)	2299.	2300. Service contract	2301. GEF budget	2302. $13,000	2303. 72100
2304. Micro-finance outreach (3.2)	2305.	2306. Service contract	2307. GEF budget	2308. $15,000	2309. 72100
2310. Business operations assessment (4.1)	2311.	2312. Service contract	2313. GEF budget	2314. $25,000	2315. 72100
2316.	2317.	2318.	2319.	**2320. $208,000**	**2321.**
2322.	2	2324.	2325.	2326.	2327.
2328.	2329.	2330.	2331.	2332.	2333.
2334. Other expenses		**2335.** 2336.	2337.	2338.	2339.
2340. Travel	2341.	2342.	2343.	2344.	2345.

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2119. PROJECT RESOURCES AND BUDGET CATEGORIES (GEF BUDGET)		2120.	2121.	2122.	2123.
2124.	2	2126.	2127.	2128.	2129.
			GEF budget	$56,000	71600
2346. Equipment and Furniture	2347.	2348.	2349. GEF budget	2350. $1,559	2351. 72200
2352. Information Technology Equipment	2353.	2354.	2355. GEF budget	2356. $2,000	2357. 72800
2358. Audio Visual & Printing	2359.	2360.	2361. GEF budget	2362. $23,000	2363. 74200
2364. Professional Services – Audits	2365.	2366.	2367. GEF budget	2368. $22,000	2369. 74100
2370. Miscellaneous (supplies, insurances, communication costs)	2371.	2372.	2373. GEF budget	2374. $5,000	2375. 74500
2376.	2377.	2378.	2379.	**2380. $109,559**	**2381.**
2382.	2	2384.	2385.	2386.	2387.
2388. TOTAL PROJECT	**2389.**	2390.	2391.	**2392. $1,326,147**	**2393.**

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2397. X. LEGAL CONTEXT

165. This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of (country) and UNDP, signed on (date). All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

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166. This project will be implemented by Ministry of Energy / Directorate-General for Energy Resources (ME / DGRE) ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

2399. XI. RISK MANAGEMENT

2400. Implementing Partner is a Government Entity (NIM)

167. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:

a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;

b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

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168. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

2402.

169. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the United Nations Security Council Consolidated Sanctions List, and that no UNDP funds received pursuant to the Project Document are used for money laundering activities. The United Nations Security Council Consolidated Sanctions List can be accessed via https://www.un.org/securitycouncil/content/un-sc-consolidated-list.

2403.

170. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

2404. (a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

2405. (b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment. SH may occur in the workplace or in connection with work. While typically involving a pattern of conduct, SH may take the form of a

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single incident. In assessing the reasonableness of expectations or perceptions, the perspective of the person who is the target of the conduct shall be considered.

171. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

v. Promptly and confidentially, record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

172. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

173. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

174. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

175. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds.

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176. In the implementation of the activities under this Project Document, UNDP places reasonable reliance upon the Implementing Partner for it to apply its laws, regulations and processes, and applicable international laws regarding anti money laundering and countering the financing of terrorism, to ensure consistency with the principles of then in force the UNDP Anti-Money Laundering and Countering the Financing of Terrorism Policy.

2407.

177. The Implementing Partner will ensure that its financial management, anti-corruption, anti-fraud and anti-money laundering and countering the financing of terrorism policies are in place and enforced for all funding received from or through UNDP.

2408.

178. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

2409.

179. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

2410.

180. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

2411.

2412. Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

2413.

181. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

2414.

2415. Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud, corruption or other financial irregularity, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

2416.

2417. *Note:* The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

2418.

182. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

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183. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

2420.

184. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, mutatis mutandis, in all sub-contracts or sub-agreements entered into further to this Project Document.

2421. XII. MANDATORY ANNEXES

2422.

1. GEF Budget Template (available from BPPS NCE-VF)
2. GEF Execution Support Letter
3. Project Map and geospatial coordinates of the project area
4. Multiyear Workplan
5. Social and Environmental Screening Procedure (SESP)
6. UNDP Atlas Risk Register
7. Overview of technical consultancies/subcontracts

2423.

8. Stakeholder Engagement Plan
9. Environmental Social Management Framework (ESMF)
10. Gender Analysis and Gender Action Plan
11. Procurement Plan
12. GEF focal area specific annexes (e.g. METT, GHG calculations, target landscape profile, feasibility study, other technical reports)
13. Co-financing letters
14. Additional agreements: such as cost sharing agreements, project cooperation agreements signed with NGOs (where the NGO is designated as the "executing entity").
15. Signed LOA between UNDP and IP requesting UNDP Support Services (if required on exceptional basis and authorized by the GEF)

2424.

16. GEF and/or LDCF/SCCF Core indicators (see template below)
17. GEF Taxonomy (see template below)

2425.

18. Partners Capacity Assessment Tool and HACT assessment *(Please refer to the HACT policies for guidance on applicability and financial thresholds -*
19. UNDP Project Quality Assurance Report (to be completed in UNDP online corporate planning system)

2426.

2427. 20. TOR International Consultant for a Full Derisking Renewable energy Investment (DREI) analysis in Benin

2428. 21. TOR national Consultant for a Full Derisking Renewable energy Investment (DREI) analysis in Benin

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2431. Annex 1: GEF Budget Template

2432. To be provided by MPSU after TBWP clearance.

2433.

2434. Expenditure Category	2435. Detailed Description	2436. Component (USDeq.)							2437. Total (USDeq.)	2438. Responsible Entity
		2439. Component 1	*2440. Component 2*	*2441. Component 3*	*2442. Component 4*	*2443. Sub-Total*	*2444. M&E*	*2445. PMC*	2446.	2447. (Executing Entity receiving funds from the GEF Agency)[1]
2448. Equipment	2449. 1.559k$ - Office furniture for PMU	2450.	2451.	2452.	2453.	2454. -	2455.	2456. 1,558.00	2457. 1,558.00	2458. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2459. Equipment	2460. 371k$ (2.2) - Equipment minigrid pilots as per technical specifications (PV panels, racks, batteries, T&D infrastructure and ancillary systems).	2461.	2462. 371,000.00	2463.	2464.	2465. 371,000.00	2466.	2467.	2468. 371,000.00	2469. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2470. Equipment	2471. 2k$ - Three (3) laptops, printer and digital camera for PMU.	2472.	2473.	2474.	2475.	2476. -	2477.	2478. 2,000	2479. 2,000	2480. Direction Générale des Ressources

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2481. Equipment	2482. 82k$ (4.2) – Procurement of AMP Digital Platform including: (i) hardware and software and ancillary systems and devices; (ii) installation and configuration; (iii) initial training for end-users; and (iv) after-sales services and warrantees as per specifications.	2483.	2484.	2485.	2486. 82,000.00	2487. 82,000.00	2488.	2489.	2490. 82,000.00	2491. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2492. Contractual services-Individual	2493. 31k$ - Project Manager (PM) for project management activities, as per terms of reference.31k$ - Project Finance and Administrative Officer (PA), as per Terms of Reference (4 years, 2/5 part-time).31k$ - Project Procurement Specialist (PS), as per Terms of Reference (4 years, 2/5 part-time).	2494.	2495.	2496.	2497.	2498. -	2499.	2500. 93,000.00	2501. 93,000.00	2502. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2503. Contractual services-Individual	2504. 32k$ (3.1: 24k$; 3.2: 8k$) - Project Manager (PM) for: (i) liaison with financial sector entities, development of workplans and participation in analytical work, promotion and communication events; (ii) drafting of Terms of Reference for consultancies; and (iii) quality assurance and overall supervision of contracted activities.	2505.	2506.	2507. 32,000.00	2508.	2509. 32,000.00	2510.	2511.	2512. 32,000.00	2513. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2514. Contractual services-Individual	2515. 33k$ (4.1: 6k$; 4.3: 13k$; 4.4: 14k$) - Project Manager (PM) for: (i) lead of Digital Strategy drafting process including embedding in IP business context; (ii) drafting of Terms of Reference for consultancies; (iii) quality assurance and overall supervision of contracted activities; (iv) responsibility for QAMF implementation; and (v) for participation in AMP Communities of Practice.15 k$ (4.2) - Project Engineer	2516.	2517.	2518.	2519. 48,000.00	2520. 48,000.00	2521.	2522.	2523. 48,000.00	2524. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)

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		(PE) for: (i) development of functional and technical specification for Digital Platform in coordination with international AMP digital expert and suppliers.									
2525. Contractual services-Individual		2526. 52k$ (2.1: 27k$ - 2.2: 25k$) Project Engineer (PE) for: (i) team leader for minigrid pilot design and implementation in collaboration; (ii) drafting of Minigrid Pilot Plan; (iii) TOR for contracted services (studies related to feasibility analysis and ESIA); (iv) functional specification of minigrid equipment and ancillary systems; (v) participation in procurement and supplier selection process; (vi) supervision of products, goods and services delivered by contractors; (vi) lead consultant for monitoring of pilots and analysis of operational data; (vii) identification of operational issues and initiation of remedial actions; (viii) identification of opportunities for enhancement and/or upscaling of the pilots; and (ix) progress reporting to Project Board.32k$ (2.3) - Project Manager (PM) with responsibilities including: (i) lead consultant to the Executive; (ii) organization and execution support to IP for tendering and selection process of minigrid pilots; (iii) drafting of Minigrid Pilot Plan with support from PR; (iv) drafting of Terms of Reference for consultancies and procurement of services; and (v) quality assurance and overall supervision of contracted activities.	2527.	2528. 84,000.00	2529.	2530.	2531. 84,000.00	2532.	2533.	2534. 84,000.00	2535. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2536. Contractual services-Individual		2537. 54k$ (1.1: 47k$ - 1.3: 5k$ - 1.5: 2k$) - One minigrid policy expert (P3-level for 4-yr period) to assume the role of Project Manager (PM) with responsibilities including: (i) lead consultant to the Executive (ME); (ii) lead consultant on topics related to regulation, institutional capacity, finance; (iii) engagement with GOB stakeholders, market actors and CSOs; (iv) drafting of Terms of Reference for consultancies and procurement of services; (v) quality assurance and	2538. 72,000.00	2539.	2540.	2541.	2542. 72,000.00	2543.	2544.	2545. 72,000.00	2546. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)

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		overall supervision of contracted activities; (vi) engagement with AMP Regional Project partners for peer review of proposals, analysis of project approaches, and participation in AMP events in Benin and abroad; and (vii) compilation of proposals and presentations to stakeholders.18k$ (1.2: 10k$ - 1.4: 8k$) - One national expert to act as Project Engineer (PE) for: (i) act as the national DREI consultant in tandem with the international DREI consultant provided by the AMP Regional Project; (ii) coordination of, and participation in, technical standards development process.									
2547. Contractual services-Company		2548. 13k$ (3.1) – One contract with specialized firm or institute for design of proposal for innovative financing mechanisms for minigrids in Benin.15k$ (3.2) - One contract with specialized firm or institute for: (i) design of training programme on minigrid finance for national banking sector; (ii) identification of microfinance opportunities addressing productive energy uses by women; and (iii) implementation of training activities (on-site, webinars).	2549.	2550.	2551. 28,000.00	2552.	2553. 28,000.00	2554.	2555.	2556. 28,000.00	2557. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2558. Contractual services-Company		2559. 25 k$ (4.2) – One contract with specialized firm for: (i) analysis of Digital Platform host's institutional context and business operations; (ii) assessment of existent digital infrastructure and systems; and (iii) drafting of recommendations and road map for implementation of AMP Digital Platform.	2560.	2561.	2562.	2563. 25,000.00	2564. 25,000.00	2565.	2566.	2567. 25,000.00	2568. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2569. Contractual services-Company		2570. 25k$ (2.2) - One or more contracts with specialized firm for final engineering and installation works for envisioned minigrid pilots.26k$ (2.3) - One or more contracts with national institutions or firms for community development and implementation of training and awareness raising activities including on business development, productive uses of electricity and renewable energies.	2571.	2572. 51,000.00	2573.	2574.	2575. 51,000.00	2576.	2577.	2578. 51,000.00	2579. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)

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2580. Contractual services-Company	2581. 30k$ (1.1) - One or more contracts with specialized consultancy firm or institution for drafting of pertinent regulation for minigrids and rural electrification in congruence with Electricity Law 2020.14k$ (1.3) – One or more contracts with specialized firm or institution for tailored capacity building (courses, events, workshops, online webinars etc.)15k$ (1.4) – One contract with specialized firm for drafting of technical standards proposals and recommendations.45k$ (1.4) – One contract with specialized consultancy firm to carry out in-depth technical analysis of components, installation quality and practices, functional design, technical specifications and material choices, operation and maintenance practices, compliance with specification and procedures, and potential weaknesses during the commissioning and reception process.	2582. 104,000.00	2583.	2584.	2585.	2586. 104,000.00	2587.	2588.	2589. 104,000.00	2590. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2591. International Consultants	2592. 10k$ (3.1) - One contract with international consultant (shortlisted by AMP) for advice on financial mechanism design and banking sector development.	2593.	2594.	2595. 10,000.00	2596.	2597. 10,000.00	2598.	2599.	2600. 10,000.00	2601. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2602. International Consultants	2603. 11 k$ (4.1) - One contract with international consultant (shortlisted by AMP) for advice on digital technologies and system integration.	2604.	2605.	2606.	2607. 11,000.00	2608. 11,000.00	2609.	2610.	2611. 11,000.00	2612. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2613. International Consult	2614. 16k$ (2.3) - One contract with international consultant (shortlisted by AMP) for technical backstopping during the minigrid pilot.	2615.	2616. 16,	2617.	2618.	2619. 16,000	2620.	2621.	2622. 16,000	2623. Direction Générale des Ressources

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ants			0 0 0 . 0 0			0. 0 0			0. 0 0	Energétiques (DGRE - Ministry of Energy)
2624. International Consultants	2625. 62k$ (1.1: 10k$ - 1.2: 42k$ - 1.3: 5k$ -1.4: 5k$) - One or more contracts with international consultant (shortlisted by AMP) for (i) technical backstopping on policy and regulation; (ii) implementation DREI framework; and (iii) advice for technical standards process.	2626. 62 , 0 0 0 . 0 0	2627.	2628.	2629.	2630. 6 2, 0 0 0. 0 0	2631.	2632.	2633. 6 2, 0 0 0. 0 0	2634. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2635. International Consultants	2636. 9k$ (5.1.1) - One international expert to support the IP during the Project's inception phase including: (i) detailing Project M&E Plan including indicators and milestones; (ii) update the first annual work plan (AWP) and procurement plan; and (iii) support IW preparation process. 15.588k$ (5.1.2) - One independent international expert to conduct the Mid-Term Evaluation.26k$ (5.1.3) - One independent international expert to conduct the GEF Terminal Evaluation.	2637.	2638.	2639.	2640.	2641. -	2642. 5 0 , 5 8 9 . 0 0	2643.	2644. 5 0, 5 8 9. 0 0	2645. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2646. Local Consultants	2647. 13k$ (1.5) – One national Social and Environmental Safeguards Expert for: (i) periodic supervision of ESMF implementation; (ii) periodic SESP rescreening; and (iii) systematization of lessons learnt and recommendations for enhancement. 14k$ (1.5) – One national gender expert for: (i) periodic supervision of Gender Action Plan implementation; and (ii) identification of issues and recommendations for enhancement.	2648. 27 , 0 0 0 . 0 0	2649.	2650.	2651.	2652. 2 7, 0 0 0. 0 0	2653.	2654.	2655. 2 7, 0 0 0. 0 0	2656. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2657. Local Consultants	2658. 16k$ (2.3) - One contract with international consultant (shortlisted by AMP) for technical backstopping during the minigrid pilot.	2659.	2660. 2 8 , 0 0 0	2661.	2662.	2663. 2 8, 0 0 0. 0 0	2664.	2665.	2666. 2 8, 0 0 0. 0 0	2667. Direction Générale des Ressources Energétiques (DGRE - Ministry of

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2668. Local Consultants	2669. 18k$ (3.2) – Once contract with national finance expert for: (i) analysis and recommendations to enhance access to finance for minigrid investors and end-users; (ii) engagement with national banking institutes; and (iii) participation in workshops and training events.	2670.	2671.	2672. 18,000.00	2673.	2674. 18,000.00	2675.	2676.	2677. 18,000.00	2678. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2679. Local Consultants	2680. 4k$ (5.2: 2k$; 5.3: 2k$) – Two local consultants for data collection and assistance during MTR and TE process.	2681.	2682.	2683.	2684.	2685. -	2686. 4,000.00	2687.	2688. 4,000.00	2689. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2690. Travel	2691. 10 k$ - Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (land travel, fuel, DSA).	2692.	2693.	2694.	2695.	2696. -	2697. 10,000.00	2698.	2699. 10,000.00	2700. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2701. Travel	2702. 16 k$ - Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (land travel, fuel, DSA).	2703.	2704. 16,000.00	2705.	2706.	2707. 16,000.00	2708.	2709.	2710. 16,000.00	2711. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2712. Travel	2713. 17k$ - Mission costs (international travel and DSA) for international consultants. Costs of	2714. 17,	2715.	2716.	2717.	2718. 17,	2719.	2720.	2721. 17,	2722. Direction Générale

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	domestic travel (land travel, fuel, DSA).	0 0 0 . 0 0				0 0 0. 0 0			0 0 0. 0 0	des Ressources Energétiques (DGRE - Ministry of Energy)
2723. Travel	2724. 2k$ - Costs of domestic travel (land travel, fuel, DSA)	2725.	2726.	2727.	2728.	2729. -	2730.	2731. 2, 0 0 0. 0 0	2732. 2, 0 0 0. 0 0	2733. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2734. Travel	2735. 5k$ - Costs of domestic travel (land travel, fuel, DSA).	2736.	2737.	2738. 5 , 0 0 0 . 0 0	2739.	2740. 5, 0 0 0. 0 0	2741.	2742.	2743. 5, 0 0 0. 0 0	2744. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2745. Travel	2746. 6k$ - Mission costs (international travel and DSA) for participation in AMP Communities of Practice. Costs of domestic travel (land travel, fuel, DSA).	2747.	2748.	2749.	2750. 6 , 0 0 0 . 0 0	2751. 6, 0 0 0. 0 0	2752.	2753.	2754. 6, 0 0 0. 0 0	2755. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2756. Other Operating Costs	2757. 22k$ - Professional services for annual auditing of project financial status, delivered outputs, and financial, asset and human resources management.	2758.	2759.	2760.	2761.	2762. -	2763.	2764. 2 2 , 0 0 0 . 0 0	2765. 2 2, 0 0 0. 0 0	2766. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)

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2767. Other Operating Costs	2768. 15k$ (4.4) – One or more contracts with national publishing company for preparation of content (including video material) as input for AMP “Insight Briefs”.	2769.	2770.	2771.	2772. 15,000.00	2773. 15,000.00	2774.	2775.	2776. 15,000.00	2777. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2778. Other Operating Costs	2779. 1k$ - Miscellaneous expenses (communication costs, insurances, supplies).	2780.	2781.	2782. 1,000.00	2783.	2784. 1,000.00	2785.	2786.	2787. 1,000.00	2788. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2789. Other Operating Costs	2790. 2k$ - Printing of technical design and studies; printing of communication leaflets and electronic media.	2791.	2792. 2,000.00	2793.	2794.	2795. 2,000.00	2796.	2797.	2798. 2,000.00	2799. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2800. Other Operating Costs	2801. 5k$ - Printing of policy and regulation proposals; AV material for presentation to stakeholders; printing of technical standard proposals documents.	2802. 5,000.00	2803.	2804.	2805.	2806. 5,000.00	2807.	2808.	2809. 5,000.00	2810. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2811. Other Operating Costs	2812. 1k$ - Miscellaneous expenses (communication costs, insurances, supplies).	2813.	2814.	2815. 1,000.	2816.	2817. 1,000.00	2818.	2819.	2820. 1,000.00	2821. Direction Générale des Ressources Energétiques (DGRE - Ministry of

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				0 0						Energy)
2822. Other Operating Costs	2823. 2k$ - Miscellaneous expenses (communication costs, insurances, supplies).	2824.	2825. 2,000.00	2826.	2827. 2,000.00	2828. 4,000.00	2829.	2830.	2831. 4,000.00	2832. Direction Générale des Ressources Energétiques (DGRE - Ministry of Energy)
2833.	*2834. Project Total*	**2835. 287,000,000.00**	**2836. 570,000,000.00**	**2837. 95,000,000.00**	**2838. 189,000,000.00**	**2839. 1,141,000.00**	**2840. 64,589.00**	**2841. 120,558.00**	**2842. 1,326,147.00**	2843.

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2845. Annex 2: GEF execution support letter Services

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2847. Does not apply

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2851. Annex 3: Project map and Geospatial Coordinates of project sites

1. Upstream interventions address sector policies and capacities for off-grid electrification, which are applicable in the entire territory of Benin. Specific minigrid pilot sites will be selected during project implementation. At this stage, no detailed coordinates are available for these. The continental territory of Benin is comprised within a rectangle stretching between: 12°24'51.8"N (North) and 6°14'08.0"N (South); 0°46'29.4"E (West) and 3°50'35.9"E (East).

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2853. Map source: https://www.un.org/geospatial/content/benin-0

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2855. Annex 4: Multi Year Work Plan

2856. | 2857. Policy de-risking | 2858. | 2859. Minigrid investment pilot(s) | 2860. | 2861. Institutional and individual capacity building | 2862. | 2863. M&E, QA and KM

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2865. Components	**2866. Outcomes**	**2867. Outputs**	**2868. Year 1 2869. (2023/2024)**				**2870. Year 2 2871. (2024/2025)**				**2872. Year 3 2873. (2025/2026)**				**2874. Year 4 2875. (2026/2027)**			
			2876. Q	2877. Q	2878. Q	2879. Q	2880. Q	2881. Q	2882. Q	2883. Q	2884. Q	2885. Q	2886. Q	2887. Q	2888. Q	2889. Q	2890. Q	2891. Q
2892. Component 1: 2893. Policy and regulation	**2894. Outcome 1: 2895. Stakeholder ownership in a national mini-grid delivery model is advanced, and appropriate policies and regulations are adopted to facilitate investment in low□carbon mini-grids.**	2896. Output 1.1. An inclusive national dialogue to identify minigrid delivery models is facilitated, clarifying priority interventions for an integrated approach to off-grid electrification.	2897.	2898.	2899.	2900.	2901.	2902.	2903.	2904.	2905.	2906.	2907.	2908.	2909.	2910.	2911.	2912.
		2913. Output 1.2. Minigrid DREI techno-economic analyses carried out to propose most cost-effective basket of policy and financial de-risking instruments and contribute to AMP Flagship Report on Cost Reduction.	2914.	2915.	2916.	2917.	2918.	2919.	2920.	2921.	2922.	2923.	2924.	2925.	2926.	2927.	2928.	2929.
		2930. Output 1.3. Capacity building provided to public officials (regulator, ministries) specifically to design procurement/tender processes that incorporate cost-reduction levers and innovative business models.	2931.	2932.	2933.	2934.	2935.	2936.	2937.	2938.	2939.	2940.	2941.	2942.	2943.	2944.	2945.	2946.
		2947. Output 1.4. Domestication of quality standards for solar mini-grid components, and institutional capacity of national standards organizations/bureau strengthened.	2948.	2949.	2950.	2951.	2952.	2953.	2954.	2955.	2956.	2957.	2958.	2959.	2960.	2961.	2962.	2963.
		2964. Output 1.5. Support provided to establish the environmental and social policies and plans to ensure mini-grid risks are properly handled.	2965.	2966.	2967.	2968.	2969.	2970.	2971.	2972.	2973.	2974.	2975.	2976.	2977.	2978.	2979.	2980.
2981. Compo	**2983. Outcome 2:**	2985. Output 2.1. Pilots developed (preparatory phase), including on	2986.	2987.	2988.	2989.	2990.	2991.	2992.	2993.	2994.	2995.	2996.	2997.	2998.	2999.	3000.	3001.

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2865. Components	2866. Outcomes	2867. Outputs	2868. Year 1 2869. (2023/2024)				2870. Year 2 2871. (2024/2025)				2872. Year 3 2873. (2025/2026)				2874. Year 4 2875. (2026/2027)			
			2876. Q	2877. Q	2878. Q	2879. Q	2880. Q	2881. Q	2882. Q	2883. Q	2884. Q	2885. Q	2886. Q	2887. Q	2888. Q	2889. Q	2890. Q	2891. Q
nent 2: 2982. Business Model innovation with private sector	**2984. Innovative business models based on cost reduction operationalized, with strengthened private sector participation in low-carbon mini-grid development.**	productive use and innovative appliances and modular hardware and system design, leading to cost-reduction in minigrids.																
		3002. Output 2.2. Commissioning and monitoring of selected pilots in alignment with AMP principles.	3003.	3004.	3005.	3006.	3007.	3008.	3009.	3010.	3011.	3012.	3013.	3014.	3015.	3016.	3017.	3018.
		3019. Output 2.3. Enhancement of minigrid business model by strengthening operator and community capacities, development of PUE and other energy nexus, and the integration of local RE sources.	3020.	3021.	3022.	3023.	3024.	3025.	3026.	3027.	3028.	3029.	3030.	3031.	3032.	3033.	3034.	3035.
3036. Component 3: 3037. Scaled-up financing	**3038. Outcome 3: 3039. Financial sector actors are ready to invest in a pipeline of low-carbon mini-grids and concessional financial mechanisms are in place**	3040. Output 3.1. Innovative financing solutions for minigrid development are identified and implemented with supporting human and institutional strengthening.	3041.	3042.	3043.	3044.	3045.	3046.	3047.	3048.	3049.	3050.	3051.	3052.	3053.	3054.	3055.	3056.
		3057. Output 3.2. Domestic financial sector capacity-building on business and financing models for minigrids.	3058.	3059.	3060.	3061.	3062.	3063.	3064.	3065.	3066.	3067.	3068.	3069.	3070.	3071.	3072.	3073.

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2865. Components	2866. Outcomes	2867. Outputs	2868. Year 1 2869. (2023/2024)				2870. Year 2 2871. (2024/2025)				2872. Year 3 2873. (2025/2026)				2874. Year 4 2875. (2026/2027)			
			2876. Q	2877. Q	2878. Q	2879. Q	2880. Q	2881. Q	2882. Q	2883. Q	2884. Q	2885. Q	2886. Q	2887. Q	2888. Q	2889. Q	2890. Q	2891. Q
	to incentivize scaled-up																	
3074. Component 4: 3075. Digital, and Knowledge Management (KM)	3076. Outcome 4: 3077. Digital and data are mainstreamed, across stakeholders, into local mini-grid market development. Increased knowledge, awareness and network opportunities in the mini-grid market and among stakeholders, including benefitting from linkages to international good practice.	3078. Output 4.1. A project digital strategy is developed and implemented, including linkages to and following guidance from the AMP Regional Project.	3079.	3080.	3081.	3082.	3083.	3084.	3085.	3086.	3087.	3088.	3089.	3090.	3091.	3092.	3093.	3094.
		3095. Output 4.3. Specification and implementation of Minigrids Digital Platform to track minigrid pilots and support scale-up and cost-reduction.	3096.	3097.	3098.	3099.	3100.	3101.	3102.	3103.	3104.	3105.	3106.	3107.	3108.	3109.	3110.	3111.
		3112. Output 4.4. Adoption and operationalization of the project's Quality Assurance and Monitoring Framework (QAMF).	3113.	3114.	3115.	3116.	3117.	3118.	3119.	3120.	3121.	3122.	3123.	3124.	3125.	3126.	3127.	3128.
		3129. Output 4.5. Engage with regional project by participating in Communities of Practice and capturing and sharing of lessons learnt.	3130.	3131.	3132.	3133.	3134.	3135.	3136.	3137.	3138.	3139.	3140.	3141.	3142.	3143.	3144.	3145.
3146. Monitoring	3147. Ensuring compliance with all	3148. Output 5.1: Inception workshop is conducted and M&E plan is implemented.	3149.	3150.	3151.	3152.	3153.	3154.	3155.	3156.	3157.	3158.	3159.	3160.	3161.	3162.	3163.	3164.
		3165. Output 5.2: Project Mid-Term	3166.	3167.	3168.	3169.	3170.	3171.	3172.	3173.	3174.	3175.	3176.	3177.	3178.	3179.	3180.	3181.

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2865. Components	2866. Outcomes	2867. Outputs	2868. Year 1 2869. (2023/2024)				2870. Year 2 2871. (2024/2025)				2872. Year 3 2873. (2025/2026)				2874. Year 4 2875. (2026/2027)			
			2876. Q	2877. Q	2878. Q	2879. Q	2880. Q	2881. Q	2882. Q	2883. Q	2884. Q	2885. Q	2886. Q	2887. Q	2888. Q	2889. Q	2890. Q	2891. Q
		Review is conducted.																
and Evaluati	mandatory monitoring and reporting requirements	3182. Output 5.3: GEF Terminal Evaluation is conducted	3183.	3184.	3185.	3186.	3187.	3188.	3189.	3190.	3191.	3192.	3193.	3194.	3195.	3196.	3197.	3198.

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3200. Annex 5: UNDP Social and Environmental Screening Procedure (SESP)

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3202. Provided in separate file.

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3206. Annex 6: UNDP (Atlas) Risk Register

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
3215. Social and Environmental Risks and Mitigation Measures					
3216. 1	3217. Discrimination or marginalization of vulnerable communities through the proposed tariff model in the proposed minigrid regulation. 3218. 3219. Income levels in project sites are generally low. This creates a risk that low-income households would not be able to benefit from the project pilot and from minigrids that may be established as a result of upstream activities. 3220. 3221. Under Benin's concession model, minigrid operators must propose a competitive, yet inclusive, differentiated tariff scheme, to be ratified by the Regulator. The Project's demonstration pilot will abide by this model. If not taken into consideration, this model may lead to setting tariffs that would discriminate against vulnerable communities (including low-income households) and prevent them from having access to essential energy services.	3222. Social and Environmental - Human Rights	3223. I = 4, L = 3 3224. (substantial)	3225. As the project risk category has been rated as Substantial, an Environmental and Social Management Framework (ESMF) has been prepared and annexed to the ProDoc. The ESMF requires that the Activity 1.1.3 (Update pertinent regulation, guidelines, and processes for minigrid concessions, in line with the new Electricity Law of 2020), undergo a Strategic Environmental and Social Assessment (SESA) that would take this risk into consideration in the decision-making process. In addition, a Stakeholder Engagement Plan (SEP) has been prepared to ensure that stakeholders have an opportunity to provide feedback on decisions that may affect them. Through this SEP, the Project will devise strategies to reach out to low-income families. Depending on the identified impacts, a Livelihood Action Framework may also be developed if needed. 3226. 3227. As per Electricity Law 2020, the service shall be metered and by default, billing is prepaid through convenient amounts adapted to the purchase capacity and needs of a variety of customers. Assistance shall be provided to customers who face difficulties to understand and use smart meters. Mechanisms to settle disputes and billing errors are foreseen; and the Concessionaire has the right to suspend the service to customers who decline to pay or tamper with their connection. 3228. 3229. The project will also put in place a project-level GRM to provide meaningful means for local communities and affected populations to raise concerns and/or grievances when activities may adversely impact them.	3230. Project Manager
3231. 2	3232. Risk on lack of ability for people to claim their rights within the areas served by the pilot minigrids. 3233. 3234. Lack of transparency and tedious or costly procedures of people/customers to claim their rights	3235. Social and Environmental - Human Rights; Accountability	3236. I = 3, L = 4 3237. (modera	3238. Through the Stakeholder Engagement Plan, the Project shall give priority to community engagement to ensure that No-one is Left Behind. This will imply a proactive attitude to reach out to vulnerable people and groups and treat people equally.	3241. Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
	may exist within the pilot areas and often the legal or contractual basis for claiming these rights is not well defined or even absent.		te)	3239. 3240. The project will also put in place a project-level GRM to provide meaningful means for local communities and affected populations to raise concerns and/or grievances when activities may adversely impact them.	
3242. 3	3243. Marginalization of vulnerable groups when developing standards and selecting the pilot minigrids 3244. 3245. Domestication of quality standards for solar mini-grid components may marginalize stakeholders from participating in this sector, or from having access to energy by setting stringent technical criteria. Selection of pilot minigrids, if not done in collaboration with all stakeholders also risks marginalizing certain groups, including indigenous peoples.	3246. Social and Environmental - Human Rights; Accountability	3247. I = 3, L = 3 3248. (moderate)	3249. A Stakeholder Engagement Plan has been prepared to manage this risk, which is associated with Output 1.4, through engaging stakeholders to ensure that standards do not marginalize any specific group and exclude them from the decision-making process on issues that affect them. It will also provide input for selection of the pilot minigrid site (Output 2.1). 3250. 3251. If the minigrid pilot site or its associated infrastructure is located in close proximity to or on land used by Indigenous Peoples, Free and Prior Informed Consent (FPIC) will be obtained prior to validation of the site. Towards that end, an Indigenous Peoples Policy Framework (IPPF) has been prepared to tackle this issue. During preparation of the Environmental and Social Impact Assessment (ESIA) for the pilot minigrid and based on the developed IPPF, an Indigenous Peoples Plan (IPP) (or its equivalent) will be developed (at the site-level or other level, as deemed appropriate) and implemented, along with measures for FPIC, as needed for compliance with the SES. 3252. 3253. The project GRM will also help in managing this risk by providing a means for affected stakeholders to raise concerns and/or grievances.	3254. Project Manager
3255. 4	3256. Reproducing existing discriminations against women through excluding them from decision-making on project activities, benefiting from project outputs and capacity building initiatives. 3257. 3258. Social and cultural factors leading to different roles between men and women in Benin and the current prevalence of men in the electricity sector may pose a challenge to ensure that women will have the chance to participate at the decisions-making level.	3259. Social and Environmental - Gender Equality and Women Empowerment	3260. I = 4, L = 4 3261. (substantial)	3262. Measures have been established through the Gender Analysis and Action Plan established at the PPG phase, to manage and reduce the risks identified on women. 3263. 3264. Examples involve the following: 3265. • Inclusion of women in the national dialogue 3266. • Elevating the importance/role of domestic appliances in alleviating women's time poverty and reducing the division of labor according to gender 3267. • Supporting gender equality in the	3273. Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
				minigrid supply chain and enabling environment 3268. • Mitigating the chances that new minigrid management and data tools have unforeseen negative consequences for women 3269. • Building financial inclusion for women as electricity suppliers and users 3270. • Sharing insights and best practices across projects. 3271. 3272. In addition, this risk will be further assessed in the SESAs and Environmental and Social Impact Assessments (ESIAs) that will be undertaken during project implementation as described in the ESMF.	
3274.5	3275. Damage to biodiversity, natural resources and cultural heritage sites due to installation and operation of pilot minigrids. 3276. 3277. Pilot minigrids may be located within or near critical habitats, environmentally sensitive areas or cultural heritage sites. However, as the pilot will only entail rehabilitation of an existing MG changes to the use of lands and resources, affecting natural ecosystems may only result from associate infrastructure such as extension of the network. Furthermore, mini-grids with a productive use entail unforeseen impacts should be expected according to the type of sector and activity to develop.	3278. Social and Environmental - : Biodiversity Conservation and Natural Resource Management; Cultural Heritage	3279. I = 3, L = 2 3280. (moderate)	3281. The pilot minigrid (Outputs 2.1 and 2.3) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that is found in the ESMF. After selection and before commencement of the pilot activity, the pilot minigrid will undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific Environmental and Social Management Plan (ESMP). Details of this process can be found in the ESMF.	3282. Project Manager
3283.6	3284. Exposure to electrocution risks for humans and any fauna (ex. animals or birds) using the minigrid area. 3285. 3286. All mini-grids involve electrical equipment. At the operational stage, the electrical structure alien to pre-existing conditions in the area, may cause the damage/death/fire/et due to the interaction with people living nearby, fauna and flora.	3287. Social and Environmental - Biodiversity Conservation and Natural Resource Management; Community Health, Safety and Security	3288. I = 3, L = 2 3289. (moderate)	3290. Selection of the pilot will include a screening on all relevant criteria of the SESP. Before commencement, the pilot will undergo a scoped ESIA or targeted assessment that will analyze all relevant risks. Mitigation measures, such as fencing of the PV plant is essential as well as guarding through community official guards, will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF.	3291. Project Manager
3292.7	3293. Climate events and disasters (including floods) on new and existing infrastructure. 3294.	3298. Social and Environmental -	3299. I = 3, L = 3	3301. The pilot minigrid (Outputs 2.1 and 2.3) will undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted	3302. Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
	3295. Benin is considered highly vulnerable to global climate change, ranking 155 out of 181 countries in the ND-GAIN index for climate vulnerability. Current trends include intensification of droughts and rains (by 100 mm/h) exacerbating soil erosion and leading to floods. Climate projections expect sea level rise by 0.4 to 0.7 meters by 2100, probably resulting in coastal disasters (complete coastal erosion, floods, and storm waves). Overall, the coastal, north-western, and far northern zones of Benin are considered to be particularly vulnerable to the impacts of climate change. 3296. 3297. As mini-grids are open air structures, they are exposed to climate events and involve build structures that may be vulnerable to the impacts of climate change or disasters.	Climate Change and Disaster Risks; Community Health, Safety and Security	3300. (moderate)	as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. The design of structures and equipment shall consider local conditions including heavy rainfall affecting electric insulation classes and pole foundations, among others.	
3303. 8	3304. Risk on the community due to domestic connections and electricity usage, and presence of hazardous materials (mainly batteries, e-waste). 3305. 3306. The novelty of some structures and practices brought about by the project may become a source of harm if not accompanied with concomitant awareness of risks and safe practices. More specifically, the use of hazardous materials by the project, domestic electrical wiring and connection activities and subsequent domestic usage of electricity.	3307. Social and Environmental - Community Health, Safety and Security	3308. I = 3, L = 2 3309. (moderate)	3310. The pilot minigrid (Outputs 2.1 and 2,3) will undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. In particular, operators, contractors and owners of sites shall be required to abide by the ESMP's requirements on safety measures and minimum qualifications for the handling of hazardous materials. 3311. National legislation requires qualified electricians. Consumer awareness campaigns should also be performed, including through local workshops, clear signage (pictograms and local language indications) and awareness-raising activities in schools and public spaces to inform communities of risks associated with installations (e.g. prevention of trespassing and/or makeshifts connections attempts, etc.) and of the safe usage of electricity domestically.	3312. Project Manager
3313. 9	3314. Community health and safety risks due to construction of the pilot minigrids and relevant infrastructure and new economic activities subsequent from productive use of the energy	3315. Social and Environmental - Community Health, Safety and Security	3316. I = 3, L = 2 3317. (moderate)	3318. The pilot minigrid (Outputs 2.1 and 2.3) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that is found in the ESMF. After selection and before commencement of the pilot activity, the pilot minigrid will undergo a scoped ESIA or target assessment that will analyze this risk. Mitigation	3319. Project Manager

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#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
				measures will then be adopted as described in the pursuant site-specific Environmental and Social Management Plan (ESMP), which shall include a Pollution Prevention and Management Plan and a Traffic Management Plan. Details of this process can be found in the ESMF.	
10	Risk on community health, safety and/or security due to the influx of people, mainly project workers and other newcomers subsequent to the new economic activities resulting from the productive use of the energy. New activities in the project's area of influence may attract newcomers affecting community health, safety and/or security as this new influx of people, expected to be mainly men, may interact with the local residents and/or involve the alteration of the normal functioning of the community leading to new diseases and/or gender safety concerns.	Social and Environmental - Community Health, Safety and Security	I = 3, L = 3 (moderate)	The pilot minigrid (Outputs 2.1 and 2.3) will undergo a scoped ESIA that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF. Contractors including any security personnel shall abide to UNDP's Standards of Conduct and apply best practices at all times. The project GRM will provide a means for affected community to report on any incidents that may occur as a result of this risk.	Project Manager
11	Physical or economic displacement and loss of livelihood due to eviction from land on which pilot minigrids may be installed. All minigrids involve the construction of new infrastructure. New built structures occupy land, and access to the area may be restricted. Expected impacts include the displacement of existing legal or illegal inhabitants to allow the new structures to be built. However, as the only pilot that will be selected will involve rehabilitation of an existing MG, this risk is expected to be Low.	Social and Environmental - Resettlement and Displacement	I = 2, L = 1 (low)	The pilot minigrid (Outputs 2.1 and 2.3) will incorporate SES criteria during the site selection process and adopt the list of exclusion criteria that is found in the ESMF. Before commencement, the pilot minigrid will undergo a scoped ESIA or targeted assessment that will analyze these risks. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF.	Project Manager
12	Loss of income for fuel sellers once pilot minigrids are operational. Traditional fuels supplied by local providers, including those from the informal/traditional sectors see their market diminished. Some mini-grid systems and project appliances to be implemented may replace an activity that was fueled with other energy sources such as diesel, charcoal and fuelwood. The decrease in fuel demand will lead to the loss of income for fuel	Social and Environmental - Human Rights	I = 3, L = 3 (moderate)	Pilot minigrids (Outputs 2.1 and 2.3) will each undergo a scoped ESIA or targeted assessment that will analyze this risk. Mitigation measures will then be adopted as described in the pursuant site-specific ESMP. Details of this process can be found in the ESMF.	Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
	suppliers, some of whom may be vulnerable people working in the informal market. Due to the fact that the pilot site has not yet been selected, the likelihood of this risk is not known but is not expected to be significant.				
3351. 13	3352. Working conditions not in line with national and international standards (by contractor or other entities involved in the project) 3353. 3354. All stages of the pilot minigrids will require labor, some of which may be sourced to unskilled/manual laborers who could be less familiar with the type of installations considered for this project and the concomitant occupational health and safety requirements and risks. Maintenance of the right-of-way and bush-clearing under transmission lines by manual labor is especially relevant in this context. This may lead to the use of child, forces, discriminatory, under-minimum practices and/or occupational health and safety accidents/incidents.	3355. Social and Environmental - Labor and Working Conditions	3356. I = 4, L = 2 3357. (moderate)	3358. For each pilot minigrid (Outputs 2.1 and 2.3), Labor Management Procedures and an Occupational Health and Safety Plan will be prepared and applied for the project to ensure labor standards and rights are upheld for project workers. 3359. 3360. The ESIA or targeted assessment will assess the likelihood of this risk and prevalence of child labor within the energy sector in the target area and propose measures to reduce it and find working persons under the age of 18 perform tasks appropriate to their age. 3361. 3362. Awarded project developers will be required to certify that the production and supply chain of used hardware, including PV panels, batteries and other electronic devices, abides to applicable international standards and best practices, included those targeting the prevention of forced and/or child labor as well as other forms of exploitation.	3363. Project Manager
3364. 14	3365. Generation of hazardous waste (specifically e-waste) from the pilot minigrids that have been installed. 3366. 3367. While minigrids are small-scale technology, construction and maintenance involves the use of minor amounts of chemicals (paints, solvents, cleaning liquids, solder). Montreal Protocol chemicals can be present in appliances power by minigrids (i.e., cooling equipment). Persistent organic pollutants will not be used under this project. However, proper work procedures and equipment handling are sufficient measures to prevent releases into the environment. 3368. 3369. In addition, modest amounts of waste will be generated during construction (ground movement and concrete residues); electric wiring and insulator ends;	3372. Social and Environmental - Biodiversity Conservation and Natural Resource Management; Pollution Prevention and Resource Efficiency	3373. I = 3, L = 3 3374. (moderate)	3375. This risk will be assessed in the ESIA or targeted assessment that will be undertaken for each pilot minigrid (Outputs 2.1 and 2.3), such that the ESMP will include a Waste Management Plan detailing the procedures for disposal of all types of waste associated with construction and operation of the pilot minigrids. The plan shall hook on and implement already-existing closed chains with Extended Supplier Responsibility. It is worth noting that according to Law, MG concessionaires after finalization should restore sites to their original conditions.	3376. Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
	broken or rejected parts and components. It is important to note that waste management in Benin is making progress, trying to eliminate open dumps, collect and transfer waste to specific spaces for treatment and revalorization 3370. 3371. Operation of minigrids will lead to the generation of different types of waste, in particular electronic waste ("e-waste") in the form of solar panels and/or batteries at the end of their useful lives will be generated. Without proper handling directives, disposal and/or recycling mandate for obsolete equipment, this could result in additional waste generation, including of hazardous/phase-outs materials, chemicals or other pollutants (e.g. from batteries). Failure to recycle non-hazardous waste could also contribute to additional waste generation.				
	3377. DREI Risk Categories				
3378. 1	3379. Energy market risk. 3380. 3381. Key challenges or Benin's electricity sector include: (i) weak public expenditure linked to Benin's position as an LDC; (ii) low purchase capacity affecting customers' capacity to pay for electricity service; (iii) nascent industrial sector including agri-processing; (iv) large unserved/underserved territory where national grid expansion is not economically feasible; and (v) insufficient collateral and/or guarantee schemes to secure (private) investment beyond electricity generation (IPP). By consequence, capital investments require careful planning and scenario analysis to ensure economic viability and reduce risks. 3382. 3383. All equipment used in Benin shall be compliant with technical standards. This is work in progress, although current projects will follow international standards. Household appliances on the market are often from informal imports and may not meet applicable standards, and may lead to consumer dissatisfaction and safety issues.	3384. Strategic, Political, 3385. Regulatory	3386. I = 3; L = 2 3387. (moderate)	3388. The Project will apply the DREI framework to support the national dialogue and provide specific recommendations for derisking policy. Data collected by the AMP can help defining financial and technical parameters for minigrid concessions and contracts. The Project will further contribute to the definition and adoption of technical standards for minigrid equipment and installations. 3389.	3390. Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
3391. 2	3392. Social Acceptance risk 3393. 3394. Many people in Benin are aware of minigrids and RE technologies. Projects including PANA Energie aim to ensure operational feasibility of distributed RE technologies and promote market uptake. Previous programs led by Government have increased awareness of the minigrid solution for off-grid electrification. 3395. 3396. However, performance issues have caused some households to seek alternative solutions such as gasoline generator or PV panels for self-supply (while others fell back to kerosene and batteries). Confidence in the minigrid model therefore has to be lifted again in these communities. 3397. 3398. Surveys are needed to determine people´s willingness to pay, price elasticity, and acceptable tariff levels. Detailed studies and recommendations can help minigrid developers to prepare financially feasible proposals for concession.	3399. Social and Environmental	3400. I = 3; L = 2 3401. (moderate)	3402. The AMP pilot is aimed at demonstrating the technical and financial performance of minigrids in Benin, define the conditions for sustainability and boost trust among all stakeholders including end-users. 3403.	3404. Project Manager
3405. 3	3406. Hardware Risk 3407. 3408. Earlier minigrids have been identified which are in a deteriorated state and underperform, the causes thereof not being clarified. Caution is therefore needed to ensure long-term technical sustainability.	3409. Other (technical)	3410. I = 4; L = 3 3411. (substantial)	3412. Although technically proven, procurement of minigrid equipment and installation, as other capital goods, shall consider warranties against defects, malfunctioning and underperformance to defer those risks that cannot or should not be borne by the operator. Approaches include extended warranties, service contracting, engineering, procurement and construction (EPC). 3413.	3414. Project Manager
3415. 4	3416. Digital Risk 3417. 3418. There exists a risk that electricity sector entities would not be able to take full benefit of digital solutions or that systems would not be continued post-project. Gaps at the beneficiary level may impede achieving the potential impact of digital technologies. 3419. 3420. Digital technologies and telecommunication are governed by the *Ministère de l'Economie Numérique et de la Digitalisation (MEND)*, which seeks to increase	3423. Strategic	3424. I = 3; L = 2 3425. (moderate)	3426. At GOB (IP) level the Project will carry out an assessment of the business processes related to the scope of the Digital Platform as a basis for functional specification. It will further assess the current digital infrastructure to propose a solution that is adequate and cost-effective, and can be maintained post-project (sustainability). 3427. 3428. In the pilot area, the Project will assess current uses of digital (telecom) services and seek synergies for expanding their use with minigrid electricity acting as an enabling technology.	3431. Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
	equitable and inclusive access to communication services to all people and businesses. Most people can use text (SMS) communication protocols, for example to access market information, or transmit system status data. Sector policies have been developed, for example in the field of e-agriculture including the implementation of digital platforms managing sector data and agents. Notably, e-agriculture is viewed by public and private sector as a gamechanger for developing Benin's agricultural potential. 3421. 3422. As the framework and procedures for digital tendering and monitoring of minigrids are not fully developed in Benin, project partners need to become familiar with aspects such as data quality, analysis and aggregation of data, confidentiality, and institutional set-up for data management. Coordination and capacity gaps expectedly exist, which need to be assessed, remedied and/or worked around.			3429. 3430. Training in the use of digital systems is considered to maximize proficiency by all beneficiaries.	
3432.5	3433. Labor Risk 3434. 3435. Operation and maintenance of minigrid systems requires fast responses to maintain service standards. Earlier minigrids under public programmes experienced difficulties to keep the systems up and running, probably due to a combination of factors including insufficient technical skills, lack of supplies and spare parts, and insufficient institutional back-up (including funding constraints). 3436. 3437. Professional skills including background knowledge and proficiency with O&M of minigrids need to be maintained and updated regularly; similarly, management skills and competences need to be adequate to meet the challenges faced by minigrid operators. Contract terms require minigrid concessionaires to have a local representative; it may prove challenging to find such person.	3438. Organizational	3439. I = 3; L = 2 3440. (moderate)	3441. RE projects must be executed by certified staff according to the Electricity Law. In addition, the Project's ESMF shall be adhered to as a safeguard to ensure professional quality and controlling labor risks to the extent possible. 3442. 3443. Complementary to baseline initiatives into this direction, the Project will further engage with vocational training institutes and programmes to foster curriculum development and professional certification.	3444. Project Manager
3445.6	3446. Developer Risk 3447.	3453. Strategic	3454. I = 4; L =	3456. Activity: 3457. The business model for minigrid concessionaires	3462. Project

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
	3448. Government has limited experience with tendering of private minigrid concessions. Systemic challenges include: (i) public sector budget and capacity constraints which pose at risk consolidation of know-how and expertise within the public sector (ABERME); (ii) transparency to assure of tender process quality; and (iii) limited experience and procedures for monitoring of performance-based contracts. Automatization of the tendering process by means of digital technologies can help reduce the administrative burden on incumbent authorities, while improving transparency and accountability. 3449. 3450. Since income levels in rural Benin are lower than in some other AMP countries, operators may face lower revenue streams at a similar CAPEX and OPEX level. A local anchor tenant can improve financial sustainability and subsidize other consumers during MG growth phase; however, such larger consumers are not widespread in rural Benin. 3451. 3452. The ownership risk under the concession model appears well controlled. All assets are depreciated annually, and in case of public fixed assets, the Concessionaire is hold to reinvest to maintain their value/lifetime. Arbitrage is foreseen in case both parties do not agree on asset value.		3 3455. (substantial)	in Benin is typical for the AMP. As the delivery model for minigrids in Benin is clearly defined, the main problem for project developers is financial sustainability. Careful system design, including a modular approach, cost-saving measures and PUE can help reduce financial exposure. 3458. 3459. A second potential risk concerns delays in project development and permitting processes, which may deter investors. The AMP aims to improve tendering, permitting and monitoring processes by enhancing GOB capacities including through digital solutions. Experience with private minigrids in Benin is currently being gained with funding from the MCA Benin-II and the SEforAll UEF. As such, the developer risk can be closely monitored over the next years and experiences incorporated into DREI analyses and resulting mitigation activities 3460. 3461.	Manager
3463.7	3464. End-user Credit Risk 3465. 3466. Many people in rural areas are not credit-worthy as a result of poor purchase capacity and absence of collateral. However, they may wish to acquire electric appliances. 3467. 3468. Credit lines from commercial banks are available in the country. There are no tools in place to assess end-users' ability to pay for electricity services and required appliances (such as credit score cards).	3469. Financial	3470. I = 2; L = 3 3471. (moderate)	3472. Prepaid meters and SMS payment tools can be used to avoid end-user credit risks. The house wiring, according to a specified user profile, shall be included in the connection. Promising opportunities to boost economic development in communities and leverage customers' money for new services and appliances, can be obtained from synergies between PAYG minigrids and telecom operators. The Project seeks to engage with partners to accelerate the uptake of smart payment solutions. 3473. 3474. In addition, the Project will provide training to domestic financial institutions enabling them to serve rural customers.	3475. Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
3476. 8	3477. Financing Risk 3478. 3479. Low payment capacity in many communities will create challenges for minigrid investors to transit the "valley of death" and become financially sustainable. Private minigrids have just started to build a track record in Benin. This may translate into a high perceived risk concerning the technical performance and the business model. 3480. 3481. National financial institutions are not prepared to approve loans to private minigrid developers. Project proposals submitted in recent years recurred to international financiers and programs, not to the local bank sector. Government spending is backed up by multilateral and bilateral development banks and funds.	3482. Financial	3483. I = 4; L = 1 3484. (low)	3485. The Project will assess this risk in more detail through the DREI analysis. As tariffs will be subsidized to cover operational costs and allow for a profit margin, a financial mechanism must be in place to ensure long-term payment thereby reducing investors' risks. 3486. . 3487. Given the large presence of multilateral financiers in the sector (MCA, AfDB, SEforALL), the Project will foster a dialogue to establish such mechanism.	3488. Project Manager
3489. 9	3490. Currency Risk 3491. 3492. The national currency in Benin is the African Financial Community Franc (FCFA). In use in the UEMOA, it is pegged to the Euro (1 FCFA = 0.0015 EUR). As such the exchange rate risk is the same as for USD (GEF grant currency) to EUR. Under normal market conditions, the UEMOA would not like devaluate the FCFA during the Project's time horizon and the lifetime of energy assets. Yet, the weakening of the EUR to USD exchange rate due to the war between Russia and Ukraine may induce EUMOA to reconsider its policy. 3493. 3494. The country is relatively stable and responsive to the recommendations issued by its multilateral partners. Security issues have recently appeared including government overturns in Burkina Faso (2022) and in Mali (2021), and incursions from violent groups into the north of Benin. Security aspects may affect the deployment of AMP activities in the northern part of the country. They may also deter private companies from investing and give rise to a risk premium on capital.	3495. Financial	3496. I = 3, L = 2 3497. (moderate)	3498. No mitigation measures are proposed. 3499. 3500. UNDP and the IP will permanently monitor the situation to protect assets and people	3501. Project Manager

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3209. #	3210. Description	3211. Risk Category	3212. Impact & Probability	3213. Risk Treatment / Management Measures	3214. Risk Owner
3502. 10	3503. Sovereign Risk 3504. 3505. Benin's systemic trade deficit is the result of the country's dependence on imports for energy needs and high value-added goods. Moreover, Benin's exports of agricultural products such as cotton and maize are highly vulnerable to the weather and price fluctuations. The economy is also affected by the COVID-19 pandemic and the higher living costs resulting from the war in Russia-Ukraine. This condition undermines Benin's ability to take loans to finance additional debt. 3506. 3507. The sovereign risk is mitigated by the international lending community. Benin has accessed the IMF's Extended Fund Facility (EFF) and the Extended Credit Facility (ECF) with a financial package to the tune of nearly US$650 million. Without such support, public expenditures in Benin could not be continued.	3508. Financial	3509. I = 4, L = 3 3510. (substantial)	3511. No direct mitigation measures are proposed.	3512. Project Manager
	3513. Other Risks to Project Success and Mitigation Measures				
3514.	3515. (none)	3516.	3517.	3518.	3519.

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3527. Annex 7: Overview of Project Staff and Technical Consultancies

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
		3532. CONSULTANTS
3533. PMU - 3534. Project Manager/Coordinator 3535. (National)	*3536. Full-time (30% of time dedicated to PM)*	3537. The Project Manager has the authority to run the Project on behalf of the PSC and within the constraints laid down by the PSC. The PM is responsible for day-to-day management and decision-making for the Project. The PM's primary responsibility is to ensure that the Project produces the results specified in the Project Document, to the required standard of quality and within specified time and cost constraints. 3538. The PM will be responsible for the overall management of the Project, including the mobilization of all project inputs, supervision of project staff, consultants and sub-contractors. The PM will report to the NPD in close consultation with the assigned UNDP Programme Manager for all of the Project's substantive and administrative issues. From the strategic point of view of the Project, the PM will report on a periodic basis to the PSC in response to NPD instructions. 3539. The PM will provide general support to the NPD who is ultimately responsible for meeting IP obligations under the Project. The PM will perform a liaison role with the government, UNDP and other UN agencies, CSOs and project partners, and with other cooperation agencies providing co-financing. 3540. 3541. <u>Specific responsibilities of the PM include:</u> • To provide direction and guidance to the Project Management Unit (PMU), other project consultants and Responsible Party(ies), if any; • To liaise with the PSC to assure the overall direction and integrity of the Project; • To assume responsibility for Project administration and update the Atlas Project Management module (if external access is made available); • Plan the activities of the project and monitor progress against the approved workplan; • To prepare the project Annual Work Plans (AWPs); • To seek advice and support as needed for Project management, planning and control; • To plan Project activities and monitor progress against the Results Framework and the approved AWP; • To mobilize personnel, goods and services, training and micro-capital grants to initiate activities; • To monitor events as determined in the project monitoring schedule plan/timetable, and update the plan as required; • To provide support for completion of assessments required by UNDP, spot checks and audits; • To monitor financial resources and accounting to ensure the accuracy and reliability of financial reports; • To prepare and submit financial reports to UNDP on a quarterly basis; • To manage requests for the provision of financial resources by UNDP, through advance of funds, direct payments or reimbursement using the Fund Authorization and Certificate of Expenditures (FACE); • To monitor financial resources and accounting to ensure the accuracy and reliability of financial reports; • To manage and monitor the Project risks initially identified and update the status of these risks by maintaining the Project Risk Log;

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
		• To prepare the GEF PIR and submit the final report to the PSC; • To ensure the MTR process is undertaken as per UNDP guidance and submit final MTR report to the PSC; • To identify follow-on actions and submit them for consideration to the PSC; • To ensure the TE process is undertaken as per UNDP guidance and submit the final TE report to the PSC. 3542. 3543. Note: approx. 10% of PM time wall be dedicated to AMP Regional Program activities. 3544. 3545. <u>Required skills and expertise</u> • P-3 level national expert to act as the Project Technical Coordinator for the Project; • A university degree (MSc or PhD) in a subject related to development studies, political sciences, public administration, civil engineering, social sciences, economics, or environmental sciences; • At least 5 years of demonstrable project/programme management experience; • Experience with international cooperation projects; GEF project implementation; or UNDP project management, is an asset; and: • Previous experience with public administration procedures in Benin is an asset. 3546. 3547. <u>Competencies</u> • Excellent leadership, managerial and coordination skills, with demonstrated ability to effectively coordinate large multi-stakeholder projects, including financial and technical aspects; • Ability to engage effectively with a variety of stakeholders across at all levels; • Ability to develop and sustain robust partnerships with project partners and collaborating agencies; • Team player with excellent abilities to effectively manage technical and administrative teams; • Effective supervision skills, specifically of working groups and external consultants in their implementation of technical activities in partnership with stakeholders from GOB and society; • Proven outstanding project management and accounting skills; • Strong drafting, presentation and reporting skills; • Strong communication skills both in presence as through conferencing media, skype, email, etc; • Strong computer proficiency, in particular the use of office applications; internet search; conferencing tools; and document management; • Good knowledge of the political and socio-economic context of Benin; and: • Excellent command of French and English language. *3548.*
	3549. Full-time (70% of time dedicated to Techni	3550. Specific technical activities for this function include: • To engage with GOB authorities and other stakeholders on minigrid development and regulation; • To promote inter-institutional coordination and partnerships to support deployment of minigrid solutions in Benin; • To draft terms of reference and work specifications, and overseeing all contractors' work; • To supervise contracted activities and verify the quality of delivered services and goods; • To provide technical backstopping for project teams and working groups, specifically for project components 1, 3 and 5;

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
	cal Activities)	• To participate in project activities including: • policy analysis and collaboration in drafting of electrification policy; • participation in donor agency coordination groups and platforms; • participation in activities to develop financing schemes for minigrids under guidance of UNDP country office; • participation in design of AMP DREI exercises including design of policy and regulatory instruments; • participation in working groups with stakeholders from education and academic sector; • engage with AMP Communities of Practice and other regional for a and groups; • organization of events and workshops and purview of contracted services; • To assure the quality of project activities, including consultancies and procurement; • To identify new risks as appropriate for consideration of the PSC to decide on possible action, as and if required; • To capture lessons learned during project implementation; • To act as focal person for coordination and oversight of minigrid pilot projects including stakeholder engagement, communication plans, and timing, and provide feedback to PSC if needed; • To act as focal person for the environmental and social safeguards, and gender experts; and: • To liaise with the AMP Regional Project PMU Staff to request and receive operational and technical support as needed, to participate in activities led by the AMP Regional Project, and share data and information with the AMP Regional Project as required.
3551. PMU – *3552. Project Finance and Administrative Officer (PA)* *3553. (National)*	*3554. Part-time (50%)*	3555. Under the guidance and supervision of the Project Manager, the Project Finance and Administrative Officer will carry out the following tasks: 3556. 3557. <u>Specific responsibilities of the PA include:</u> • Assist the Project Manager in day-to-day management and oversight of project activities; • Provide PMU-related administrative and logistical assistance; • Assist in the preparation of progress reports; • Ensure all project documentation (progress reports, consulting and other technical reports, minutes of meetings, etc.) are properly maintained in hard and electronic copies in an efficient and readily accessible filing system, for when required by PSC, UNDP, and PMU staff; • Keep records of project funds and expenditures, and ensure all project-related financial documentation are well maintained and readily available when required by the Project Manager; • Review project expenditures and ensure that project funds are used in compliance with the Project Document and GOJ financial rules and procedures; • Validate and certify FACE forms before submission to UNDP; • Provide necessary financial information as and when required for project management decisions; • Provide necessary financial information during project audit(s); • Review annual budgets and project expenditure reports, and notify the Project Manager if there are any discrepancies or issues; • Consolidate financial progress reports submitted by the responsible parties for implementation of project activities; and:

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
		• Liaise and follow up with the responsible parties for implementation of project activities in matters related to project funds and financial progress reports. 3558. 3559. Note: approx. 10% of PA time will be dedicated to AMP Regional Program activities. Among other elements, this includes timely reporting of data to the regional project staff per the AMP Regional Project's M&E protocols. 3560. 3561. <u>Expertise, skills and competencies:</u> • A Bachelor's degree or an advanced diploma in accounting/ financial management; • At least 5 years of relevant working experience, preferably in a project management setting involving multi-lateral/ international funding agency. Previous experience with UN projects will be an asset; • Very good inter-personal skills; • Proficiency in the use of computer software applications especially MS Word and MS Excel; and • Excellent language skills in French; proficiency in English (writing and reading).
3562. PMU – 3563. Project Procurement Specialist (PS) 3564. (National)	*3565. Part-time (50%)*	3566. Under the guidance and supervision of the Project Manager, the Project Procurement Specialist (PS) will provide administrative support and manage the project's finances as well as procurement of goods and services required by the Project, ensuring transparency and integrity. 3567. 3568. <u>Specific responsibilities include:</u> • Very good inter-personal skills; • Provide inputs to preparation of the Project Procurement Plan based on the Annual Work Plans and its timely and regular updating according to the project activities; • Organize procurement processes including preparation and conduct of RFQs, ITBs or RFPs, receipt of quotations, bids or proposals, their evaluation, negotiation of certain conditions of contracts in full compliance with UNDP rules and regulations. Support the procurement process to ensure that it is fair, competitive, transparent, ethical and provides best value for money as well as the procurement action is in accordance with UNDP rules and regulations, procedures and instructions; • Keep records of project funds and expenditures and ensure all project-related financial documentation are well maintained and readily available when required by the NPC; • Review project expenditures and ensure that project funds are used in compliance with the Project Document and government financial rules and procedures; • Validate and certify FACE forms before submission to UNDP; • Provide necessary financial information as and when required for project management decisions; • Provide necessary financial information during project audit(s); • Review annual budgets and project expenditure reports and notify the NPC if there are any discrepancies or issues; • Consolidate financial progress reports submitted by the responsible parties for implementation of project activities; • Liaise and follow up with the responsible parties for implementation of project activities in matters related to project funds and financial progress reports. 3569. <u>Expertise, skills and competencies:</u> • Very good inter-personal skills; • Degree or an advanced diploma in accounting/ financial management or a related field;

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
		• At least five years of relevant work experience preferably in a project management setting involving multi-lateral/ international funding agency. Previous experience with UNDP or UN project will be a definite asset; • Proficiency in the use of computer software applications particularly MS Excel; • Excellent command of French, with English skills preferred.
3570. Technical Advisor *3571. (International)*	*3572. As needed*	3573. The Project Technical Advisor (TA) will provide orientation to the IP, UNDP and the other stakeholders by bringing in specific knowhow and expertise on minigrid development including design aspects. The TA will act as a resource person for GOB stakeholders, private sector, CSOs and academic staff during the design and execution of project activities. The TA will form a strong tandem with the PM to push forward minigrid development in Benin. To this purpose, the TA will engage closely with GOB stakeholders to initiate and develop specific topics including minigrid business model development, digital technology approaches, technical standards, tax policy, financial incentives, training needs, opportunities for business development, and waste management strategies. 3574. 3575. Specific responsibilities of the TA include: • To engage with authorities and other stakeholders on minigrid development and regulation; • To promote partnerships to support deployment of minigrid solutions in Benin; • To draft terms of reference and work specifications, and overseeing contractors' work; • To supervise contracted activities and verify the quality of delivered services and goods; • To provide technical backstopping for project teams and working groups; • To ensure a due diligence process during the pilots including application of ESIA measures and gender action plan, and quality assurance (QA) as it relates to the provision of goods and services; • To be responsible for design of data collection programme in alignment with the AMP guidelines and protocols; • To supervise solar PV battery and waste management activities; • To act as the direct superior for the Project Engineer (PE) assigned to the pilots; • To act as resource person during Project events and workshops; • To be deployed in project activities including: o collaborate in drafting of minigrid and off-grid electrification policy; o participation in design of technical standards and regulation; o participation in design of financial instruments and incentives; o participation in missions and analytical work as part of the AMP DREI exercise; o lead the installation and operationalization of the data acquisition/repository package on behalf of the Project and its national counterparts; o specification of ICT equipment and software and participation in procurement process; o initiate studies for the pilots and integrate the results into procurement documents or calls for tender; o functional specification of equipment and participation in procurement process;

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
		○ reception of minigrid equipment and related services and installations, in accordance with established UNDP and/or GOB procedures and rules; ○ identify operational issues and risks during minigrid construction and operation, and development of solutions in collaboration with involved stakeholders; ○ organize data analysis activities and daily monitoring of indicators; and: ○ participate in Communities of Practice; • To assure the quality of project activities, including consultancies and procurement; • To act as focal person for the environmental and social safeguards and gender experts; and • To capture lessons learned during project implementation. 3576. 3577. Required skills and expertise • P-3 level international expert to act as the Project eMobility Expert for the Project; • University degree (MSc or PhD) in engineering (industrial, civil, electrical or mechanical engineer), or similar; • At least 3 years' experience with applied research of electric minigrid systems at public or private organizations; or: • At least 3 years' experience with the design and execution of minigrid project development for the public or private sector; • Experience with international cooperation projects; GEF project implementation; or UNDP project management is an asset; and: • Experience with public administration procedures in Benin is an asset. 3578. 3579. Competencies • Strong managerial skills, with demonstrated ability to effectively coordinate technical, environmental and financial aspects of mobility projects; • Ability to engage effectively with a variety of stakeholders across at all levels; • Team player with excellent abilities to effectively manage and participate in technical teams; • Ability to translate identified requirements into functional and technical specifications for external suppliers; • Leadership skills to steer contractors and suppliers; enforce agreed terms and quality standards; and agility to manage and resolve issues and disputes; • Strong drafting, presentation and reporting skills; • Strong communication skills both in presence as through conferencing media, skype, email, etc; • Strong computer proficiency, in particular the use of office applications; internet search; conferencing tools; and document management; • Good knowledge of the political and socio-economic context of Benin; and: • Excellent command of French and English language.
3580. Project Engineer	*3582. Full-time (2*	3583. The Project will hire an international project engineer (PE) to assist during the design and implementation of the minigrid pilots. The PE will work under the direct guidance and supervision of Project Technical Advisor (TA). National staff designated by the

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
3581. (international)	*years)*	Implementing Partner, the TA and PE together form the Technical Unit to prepare, guide, and monitor the minigrids targeted by the Project. 3584. 3585. <u>The tasks and responsibilities of the Project Engineer include:</u> • Technical assistance to project partners for minigrid development and design (GEF Project Component 2); • Development of conceptual and technical studies as input for scoping, design and engineering of the pilots; • Review of studies and reports submitted by external suppliers and project partners; • Participate in meetings with technology suppliers and project partners; • Participate in working sessions for minigrid technical standard development; • Under supervision of the Lead Technical Advisor: o support drafting of functional and technical specifications; o participate in reception process of equipment; o supervise installation activities by contractors; o verify compliance with risk mitigation plans including ESIA, Gender Action Plan, waste management plan, and report and address discrepancies; o design the data collection methodology for the envisioned minigrids; o carry out daily monitoring of technical performance of the minigrids in agreement with the respective operators; o analyze obtained data and identify anomalies and events; o act as a resource person during project events and workshops; o capture lessons learned during project implementation. 3586. 3587. <u>Required skills and expertise</u> • International expert to act as the Project Engineer (PE) for the Project; • University degree (preferably MSc level) in engineering (electrical, computer sciences, mechanical engineer, physics), or similar; • At least 2 years' experience with applied research of electric power systems ICT systems, energy systems, or similar, at public or private organizations; • At least 1 year experience with the design and execution of technology projects (ICT, energy, or similar) for the public or private sector; and: • Demonstrated experience and up-to-date knowledge of data acquisition technologies and communication protocols; demonstrated experience with the handling and analysis of large data sets; and: • Proficiency with data analysis software and modelling of scenarios. 3588.

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
		3589. Competencies • Strong engineering skills; • Accurate work attitude including filing of information; • Ability to break complex technical problems into smaller pieces; • Ability to translate customer needs into technical solutions; • Strong communication skills both in presence as through conferencing media, skype, email, etc; • Strong computer proficiency, in particular the use of office applications; internet search; conferencing tools; and document management; and: • Excellent command of French and English language.
3590. MG /DREI Expert *3591. (National)*	*3592. Full-time (2 years)*	3593. The Project will hire a national expert (MG) to assist during the design and implementation of the minigrid pilots. The MG will work under the direct guidance and supervision of Project Technical Advisor (TA) and the Project Engineer (PE). National staff designated by the Implementing Partner, MG, TA and PE together form the Technical Unit to prepare, guide, and monitor the minigrids targeted by the Project. The role of the MG is to assist the international experts and act as an anchor point for building competences on minigrid development in the country. This national expert will collaborate in the DREI exercise initiated by the Regional Programme. 3594. 3595. The tasks and responsibilities of the Project Engineer include: • Technical assistance to project partners for minigrid development and design (GEF Project Component 2); • Contributions to studies as input for scoping, design and engineering of the pilots; • Review of studies and reports submitted by external suppliers and project partners; • Participate in meetings with technology suppliers and project partners; • Participate in working sessions for minigrid technical standard development; • Under supervision of the Lead Technical Advisor: o support drafting of functional and technical specifications; o participate in reception process of equipment; o collaborate in the design of data collection systems for the envisioned minigrids; o carry out daily monitoring of technical performance of the minigrids in agreement with the respective operators; o analyze obtained data and identify anomalies and events; o act as a resource person during project events and workshops; o capture lessons learned during project implementation. 3596. 3597. Required skills and expertise • National expert to act as the MG/DREI expert (MG) for the Project;

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3529. Consultant	3530. Time Input	3531. Tasks, Inputs and Outputs
		• University degree (BSc level) in engineering (electrical, computer sciences, mechanical engineer, physics), or similar; • At least 2 years' work experience in a relevant branch at public or private organizations; • Knowledge of data acquisition technologies and communication protocols. 3598. 3599. Competencies • Strong engineering skills; • Accurate work attitude including filing of information; • Ability to translate community needs into technical solutions; • Strong communication skills both in presence as through conferencing media, skype, email, etc; • Computer proficiency, in particular the use of office applications; internet search; conferencing tools; and document management; and: • Excellent command of French language.
3600. SES Expert *3601. (International)*	*3602. Part-time (as needed)*	3603. Duties and Responsibilities • Monitor progress in development/implementation of the project ESMP/ESMF ensuring that UNDPs SES policy is fully met and the reporting requirements are fulfilled; • Oversee/develop/coordinate implementation of all safeguard related plans; • Ensure social and environmental grievances are managed effectively and transparently; • Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary; • Ensure full disclosure with concerned stakeholders; • Ensure environmental and social risks are identified, avoided, mitigated and managed throughout project implementation; and: • Work with the M&E officer to ensure reporting, monitoring and evaluation fully address the safeguard issues of the project.
3604. Gender Expert *3605. (National)*	*3606. Part-time (as needed)*	3607. Duties and Responsibilities • Monitor progress in implementation of the project Gender Action Plan ensuring that targets are fully met and the reporting requirements are fulfilled; • Oversee/develop/coordinate implementation of all gender-related work; • Review the Gender Action Plan annually, and update and revise corresponding management plans as necessary; and: • Work with the M&E officer and Safeguards Officer to ensure reporting, monitoring and evaluation fully address the gender issues of the project.
		3608. Contracted Services - Companies
3609.	*3610.*	• SPECIFIC TERMS WILL BE DRAFTED DURING PROJECT IMPLEMENTATION

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Annex 8: Stakeholder Engagement Plan

See separate file.

Annex 9: Environmental and Social Management Framework (ESMF).

See separate file.

Annex 10: Gender Analysis and Gender Action Plan

See separate file.

Annex 11: Procurement Plan

See separate file.

Annex 12: GEF focal area specific annexes

See separate file.

Annex 13: Co-financing letters

See separate file.

Annex 14: Additional agreements

None

Annex 15: Signed LOA between UNDP and IP requesting UNDP Support Services

None

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3642.

3643. Annex 16: GEF Core indicators

3644.

3645. Core Indicator 6	**3646. Greenhouse gas emission mitigated**					**3647. *(Metric tons of CO_2e)***
3648.	3649.		*3650.* Expected metric tons of CO_2e (6.1+6.2)			
3651.	3652.		3653. PIF stage	3654. Endorsement	3655. MTR	*3656.* TE
3657.	3658. Expected CO2e (direct)		*3659.* 13,859	*3660.* 9,056	3661.	*3662.*
3663.	3664. Expected CO2e (indirect)		*3665.* 178,402	*3666.* 179,273	3667.	*3668.*
3669. Indicator 6.4	3670. Increase in installed renewable energy capacity per technology					3671.
3672.	3673.	3674. Technology	3675. Capacity (MW/MWh)			
			3676. Expected		3677. Achieved	
			3678. PIF stage	3679. Endorsement	3680. MTR	3681. TE
3682.	3683.	3684. Solar Photovoltaic	3685. 0.280 MW	3686. 0.368 MW	3687.	3688.
3689.	3690.	3691. Energy Storage	3692. 0 MWh	3693. 0.892 MWh	3694.	3695.
3696. Core Indicator 11	**3697. Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment**					**3698. *(Number)***
3699.	3700.	3701.	3702. Number			
			3703. Expected		3704. Achieved	
3705.	3706.	3707.	3708. PIF stage	3709. Endorsement	3710. MTR	3711. TE
3712.	3713.	3714. Female	*3715.* 50.8%	*3716.* 50%	3717.	3718.
3719.	3720.	3721. Male	3722. 49.2%	3723. 50%	3724.	3725.
3726.	3727.	3728. *Total*	*3729.* 13,977	*3730.* 12,152	3731.	3732.

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3736. Annex 17: GEF 7 Taxonomy

3737.

3738. Level 1	3739. Level 2	3740. Level 3	3741. Level 4
3742. ☒ Influencing models	3743.	3744.	3745.
3746.	3747. ☒ Transform policy and regulatory environments	3748.	3749.
3750.	3751. ☒ Strengthen institutional capacity and decision-making	3752.	3753.
3754.	3755. ☒ Convene multi-stakeholder alliances	3756.	3757.
3758.	3759. ☒ Demonstrate innovative approaches	3760.	3761.
3762.	3763. ☒ Deploy innovative financial instruments	3764.	3765.
3766. ☒ Stakeholders	3767.	3768.	3769.
3770.	3771. ☐ Indigenous Peoples	3772.	3773.
3774.	3775. ☒ Private Sector	3776.	3777.
3778.	3779.	3780. ☒ Capital providers	3781.
3782.	3783.	3784. ☒ Financial intermediaries and market facilitators	3785.
3786.	3787.	3788. ☒ Large corporations	3789.
3790.	3791.	3792. ☒ SMEs	3793.
3794.	3795.	3796. ☒ Individuals/Entrepreneurs	3797.
3798.	3799.	3800. ☐ Non-Grant Pilot	3801.
3802.	3803.	3804. ☐ Project Reflow	3805.
3806.	3807. ☒ Beneficiaries	3808.	3809.
3810.	3811. ☒ Local Communities	3812.	3813.
3814.	3815. ☒ Civil Society	3816.	3817.
3818.	3819.	3820. ☒ Community Based Organization	3821.
3822.	3823.	3824. ☐ Non-Governmental Organization	3825.
3826.	3827.	3828. ☐ Academia	3829.
3830.	3831.	3832. ☐ Trade Unions and Workers Unions	3833.
3834.	3835. ☒ Type of Engagement	3836.	3837.
3838.	3839.	3840. ☒ Information Dissemination	3841.
3842.	3843.	3844. ☒ Partnership	3845.
3846.	3847.	3848. ☒ Consultation	3849.
3850.	3851.	3852. ☒ Participa	3853.

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3738. Level 1	3739. Level 2	3740. Level 3	3741. Level 4
		tion	
3854.	3855. ☒Communications	3856.	3857.
3858.	3859.	3860. ☒Awareness Raising	3861.
3862.	3863.	3864. ☒Education	3865.
3866.	3867.	3868. ☒Public Campaigns	3869.
3870.	3871.	3872. ☒Behavior Change	3873.
3874. ☒Capacity, Knowledge and Research	3875.	3876.	3877.
3878.	3879. ☒Enabling Activities	3880.	3881.
3882.	3883. ☒Capacity Development	3884.	3885.
3886.	3887. ☒Knowledge Generation and Exchange	3888.	3889.
3890.	3891. ☐Targeted Research	3892.	3893.
3894.	3895. ☒Learning	3896.	3897.
3898.	3899.	3900. ☒Theory of Change	3901.
3902.	3903.	3904. ☒Adaptive Management	3905.
3906.	3907.	3908. ☒Indicators to Measure Change	3909.
3910.	3911. ☒Innovation	3912.	3913.
3914.	3915. ☒Knowledge and Learning	3916.	3917.
3918.	3919.	3920. ☒Knowledge Management	3921.
3922.	3923.	3924. ☒Innovation	3925.
3926.	3927.	3928. ☒Capacity Development	3929.
3930.	3931.	3932. ☒Learning	3933.
3934.	3935. ☒Stakeholder Engagement Plan	3936.	3937.
3938. ☒Gender Equality	3939.	3940.	3941.
3942.	3943. ☒Gender Mainstreaming	3944.	3945.
3946.	3947.	3948. ☒ Beneficiaries	3949.
3950.	3951.	3952. ☒Women groups	3953.
3954.	3955.	3956. ☒Sex-disaggregated indicators	3957.
3958.	3959.	3960. ☒Gender-sensitive indicators	3961.
3962.	3963. ☒Gender results areas	3964.	3965.
3966.	3967.	3968. ☐Access and control over natural resources	3969.
3970.	3971.	3972. ☒Participation and leadership	3973.

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Level 1	Level 2	Level 3	Level 4
		☒ Access to benefits and services	
		☒ Capacity development	
		☒ Awareness raising	
		☒ Knowledge generation	
☒ Focal Areas/Theme			
	☒ Climate Change		
		☒ Climate Change Mitigation	
			☐ Agriculture, Forestry, and other Land Use
			☒ Energy Efficiency
			☐ Sustainable Urban Systems and Transport
			☒ Technology Transfer
			☒ Renewable Energy
			☒ Financing
			☒ Enabling Activities
		☒ United Nations Framework on Climate Change	☒ Nationally Determined Contribution
	☒ Rio Markers		
		☐ Climate Change Mitigation 0	
		☐ Climate Change Mitigation 1	
		☒ Climate Change Mitigation 2	
		☐ Climate Change Adaptation 0	
		☐ Climate Change Adaptation 1	
		☐ Climate Change Adaptation 2	

Annex 18: Partners Capacity Assessment Tool and HACT assessment

This Annex must completed/prepared as a separate document and completed early in the PPG phase to help inform the design of the project. It must be made available to the LPAC members. It does not need to be submitted to the GEF and does not need to be part of the project document that is signed by the relevant parties.

Guidance to project developer *Please refer to the HACT policies for guidance on applicability and financial thresholds (Partners Capacity Assessment Tool and HACT assessment).*

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4083. Annex 19: UNDP Project Quality Assurance Report

4084. This Annex must completed/prepared as a separate document by the Country Office. It must be made available to the LPAC members. It does not need to be submitted to the GEF and does not need to be part of the project document that is signed by the relevant parties.

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XII. Non-Mandatory Annexes

Annex 20: TORs - International Consultant for a Full Derisking Renewable Energy Investment ("DREI") Analysis in Benin

Refer to separate document

Annex 21: TORs - National Consultant for a Full Derisking Renewable Energy Investment ("DREI") Analysis in Benin

Refer to separate document

Annex 22: Project GHG ER and other targets (2022 09 16) for Benin AMP Child Project

Refer separate document